Dynamic AI Signal Policy Design

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Human-in-the-loop AI systems often decide not only whether to signal a decision-maker, but how much assistance to provide over time. More frequent or more prescriptive signals—e.g., solutions rather than hints—can substantially improve current decisions but can also increase alert fatigue and reduce cognitive effort, eroding future performance and skill retention. We study a constrained signal-policy design problem (SPD) in a finite-horizon Markov decision process in which a planner chooses signal intensity for a human operator. We show that reduced-form transition kernels are structurally insufficient for signal design because they do not identify whether downstream consequences arise from action-mediated displacement or from changes in future *efficacy*—the system's state-dependent capacity to improve subsequent decisions, conditional on the realized action. To recover these design-relevant components, we introduce a factored-kernel representation that separates behavioral response, physical transitions, and *efficacy drift*. This structure yields a clear design logic for dynamic signal design: use direct SPD under full specification, and use a structured relative-design (RD) when the planner can evaluate marginal signal upgrades without fully specifying downstream costs across states; under RD, stronger signals are used when their incremental benefit justifies their downstream cost. We formalize this principle by exactly characterizing the optimal penalized ordered-menu policy through a single shadow price. Computational studies in healthcare and engineering reveal a three-regime pattern: when efficacy drift is low, myopic rules based on immediate rewards are nearly optimal; when drift is material and the model is correctly specified, RD retains nearly all of SPD's value; under misspecification, RD can improve realized performance by regularizing the design problem. Our framework provides a disciplined basis for the design and governance of AI signal policies.

Key words: human-AI collaboration, algorithmic advice, dynamic signal design, Markov decision processes, efficacy drift

1. Introduction

Algorithmic and AI decision-support systems are increasingly embedded in operational workflows, from clinical alerts and patient monitoring (Alam et al. 2014, Piri et al. 2022, Li and Piri 2025) to predictive-maintenance recommendations (Saxena et al. 2008, Bousdekis et al. 2021) and operational triage tools (Raghu et al. 2019, Lin et al. 2023). In many of these settings, algorithms do not act autonomously; they provide signals—alerts, hints, or recommendations of varying prescriptiveness—to human operators, who interpret those signals and act (Kesavan and Kushwaha 2020, Snyder et al. 2025). Because these signals are deployed within sequential operational processes, their consequences extend beyond the focal decision and can shape downstream per-

formance (Bastani et al. 2026). The central managerial question is therefore how to design signal policies for overall process performance—when to signal and how strongly.

Assistance, however, cannot be arbitrarily frequent or overly prescriptive. A hospital may seek to improve cumulative treatment quality while keeping alert fatigue and diagnostic overload below a service-level target (Ancker et al. 2017, Li and Piri 2025); a predictive-maintenance team may seek to improve intervention timing while limiting movement toward lower-coverage or harder-to-support operating regions; a triage, routing, or escalation operation may seek to improve prioritization quality while capping operator overload and preserving downstream capability over a deployment cycle (Raghu et al. 2019, Kc et al. 2020, Lin et al. 2023). Similar considerations arise in educational and skill-development settings, where assistance can improve immediate performance while undermining learning and skill development (Bastani et al. 2025, Poulidis et al. 2025a, Chung et al. 2026, Bondi and Johnson 2026). The design problem is therefore naturally constrained: a planner must choose a signal policy that improves cumulative performance subject to an explicit or implicit tolerance for the cumulative cost that signaling imposes.

Signals can affect downstream performance not only by changing current actions and thereby altering future state trajectories (Puterman 1994, Bertsekas 2020, Joshi et al. 2023), but also by changing the system's future capacity to improve subsequent decisions, ceteris paribus, conditional on the realized action. We refer to this state-dependent capacity as *efficacy* and to signal-induced changes in it as *efficacy drift*. Advice that confirms a physician's diagnosis, for example, may preserve cognitive bandwidth or contribute to alert fatigue (Embi and Leonard 2012, Topaz et al. 2016, Ancker et al. 2017, Cha et al. 2020, Li and Piri 2025); even when the signal does not change the physician's immediate action, it may still alter future vigilance, diagnostic effort, or responsiveness (Parasuraman and Riley 1997, Parasuraman and Manzey 2010, Poulidis et al. 2025b).

Once efficacy drift is present, reduced-form transition kernels are insufficient for signal design; they cannot disentangle whether signaling's downstream consequences arise because signals changed current actions—thereby shifting the downstream trajectory—or because signal exposure directly altered future efficacy conditional on the realized action. That distinction is critical for signal-policy design. The same reduced-form continuation evidence can therefore support opposite local signal rankings, depending on how continuation effects are allocated between action-mediated displacement and efficacy drift. A planner who designs signal policies from reduced-form models may therefore choose the wrong policy even when the model fits the observed continuation data.

We study this problem as a finite-horizon Markov decision process in which a planner selects signal intensity from an ordered menu for a human operator. The underlying optimization problem is a constrained signal-policy design problem: choose a dynamic signal policy to maximize cumulative decision quality subject to a budget on the cumulative downstream cost created by

signaling. We refer to this problem as signal-policy design (SPD). Under the maintained finite-horizon finite-state assumptions, SPD is numerically tractable as a constrained Markov decision process. The central obstacle, however, is not tractability but specification. Because reduced-form transition kernels are decision-insufficient, a direct plug-in solution to SPD requires the planner to specify a downstream-cost object over all state–signal pairs. In most deployments, however, that full cost object cannot be specified credibly.

To address this specification problem, we model signal-dependent dynamics through a factored transition kernel that separates three objects: how the operator responds to the signal, how the physical state evolves given the realized action, and how future efficacy evolves given the state, action, and signal. The factorization does not alter the full-information dynamic program. Its value is diagnostic and constructive: it isolates the channels through which signaling matters, clarifies what reduced-form continuation evidence cannot identify, and provides the primitives needed for credible dynamic signal design.

The factored representation and the SPD formulation together imply a selection rule over *design objects* for dynamic AI signal design. The relevant distinction is not tractability but credible specification. When the planner can recover the design-relevant primitives and defend a sufficiently rich downstream-cost object, direct SPD is the correct benchmark. In many deployments, however, the planner can defend aggregate tolerance for signaling cost and relative upgrade judgments against silence, but not a fully flexible cardinal cost schedule over all state–signal pairs. In that regime, the economically credible constructive object is a structured relative-design formulation anchored at silence. We refer to this formulation as Relative Design (RD). RD is therefore introduced not as a tractable surrogate for SPD, but as the correct design object for partial-specification environments.

The supporting results follow that logic. On the diagnostic side, we show that reduced-form continuation evidence is insufficient for signal design because observationally equivalent reduced-form environments can imply opposite local signal rankings, and we derive a decomposition of signaling's value into immediate, displacement, and efficacy-drift channels. On the constructive side, we show that once RD is the selected object, the optimal penalized ordered-menu policy admits an exact characterization through a single shadow price. On the deployment side, we derive a safe-harbor boundary, a near-tie screen, and a one-step approximation with a full-horizon performance guarantee.

Computational illustrations then validate this design logic across calibrated healthcare and engineering benchmarks. When efficacy drift is weak, simple myopic signaling is already near-optimal. When drift is material and the model is correctly specified, direct SPD is the natural upper benchmark and RD retains nearly all of its value. Under named classes of misspecification—over-restrictive burden estimation and order-breaking observation noise—RD can outperform plug-in

SPD by regularizing the design problem. The theory and computational illustrations therefore do not imply that RD should be selected universally over SPD; rather, they show that RD is nearly costless when SPD is credibly specified and can be preferable when it is not. Table 1 summarizes the progression from diagnosis to construction to deployment; Table 3 summarizes the resulting design-object selection rule.

1.1. Related Literature

There has been a surge of interest in human–AI collaboration, that is, in using algorithmic advice to improve human decision-making (Grand-Clément and Pauphilet 2024, Baek et al. 2025). Although AI tools are becoming increasingly capable, humans often remain in the loop. Algorithmic advice may still be unreliable because of misspecification, misalignment with human objectives, cognitive limitations, or exclusion of private information available to the decision-maker (Dai and Abràmoff 2023, Boyacı et al. 2024, Balakrishnan et al. 2025). Even when algorithms are powerful and well calibrated, the human may still need to make the final decision for reasons of accountability or physical constraints (Sun et al. 2022).

Prior work also shows that the performance of human–AI systems depends not only on algorithm quality, but also on how humans respond to AI assistance over time. Advice uptake varies with expertise, context, and presentation (Fügener et al. 2021, Krakowski et al. 2025); complementary work studies how advice is processed under uncertainty, how implementation details shape compliance, and where judgment should enter operational planning (de Véricourt and Gürkan 2023, Siemsen and Spiliotopoulou 2024, DosSantos DiSorbo et al. 2025). Our paper builds on these insights, but goes beyond such formulations by isolating efficacy as a distinct design object and by studying richer advice spaces with signals of varying intensity.

A closely related body of work studies the design of algorithmic advice itself, including how prescriptive signals should be and how signal policies should be deployed. The foundational levels-of-automation taxonomy (Parasuraman et al. 2000) provides conceptual grounding for our ordered signal menu, in which the planner selects among discrete assistance intensities rather than choosing only between full automation and full human control. Related work studies action recommendations, attention-like alerts, disclosure, and selective automation in human–AI systems (Alam et al. 2014, Kesavan and Kushwaha 2020, Agarwal et al. 2023). Our distinction between more and less prescriptive forms of advice is also related to recent work comparing action and attention signals in sequential human–AI settings (Poulidis et al. 2025b). Within this literature, our paper is especially close to Agarwal et al. (2025). Their sufficient-statistic approach to information disclosure and selective automation requires less structural commitment, whereas our factorization imposes more structure but delivers distinct design leverage by separating action-mediated displacement from action-conditioned efficacy drift.

Our paper is also related to work on sequential decision-making and dynamic signal design, where current signals shape future actions, states, or beliefs. More broadly, adjacent work in information design studies how current signals influence future behavior over time (Kamenica and Gentzkow 2011, Ely 2017, Renault et al. 2017, Li and Norman 2021). In operations and control, finite-horizon stochastic-control models provide the methodological foundation for studying dynamic policies (Puterman 1994, Altman 1999), while deteriorating-system and fatigue models capture environments in which current interventions affect future performance or responsiveness (Pierskalla and Voelker 1976, Piri et al. 2022). We study human–AI decision support in which signals affect not only current actions and downstream state trajectories, but also future efficacy conditional on the realized action.

1.2. Contributions

Our paper makes the following contributions to the literature. We formulate a constrained signal-policy design problem (SPD) and show that the central obstacle to dynamic signal design is structural: reduced-form transition kernels do not separate action-mediated displacement from efficacy drift conditional on the realized action. We identify this design-relevant distinction through a factored-kernel representation that separates behavioral response, physical transitions, and efficacy drift, and we use that structure to select the appropriate design object. Direct SPD is the correct object under credible full specification. When the planner can evaluate marginal signal upgrades—providing a signal of higher intensity—without fully specifying downstream costs across states, the appropriate constructive object is the structured relative-design formulation (RD). Within that selected object, we exactly characterize the optimal penalized ordered-menu policy through a single shadow price.

The remaining results turn that constructive contribution into a deployment framework. We derive a safe-harbor boundary under signal-independent efficacy, a near-tie screen for when continuation effects can overturn myopic rankings, and a one-step approximation with a full-horizon performance guarantee. Across evidence-anchored healthcare settings and a structurally specified engineering benchmark, the computational studies then reveal a three-regime pattern that maps environments into recommended policies: when efficacy drift is low, myopic rules based on immediate rewards are nearly optimal; when drift is material and the model is correctly specified, RD retains nearly all of the value of solving SPD directly; and under named classes of misspecification, RD can improve realized performance by regularizing the design problem.

2. Model and Design Problem

We study a finite-horizon Markov decision process in which a planner chooses whether and how strongly to signal a human operator. Let $\mathcal{S}$ denote the physical state space, $\mathcal{A}$ the action space, and

$\mathcal{L} = \{0, 1, \ldots, k\}$ an ordered signal menu, where $\ell = 0$ denotes silence and higher integers correspond to more prescriptive signals.[1]

The planner's state is augmented by an efficacy state $c \in \mathcal{C}$. We therefore write $\mathcal{X} := \mathcal{S} \times \mathcal{C}$ for the augmented state space and $x = (s, c) \in \mathcal{X}$ for a generic planner state. The efficacy state summarizes the system's current capacity to improve decisions. It may capture operator-side factors such as alert fatigue, cognitive bandwidth, diagnostic readiness, or skill retention, as well as algorithm-side factors such as local model quality or coverage.[2] In the main analysis, we treat c_t as observed by the planner; Section 4.3 quantifies how the design degrades when the planner instead works with an imperfect proxy $\hat{c}_t$. Throughout, we assume a finite horizon $T < \infty$, finite spaces $\mathcal{S}$, $\mathcal{C}$, and $\mathcal{A}$, and a finite ordered signal menu $\mathcal{L}$; these finiteness assumptions underpin the occupancy-measure LP formulation, finite-dimensional LP duality, and the piecewise-affine frontier characterization developed later.

At each decision epoch, the operator chooses an action $a \in \mathcal{A}$ after observing signal intensity $\ell \in \mathcal{L}$. Three kernels govern the dynamics. The behavioral-response kernel $\mu(a \mid s, c, \ell)$ describes how the operator responds to the signal. The physical transition kernel $F(s' \mid s, a)$ governs the evolution of the physical state after action a.[3] The efficacy kernel $K(c' \mid c, s, a, \ell)$ governs how signal exposure changes future efficacy.

The planner controls signal intensity $\ell \in \mathcal{L}$, not the realized action $a \in \mathcal{A}$. Control is therefore indirect: ℓ shifts the action distribution through μ, and the realized action jointly determines physical evolution through F and efficacy evolution through K. A *signal policy* is a sequence of decision rules $\pi = (\pi_1, \ldots, \pi_T)$, where $\pi_t \colon \mathcal{S} \times \mathcal{C} \to \mathcal{L}$ specifies the signal level chosen at stage t for planner state (s, c).

2.1. The Constrained Signal-Policy Design Problem (SPD)

We begin with the planner's underlying design problem. For initial state distribution ν, the planner chooses a signal policy π to maximize cumulative decision quality subject to a cumulative downstream-cost budget:

$$\underset{\pi}{\text{maximize}} \quad \mathbb{E}_\nu^\pi \left[\sum_{t=1}^{T} \bar{R}(X_t, \ell_t) \right] \tag{SPD}$$

[1] We identify menu levels with integers, so that arithmetic expressions refer to adjacent levels in the natural order. The baseline model treats signal content as fixed within each level; an extension in which the planner chooses both content and intensity is discussed in Appendix A.5. In operator-side settings, this ordering is substantive: more prescriptive signals substitute more strongly for the operator's own diagnostic or problem-solving effort, so they can generate larger immediate gains but also greater learning loss (Bastani et al. 2025, Poulidis et al. 2025a). Appendix E formalizes this intuition by showing that, under primitive monotonicity conditions, more prescriptive signals induce weakly greater downstream unfamiliarity and weakly larger downstream cost.

[2] We use a single aggregate state to keep the design problem and estimation requirements tractable; richer decompositions into $\mathcal{C}_H \times \mathcal{C}_A$ are natural extensions deferred to Appendix A.1.

[3] The kernel $F(s' \mid s, a)$ is signal-independent conditional on the realized action, so the signal affects the physical environment only through the action it induces.

$$\text{subject to} \quad \mathbb{E}_{\nu}^{\pi}\left[\sum_{t=1}^{T} B(X_t, \ell_t)\right] \leq \overline{B}.$$

Here, $\bar{R}(x, \ell)$ denotes expected immediate decision quality at planner state $x = (s, c)$ under signal level ℓ, and $B(x, \ell)$ denotes the corresponding downstream cost. The scalar $\overline{B} \geq 0$ is the planner's horizon-wide cost budget—an exogenous cap on expected total downstream cost, such as an alert-fatigue allowance, a vigilance-loss ceiling, or a capability-erosion limit.

Problem (SPD) is the paper's primary optimization problem. It is the relevant object when the organization seeks to improve cumulative decision quality subject to an explicit or implicit limit on the cumulative cost induced by signaling. In clinical decision support, that cost may take the form of alert fatigue, vigilance loss, diagnostic overload, or capability erosion. In predictive maintenance, it may arise through movement toward lower-coverage or harder-to-support operating states. In triage, routing, or escalation systems, it may arise through operator overload and degradation of downstream judgment capability. In these settings, downstream cost is not merely a state variable; it is itself a managed organizational resource or risk budget. If, instead, the application admits a single fully monetized objective with no separate cost constraint, the relevant benchmark is the unconstrained MDP ($\overline{B} = \infty$); we study the constrained formulation because most deployments operate under explicit or implicit caps on cumulative cost.

Under the maintained finite-horizon finite-state assumptions, (SPD) is numerically tractable as a constrained Markov decision process. The unconstrained case is a standard augmented-state MDP. The constrained case admits a standard occupancy-measure LP formulation (see Altman 1999), with dual multiplier $\lambda^{\star} \geq 0$ interpreted as the shadow price of tightening the downstream-cost budget. The paper's contribution therefore lies not in solving SPD computationally, but in making dynamic signal design credible and governable.

That distinction shifts the central question from numerical solvability to credible specification. In the finite-horizon finite-state setting studied here, the issue is not how to solve a constrained Markov decision process, but whether the inputs required by direct SPD can be credibly defended. Direct SPD requires a fully flexible downstream-cost object $B(x, \ell)$ over all state–signal pairs. In many deployments, however, the planner can govern aggregate downstream cost and defend relative upgrade judgments without responsibly assigning cardinal cost values throughout the full state–signal space.

That asymmetry points to a relative-design (RD) formulation. We preview it here; the factored-kernel representation needed to formalize it is introduced in Section 2.3, and the full ordered-menu analysis is developed in Section 3.2. Once a benchmark silence level ℓ_0 and a bounded downstream unfamiliarity index $\mathcal{U}(s, c) \in [0, 1]$ are fixed, the planner can compare each upgrade

against a common, auditable reference rather than assign cardinal costs across the full state–signal space. Define

$$G(x,\ell) := \bar{R}(x,\ell) - \bar{R}(x,\ell_0) \qquad \text{and} \qquad \Psi(x,\ell) := \mathbb{E}[\mathcal{U}(S_{t+1},C_{t+1}) \mid x,\ell] - \mathbb{E}[\mathcal{U}(S_{t+1},C_{t+1}) \mid x,\ell_0].$$

Then $G(x,\ell)$ is the immediate gain from upgrading assistance relative to silence, and $\Psi(x,\ell)$ is the corresponding relative downstream cost. The resulting formulation preserves the constrained organizational logic of SPD while restricting attention to the part of the cost object that can be credibly defended. The ordered-menu geometry developed later follows from this relative-to-silence construction, and Section 4 validates the resulting deployment rule.

2.2. Why Reduced-Form Structure Is Not Enough

Under full knowledge of the transition law on (s,c), the unconstrained version of (SPD) is a standard augmented-state MDP, and the constrained version is a standard finite-horizon CMDP with an occupancy-measure LP formulation. The difficulty, however, is structural. Reduced-form continuation effects do not identify whether downstream consequences arise from action-mediated displacement—where the signal changes the current action and thereby alters the physical trajectory—or from efficacy drift conditional on the realized action—where signal exposure changes future efficacy even holding the action fixed. Without that separation, the same reduced-form evidence can support opposite design choices.

For two signal levels ℓ_0, ℓ_1, define the induced reduced-form efficacy transition at planner state $x = (s,c)$ by

$$\bar{P}_m(\cdot \mid x) := \sum_{a\in\mathcal{A}} \mu(a \mid x,\ell_m)\, K(\cdot \mid c,s,a,\ell_m), \qquad m \in \{0,1\}.$$

The proposition below is a decision-theoretic statement about *signal rankings*, not merely an identification observation. Two structural environments can generate the same reduced-form efficacy-transition evidence yet imply opposite optimal signal rankings at the same state. A planner who designs signal policies using only the reduced-form continuation kernel can therefore be optimal in one environment and strictly suboptimal in the other. The relevant reduced-form object is the efficacy transition because the observational equivalence is driven by how signal dependence is allocated between μ and K once the action-conditioned physical kernel F is held fixed.

Proposition 1 **(Reduced-form decision insufficiency)**. *Fix a planner state $x = (s,c)$ and two ordered signal levels $\ell_0 < \ell_1$ in $\mathcal{L}$. Suppose $|\mathcal{A}| \geq 2$, $|\mathcal{C}| \geq 2$, and $\bar{P}_0(\cdot \mid x) \neq \bar{P}_1(\cdot \mid x)$. Then the same reduced-form efficacy transitions $\bar{P}_0$ and $\bar{P}_1$ can arise from observationally equivalent factorizations with* opposite *optimal signal rankings at x: in one factorization the higher signal ℓ_1 is strictly beneficial; in the other, it is strictly harmful. In particular, in one factorization the efficacy-drift channel is absent (future efficacy is signal-independent conditional on action), while in the other it is active and reverses the local optimum.*

Proof. See Appendix F.1.

Corollary 1 **(Decision-insufficiency of reduced-form evidence)**. *Any signal design rule that selects signal intensities using only the reduced-form efficacy transitions in Proposition 1 and the immediate signal-level payoffs, but not the factorization into μ, F, and K, must be suboptimal in at least one of the two observationally equivalent environments constructed in Proposition 1. If the local signal ranking disagrees on a state-stage pair with occupancy $p > 0$ and Bellman gap $\eta > 0$, the induced lifecycle loss is at least $p\eta$.*

Proof. See Appendix F.2.

This is why the design problem requires a structured representation of the transition law. Whether signaling is safe or harmful depends on separating action-mediated displacement from action-conditioned efficacy drift, and reduced-form continuation objects do not reveal those channels separately.

2.3. A Factored-Kernel Representation for Signal Design

Under full knowledge of the transition law, the factored and unfactored formulations are policy-equivalent; the value of the factorization lies in the diagnostic, constructive, and operational leverage it provides.

Definition 1 (Factored-kernel representation). We say that the signal-design problem admits a *factored-kernel representation* if its augmented-state transition law can be written as

$$P_\ell(s', c' \mid s, c) = \sum_{a \in \mathcal{A}} \mu(a \mid s, c, \ell)\, F(s' \mid s, a)\, K(c' \mid c, s, a, \ell).$$

Identification and operational use. The factored representation is a modeling contract, not an identification theorem. Operationally, it requires a deployment log that records the realized action at the same decision epoch as the signal intensity, together with a planner-side information state c_t for future support quality—either a directly observed efficacy label or a proxy that preserves the relevant ordering. Under those ingredients, μ is estimable from signal–response tuples and F from state–action–next-state transitions. The difficult object is the efficacy kernel $K(c' \mid c, s, a, \ell)$, because passive observational continuation data do not by themselves separate residual efficacy drift from signal-induced selection in μ. Recovering K therefore requires additional identifying structure—designed variation in ℓ, institutionally grounded exogeneity, or parametric restrictions on the drift component of K.[4]

[4] Appendix A.1 develops the proxy construction with a worked clinical-decision-support example. Section 4.3 quantifies value retention under a noisy proxy.

Transition kernel: $P_\ell(s', c' \mid s, c) = \sum_a \mu(a \mid s, c, \ell) \cdot F(s' \mid s, a) \cdot K(c' \mid c, s, a, \ell)$

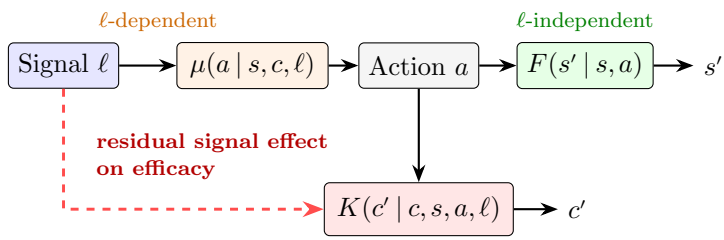

Figure 1 Factored representation of signal-dependent dynamics. Signal intensity ℓ shifts the action distribution through the behavioral-response kernel μ. The realized action then shapes physical evolution through F and future efficacy through K. The dashed arrow marks the residual signal-to-efficacy channel that reduced-form transitions do not isolate.

As Figure 1 illustrates, the factored representation reveals four design-relevant objects that the reduced form obscures: a decomposition of signaling into immediate, displacement, and drift channels; modular estimation of behavioral, physical, and efficacy primitives from distinct data slices; direct certification of safe cases in which efficacy drift is absent; and identification of environments in which reduced-form continuation effects do not recover the primitives needed for design. The deployment question is therefore rarely whether a dynamic policy can be solved in principle, but whether the available logging and variation support a credible direct design—and, if not, what disciplined design object remains available.

Operational scope. The framework selects the design object by what the deployment can credibly defend, not by what is easiest to compute. The concrete data prerequisites for each tier are stated at the close of this subsection.

Behavioral stationarity. The baseline analysis treats the behavioral-response kernel as stationary within each deployment cycle, that is, over the window during which the operator's response rule is treated as fixed. The foundational results below (Proposition 1 and Lemma 1) require no more than the factored representation itself, whereas the sharper constructive results in Section 3 invoke additional theorem-local regularity conditions stated there.

Assumption 1 **(Within-episode stationarity of the behavioral-response kernel)**. *The behavioral-response kernel $\mu(a \mid s, c, \ell)$ is stationary and Markov within each deployment cycle.*

Within a deployment cycle, the operator's response rule does not drift over time once we condition on the current state and signal; any accumulated signal-exposure effects are absorbed by the efficacy state c as intended.[5]

[5] Within-cycle stationarity is most plausible over moderate deployment horizons. Proposition A.1 in Appendix A.6 provides a quantitative robustness bound for deviations, and Appendix A.6 discusses behavioral loops and the scope of the approximation.

Identification in practice. The incremental identification demand of the factored representation lies in the signal-dependence of K conditional on the realized action; μ and F face the standard challenges of models built on observed actions and states. Three routes are available. *Designed variation*—randomizing or systematically varying signal intensity while logging actions and downstream states (e.g., A/B tests of alert thresholds in hospital CDS)—identifies K from efficacy-state transitions across signal levels conditional on the realized action. *Institutional exogeneity*—cross-provider or cross-period variation in signal configurations orthogonal to operator heterogeneity—provides a natural experiment for the drift component. *Parametric restrictions*—modeling drift through a low-dimensional family such as monotone exposure-dependent erosion at rate β_d—permit calibration from a single field estimate of cumulative signal-response decay (Li and Piri 2025). These routes are not mutually exclusive; the choice among them determines which tier of the framework is operationally available (Appendix A records the full estimation path). The computational studies in Section 4 use the third route for the healthcare benchmarks and full structural specification for the engineering benchmark; demonstrating structural estimation under the first two routes is a natural next step.[6]

In practice, the distinction is simple. Direct SPD requires realized-action logs, next-state logs, efficacy-state or proxy labels, enough variation to recover K, and a credible full cost schedule. RD requires only an ordered efficacy proxy, a silence benchmark, relative cost judgments for upgrades, and an ordered signal ladder, without a fully flexible cardinal cost schedule. When neither set of prerequisites is met, the framework supports diagnostic use only.

3. Structural Design Methodology

This section moves from diagnosis to construction to deployment. The diagnostic results show why reduced-form evidence cannot solve the design problem and which channels any credible solution must track. The paper's main constructive result is Theorem 1, which shows that once RD is the selected design object, the optimal penalized ordered-menu policy admits an exact characterization through a one-dimensional shadow-price system over adjacent upgrades. Proposition 2 provides the supporting frontier-side counterpart. The remaining propositions translate that constructive result into deployment guidance. Table 1 summarizes this progression.

Throughout, uppercase letters denote random states and lowercase letters their realizations: at epoch $t = 1, \ldots, T$, $X_t = (S_t, C_t)$ denotes the planner state and $x = (s, c) \in \mathcal{X}$ a generic realization.[7] Under the factored-kernel representation, expected immediate decision quality satisfies

$$\bar{R}(x, \ell) = \sum_a \mu(a \mid s, c, \ell)\, R(s, a).$$

[6] Such an exercise would provide an end-to-end empirical test of the design logic on primitives that are identified rather than calibrated.

[7] We assume $X_t = (S_t, C_t)$ is a sufficient statistic for the relevant history, so the process is Markov in X_t.

Result	Design question answered	Managerial implication
A. Foundational diagnosis		
Proposition 1 + Corollary 1	Why is reduced-form evidence not enough for design?	The same reduced-form evidence can support opposite local designs
Lemma 1 (IDE decomposition)	What channels must any credible solution track?	Immediate gains, displacement, and efficacy drift must be separated
B. Main constructive result		
Theorem 1 (index-threshold characterization)	Once RD is the selected object, what policy does it imply?	Ordered-menu policy obtained by thresholding adjacent-upgrade indices with a single shadow price
C. Supporting frontier implication		
Proposition 2 (value–cost frontier)	How does the selected policy move with the budget?	$V^\star(\bar{\Psi})$ is weakly increasing and concave in $\bar{\Psi}$; slope $\lambda^\star(\bar{\Psi})$ where differentiable
D. Deployment corollaries		
Proposition 3 (safe harbor)	Where does the harmful channel vanish?	Signal-independent efficacy $\Rightarrow$ drift channel absent
Proposition 4 (near-tie screen)	Where can continuation effects overturn myopic rankings?	Only near myopic ties
Proposition 5 (performance guarantee)	How can the exact policy be approximated tractably?	One-step score rule with bounded loss

Table 1 Roadmap from diagnosis to constructive design object to deployment.

Here and throughout, $\mu(a \mid x, \ell)$ and $\mu(a \mid s, c, \ell)$, and similarly $P_\ell(\cdot \mid x)$ and $P_\ell(\cdot \mid s, c)$, are used interchangeably since $x = (s, c)$. For $t = 1, \ldots, T$, define the one-step action values

$$Q_t(x, \ell) = \bar{R}(x, \ell) + \mathbb{E}[V_{t+1}(X_{t+1}) \mid X_t = x, \ell_t = \ell], \qquad \ell \in \mathcal{L},$$

and the value function

$$V_t(x) = \max_{\ell \in \mathcal{L}} Q_t(x, \ell), \qquad t = 1, \ldots, T,$$

with $V_{T+1} \equiv 0$.

The optimal policy maximizes $Q_t(x, \ell)$. By contrast, the *myopic benchmark*—the signal rule that maximizes immediate expected decision quality $\bar{R}(x, \ell)$ and ignores continuation effects—selects the best signal for the current decision alone. Confidence-gating, accuracy-based thresholding, and selective-automation rules that condition only on current-period information are all special cases of this benchmark. The gap between the optimal and myopic criteria is the paper's central policy object. When signaling affects future efficacy, optimal signal design can no longer be determined from immediate decision quality alone.

3.1. What Signal Design Must Account For

Advice that improves the current decision can still reduce long-run welfare if it degrades future decision-support quality. We formalize this possibility through *locally beneficial signaling*—defined

precisely after Lemma 1—which holds at a state-stage pair when the immediate and displacement channels jointly favor the more prescriptive signal ($\mathcal{I}+\mathcal{D}\geq 0$).

Under the factored-kernel representation $P_\ell = \mu \cdot F \cdot K$ (Definition 1), the next lemma decomposes the value effect of signaling into immediate gains, action-mediated displacement, and efficacy drift. The decomposition separates ordinary dynamic look-ahead from the action-conditioned change in future efficacy that is central to the paper.[8]

LEMMA 1 **(IDE decomposition)**. *Under the factored-kernel representation, fix (s,c), stage t, and any two signal levels $\ell',\ell\in\mathcal{L}$. Define the action-contingent continuation value*

$$w_t(s,c,a,\ell) := \sum_{s',c'} F(s' \mid s,a)\, K(c' \mid c,s,a,\ell)\, V^\star_{t+1}(s',c')$$

and

$$\Delta\mu_a := \mu(a \mid s,c,\ell') - \mu(a \mid s,c,\ell).$$

Then

$$Q_t(s,c,\ell') - Q_t(s,c,\ell) = \underbrace{\textstyle\sum_a \Delta\mu_a\, R(s,a)}_{\mathcal{I}} + \underbrace{\textstyle\sum_a \Delta\mu_a\, w_t(s,c,a,\ell)}_{\mathcal{D}} + \underbrace{\textstyle\sum_a \mu'_a\big[w_t(a,\ell') - w_t(a,\ell)\big]}_{\mathcal{E}}, \quad (1)$$

where $\mu'_a := \mu(a \mid s,c,\ell')$, and $w_t(a,\ell) := w_t(s,c,a,\ell)$ abbreviates the action-contingent continuation value at the fixed state (s,c). More generally, the same identity holds if $V^\star_{t+1}$ in the definition of w_t is replaced everywhere by any bounded continuation-value function V_{t+1}.

Proof. See Appendix F.3.

The first two channels, $\mathcal{I}$ and $\mathcal{D}$, can be partially recovered from observed action shifts and downstream transitions; efficacy drift $\mathcal{E}$ becomes structurally available only once the transition law is factored. Harmful signaling arises when $\mathcal{E} < -(\mathcal{I}+\mathcal{D})$. We therefore say that the environment satisfies *locally beneficial signaling* at (s,c,t) for $\ell'\geq\ell$ if $\mathcal{I}+\mathcal{D}\geq 0$.[9]

3.2. A Governable Relative-Design Problem for Partial-Specification Environments

The central obstacle to applying SPD directly is not optimization but elicitation. A direct solution requires the organization to specify a flexible cost schedule $B(x,\ell)$ that assigns a credible downstream-cost value to every state–signal pair. In many operational settings, that object is too ambitious relative to what the planner can actually defend. The organization may know that stronger signals are costlier than weaker ones, may be able to cap cumulative downstream cost over a deployment cycle, and may be able to judge whether upgrading from silence to a stronger

[8] Figure B.1 in Appendix B gives a visual mnemonic.

[9] Appendix B records the mechanism classes behind signal-dependent efficacy, the downstream unfamiliarity index used in Ψ, and practical calibration of λ.

signal is worthwhile; yet it may have no responsible basis for assigning cardinal cost values across the full state–signal space. The economically relevant question is therefore not how to solve SPD when full cost elicitation is available, but what design object remains credible when it is not.

That asymmetry points to a relative-design formulation. Anchoring at silence converts a fragile cardinal-pricing problem into an auditable upgrade problem: relative to silence, how much does stronger assistance improve immediate decision quality, and how much additional downstream cost does it create? This is also how organizations typically govern such systems in practice—through burden caps, overload tolerances, capability-preservation policies, and deployment-cycle targets rather than fully state-contingent cardinal pricing. The ordered-menu geometry derived below is therefore not an arbitrary restriction adopted for convenience; it is the constructive consequence of working with the cost object the planner can credibly defend.

Formally, fix a benchmark silence level ℓ_0 and a bounded downstream unfamiliarity index $\mathcal{U}(s,c) \in [0,1]$. Under the maintained convention of Section 2, the silence level is $\ell_0 = 0$; we write ℓ_0 here to emphasize its role as the relative-cost anchor for G and Ψ. Define the gain relative to silence by

$$G(x,\ell) := \bar{R}(x,\ell) - \bar{R}(x,\ell_0)$$

and the relative downstream cost by

$$\Psi(x,\ell) := \mathbb{E}[\mathcal{U}(S_{t+1},C_{t+1}) \mid x,\ell] - \mathbb{E}[\mathcal{U}(S_{t+1},C_{t+1}) \mid x,\ell_0].$$

The index $\mathcal{U}(s,c)$ summarizes movement toward harder-to-support states and is flexible enough to capture overload, vigilance loss, capability erosion, or movement toward lower-coverage regions.[10]

At any shadow price $\lambda \geq 0$, define the local score of signal level ℓ by

$$S_\ell(x;\lambda) := G(x,\ell) - \lambda\,\Psi(x,\ell).$$

Throughout the remainder of this section and the appendices, we write $G_\ell(x) \equiv G(x,\ell)$ and $\Psi_\ell(x) \equiv \Psi(x,\ell)$ when the subscript form is more compact. The score-line representation is useful for local menu geometry and for the one-step approximation results below, but it is not itself the exact dynamic selector. The exact ordered-menu policy is characterized through the penalized action values $Q_t^\lambda(x,\ell)$ and the adjacent-upgrade indices $\theta_{t,\ell}(x)$.

[10] The formal construction of $\mathcal{U}$ and the one-step score approximation are in Appendix B.2.

A disciplined relative-design problem. Under this relative-to-silence construction, the planner's constrained problem becomes

$$\begin{aligned} \underset{\pi}{\text{maximize}} \quad & \mathbb{E}_\nu^\pi\left[\sum_{t=1}^{T} G(X_t,\ell_t)\right] \\ \text{subject to} \quad & \mathbb{E}_\nu^\pi\left[\sum_{t=1}^{T} \Psi(X_t,\ell_t)\right] \le \bar{\Psi} \end{aligned} \qquad \text{Relative Design (RD)}$$

Here, $G(x,\ell)$ is the immediate gain relative to silence and $\Psi(x,\ell)$ the corresponding relative downstream cost, with both equal to zero at $\ell=\ell_0$. The budget $\bar{\Psi}$ can be read either as an organizational constraint on cumulative downstream cost or as a design device whose shadow price summarizes the marginal trajectory-value cost of tightening that constraint.

By standard LP duality (Appendix F.4), any optimal policy of (Relative Design (RD)) also solves an unconstrained penalized dynamic program with reward $G(x,\ell)-\lambda^\star\Psi(x,\ell)$. The multiplier $\lambda^\star$ is therefore the shadow price of tightening $\bar{\Psi}$.

The dynamic signaling index. Combined with the factored-kernel structure, RD yields an adjacent-upgrade index system that exactly characterizes the penalized selector through a single scalar object, together with the corresponding value–cost frontier. Under the implementation conditions in Appendix F.5, the same system also characterizes the constrained optimum.

For each adjacent upgrade $\ell\to\ell+1$ at state-stage (x,t), define the *upgrade advantage* at cost price λ as

$$\Delta_t^\ell(x;\lambda) \coloneqq Q_t^\lambda(x,\ell{+}1)-Q_t^\lambda(x,\ell), \tag{2}$$

where

$$Q_t^\lambda(x,\ell)\coloneqq G(x,\ell)-\lambda\Psi(x,\ell)+\mathbb{E}[V_{t+1}^\lambda(X_{t+1})\mid x,\ell]$$

is the penalized action value under the RD normalization and V_{t+1}^λ is the corresponding penalized value function. The upgrade advantage measures the net benefit of moving from level ℓ to $\ell+1$ under cost price λ; it includes immediate gains, continuation effects, and the cost penalty.

Define the *dynamic signaling index* as the critical cost price at which upgrading becomes unprofitable:

$$\theta_{t,\ell}(s,c) \coloneqq \sup\bigl\{\lambda\ge 0 : \Delta_t^\ell(s,c;\lambda)\ge 0\bigr\}. \tag{3}$$

This is the highest shadow price at which the planner still prefers to upgrade from ℓ to $\ell+1$. It is an intrinsic property of the state: it is defined from the exact penalized dynamic program, not from a particular value of $\lambda^\star$.

From diagnosis to construction. The diagnostic results above—non-identifiability and the IDE decomposition—require only the bare factored kernel; they do not require any menu regularity. A usable constructive object requires more. The ordered-menu assumptions below are the price of converting the design logic into a single shadow-price system. The next theorem shows that this price yields an exact characterization. Maintain the following conditions only for the theorem:

(R1) *Single-crossing in the cost price.* For every stage t, state $x = (s, c)$, and $\lambda \geq 0$, the map $(\ell, \lambda) \mapsto Q_t^\lambda(x, \ell)$ has decreasing differences; equivalently, $\Delta_t^\ell(x; \lambda') \leq \Delta_t^\ell(x; \lambda)$ for all $\lambda' \geq \lambda$.

(R2) *Ordered-menu curvature.* For every (t, x, λ), the adjacent increments are weakly decreasing in the menu index, $\Delta_t^0(x; \lambda) \geq \Delta_t^1(x; \lambda) \geq \cdots \geq \Delta_t^{k-1}(x; \lambda)$.

(R3) *Continuity in the cost price.* For each (t, x, ℓ), the map $\lambda \mapsto Q_t^\lambda(x, \ell)$ is continuous.

Primitive interpretation. Consider a five-level ladder: silence, alert, partial information, guided suggestion, full prescription. (R1) says that as downstream-cost pressure rises, each upgrade becomes weakly less attractive—a planner facing tighter budget pressure should not signal more aggressively. (R2) says that adjacent upgrades become weakly less valuable as one moves up the ladder—the jump from silence to alert is at least as valuable as the jump from guided suggestion to full prescription. (R3) is a technical continuity condition. Together, these conditions describe deployments in which each level is more prescriptive than its predecessor in a way the organization can assess, and each upgrade trades bounded additional downstream cost for bounded additional immediate decision quality.

THEOREM 1 **(Index-threshold characterization)**. *Consider the signal-design problem under the factored-kernel representation (Definition 1), with ordered signal menu $\mathcal{L} = \{0, \ldots, k\}$ and the constrained design formulation (Relative Design (RD)). Under the theorem-local regularity conditions (R1)–(R3):*[11]

(i) Indexability. *For each stage t, state $x = (s, c)$, and adjacent pair $\ell, \ell + 1$, define*

$$\theta_{t,\ell}(x) := \sup\{\lambda \geq 0 : \Delta_t^\ell(x; \lambda) \geq 0\}.$$

Then

$$\Delta_t^\ell(x; \lambda) \geq 0 \qquad \Longleftrightarrow \qquad \lambda \leq \theta_{t,\ell}(x).$$

(ii) Ordered indices and penalized selector. *The adjacent indices are ordered across menu levels,*

$$\theta_{t,0}(x) \geq \theta_{t,1}(x) \geq \cdots \geq \theta_{t,k-1}(x).$$

Hence, for each cost price λ, the maximal penalized selector is

$$\pi_t^{\lambda,\max}(x) = \max\{\ell \in \mathcal{L} : \theta_{t,\ell-1}(x) \geq \lambda\}. \tag{4}$$

[11] Throughout the theorem and the rest of this subsection we use the conventions $\sup \varnothing := -\infty$, $\theta_{t,-1}(x) := +\infty$, and $\theta_{t,k}(x) := -\infty$. Index inequalities are interpreted in the extended-real sense.

Proof. See Appendix F.6.

Managerial payoff. The planner need not inspect an opaque state-by-state policy table. Each adjacent upgrade—silence to alert, alert to partial information, and so on—is governed by a one-dimensional threshold. The organization's aggregate cost tolerance enters through a single scalar shadow price $\lambda^\star$, which maps directly into local upgrade-or-not decisions at each state-stage pair. Lemma 2 then shows *which channel*—immediate response, displacement, or efficacy drift—drives each marginal upgrade, providing the channel-level accountability needed for communication and audit.[12] The proof uses standard single-crossing machinery (Whittle 1988, Topkis 1998). The novelty lies not in the indexability technique itself, but in what it operates on: partial specification selects RD as the constructive object, and each adjacent-upgrade index admits an exact IDE decomposition (Lemma 2) that identifies which of the three signaling channels drives the upgrade decision.

Lemma 2 **(IDE representation of the adjacent-upgrade index)**. *Let $\mathcal{I}_t^\ell(x;\lambda)$, $\mathcal{D}_t^\ell(x;\lambda)$, $\mathcal{E}_t^\ell(x;\lambda)$ denote the immediate, displacement, and efficacy-drift channel terms obtained by applying the IDE decomposition of Lemma 1 to the adjacent upgrade $\ell \to \ell+1$ with continuation value V_{t+1}^λ, and let $\Delta\Psi^\ell(x) := \Psi(x,\ell{+}1) - \Psi(x,\ell)$. Fix a state-stage pair (x,t) and an adjacent level pair $(\ell,\ell{+}1)$, and assume only the three local conditions $\Delta\Psi^\ell(x) > 0$, $0 \le \theta_{t,\ell}(x) < +\infty$, and pointwise continuity at $\lambda = \theta_{t,\ell}(x)$ of the upgrade gap $\lambda \mapsto \Delta_t^\ell(x;\lambda)$, where $\Delta_t^\ell(x;\lambda) := Q_t^\lambda(x,\ell{+}1) - Q_t^\lambda(x,\ell)$. Then the adjacent-upgrade index satisfies*

$$\theta_{t,\ell}(x) \;=\; \frac{\mathcal{I}_t^\ell\big(x;\theta_{t,\ell}(x)\big) + \mathcal{D}_t^\ell\big(x;\theta_{t,\ell}(x)\big) + \mathcal{E}_t^\ell\big(x;\theta_{t,\ell}(x)\big)}{\Delta\Psi^\ell(x)}.$$

The index $\theta_{t,\ell}(x)$ is therefore the shadow price at which the total (immediate + displacement + efficacy-drift) benefit of moving from level ℓ to level $\ell+1$ is exactly offset by the marginal relative-cost increment $\Delta\Psi^\ell(x)$.

Proof. See Appendix B.1. The identity holds pointwise under the three local assumptions above, without invoking the global regularity (R1)–(R3).

Lemma 2 gives the index system in Theorem 1 its managerial content: at each state-stage pair, the index is not a generic priority weight but a shadow price with an explicit three-channel decomposition that identifies which channel—immediate response, displacement, or efficacy drift—drives the upgrade decision. This is precisely the kind of channel-level accountability the planner needs when design choices must be communicated or audited.

[12] The appendix records the selector-cell representation, feasibility and complementary-slackness details, and the fuller IDE representation of the index.

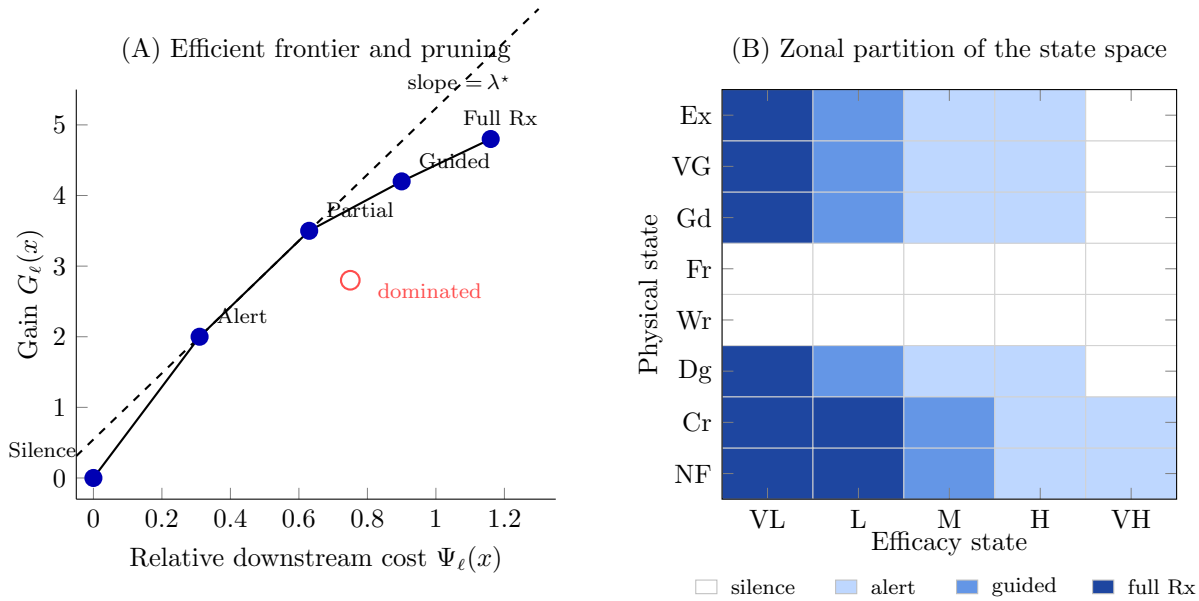

Figure 2 The geometry of optimal AI signal design. Panel A: efficient frontier and pruning at a representative state. Efficient levels (filled circles) lie on the upper concave hull; dominated levels (open circle) are pruned. The dashed line is a supporting line at the optimal shadow price, whose tangency identifies the optimal level at that state. Panel B: zonal partition of the augmented state space. The optimal policy partitions states into signal-intensity zones, and the zone boundaries shift monotonically as the cost price changes.

Frontier-side counterpart. The following supporting result records the value–cost frontier and the shadow-price meaning of the budget that indexes Theorem 1.[13]

PROPOSITION 2 **(Value–cost frontier)**. *Let $V^\star(\bar{\Psi})$ denote the optimal value of the constrained design problem* (Relative Design (RD)) *as a function of the cost budget $\bar{\Psi}$.*

(i) $V^\star(\bar{\Psi})$ is weakly increasing and concave in $\bar{\Psi}$.

(ii) The optimal dual multiplier $\lambda^\star(\bar{\Psi})$ is a supergradient of $V^\star$ at $\bar{\Psi}$; where differentiable,

$$\frac{dV^\star(\bar{\Psi})}{d\bar{\Psi}} = \lambda^\star(\bar{\Psi}).$$

Proof. See Appendix F.7.

Together, Theorem 1 and Proposition 2 deliver the oracle design payoff of RD through a single shadow price applied to a fixed index system. Figure 2 visualizes the resulting geometry.[14]

The fuller ordered-menu geometry is recorded in Appendix E. The next three subsections develop the Part D deployment corollaries of this design system.

[13] This is a standard consequence of LP duality for finite-horizon CMDPs (Altman 1999). It is stated in the main text because $\lambda^\star$ is the supergradient of $V^\star$ at the binding budget, which makes the index characterization operational as a single scalar dual.

[14] Appendix E records the interval-implementation details and menu-dominance extensions.

3.3. When Simple Signal Rules Are Enough

We now identify the boundary at which the harmful efficacy-drift channel disappears. The key condition is whether future efficacy remains signal-dependent after conditioning on the realized action. If it does not, the efficacy-drift channel vanishes and, under locally beneficial signaling, harmful signaling is impossible.

Formally, the environment satisfies signal-independent efficacy if

$$K(c' \mid c, s, a, \ell') = K(c' \mid c, s, a, \ell) \quad \text{for all } a, c, c', s, \ell, \ell'.$$

Future efficacy may still depend on the realized action, but not on the signal conditional on that action. The proposition below is pointwise: it applies to any state-stage pair and ordered signal pair satisfying locally beneficial signaling.

Proposition 3 **(Safe harbor under signal-independent efficacy)**. *Fix a state-stage pair (s, c, t) and an ordered pair of signal levels $\ell' \geq \ell$. Suppose that, for this comparison, locally beneficial signaling holds, in the sense that $\mathcal{I} + \mathcal{D} \geq 0$, where $\mathcal{I}$ and $\mathcal{D}$ are evaluated using the optimal continuation value $V^{\star}_{t+1}$. If efficacy is signal-independent conditional on the realized action, then the efficacy-drift term $\mathcal{E}$ in Lemma 1 is zero. Consequently,*

$$Q_t(s, c, \ell') - Q_t(s, c, \ell) = \mathcal{I} + \mathcal{D} \geq 0.$$

Hence harmful signaling is impossible at any state-stage pair and ordered signal pair satisfying this pointwise condition.

Proof. See Appendix F.13.

The proposition is pointwise rather than a primitive global sufficient condition: its requirement $\mathcal{I} + \mathcal{D} \geq 0$ is evaluated at the relevant state-stage pair under the corresponding continuation value.

3.4. Where Dynamic Continuation Effects Matter

If the safe-harbor boundary fails, the question is no longer whether harmful signaling can arise in principle, but whether continuation effects are large enough to overturn the myopic choice. Such reversal can occur only where the myopic ranking is fragile: continuation effects cannot overturn the myopic benchmark when the latter dominates by a wide immediate margin. Three local objects determine whether dynamic reversal is possible: the myopic gap, the change in next-state distributions induced by switching signals, and the spread of continuation values next period.

The *myopic benchmark* selects the signal level that maximizes immediate expected decision quality and ignores continuation effects. Formally,

$$\pi_t^{\text{myo}}(x) \in \arg\max_{\ell} M_\ell(x),$$

where

$$M_\ell(x) := \sum_a \mu(a \mid x, \ell)\, R(s,a) = \bar{R}(x,\ell)$$

is the myopic score, equivalently the expected immediate decision quality. Fix throughout a measurable tie-breaking rule for this argmax, and let $\pi_t^{\mathrm{myo}}(x)$ denote the selected element of $\arg\max_{\ell\in\mathcal{L}} M_\ell(x)$. We also write $\ell_t^{\mathrm{myo}}(x) \equiv \pi_t^{\mathrm{myo}}(x)$ when we wish to emphasize that the object is a signal level used as a subscript. Confidence-gating, accuracy-based thresholding, and selective-automation rules that condition only on current-period model confidence are all special cases of this benchmark.

To determine whether an alternative signal ℓ can overturn the myopic choice at a state-stage pair (x,t), define

$$\Gamma_t(x,\ell) := M_{\ell_t^{\mathrm{myo}}(x)}(x) - M_\ell(x) \geq 0,$$

$$\Delta_t^{\mathrm{TV}}(x,\ell) := \mathrm{TV}\big(P_\ell(\cdot \mid x),\, P_{\ell_t^{\mathrm{myo}}(x)}(\cdot \mid x)\big),$$ [15]

and

$$\Phi_{t+1} := \sup_{x'} V_{t+1}^\star(x') - \inf_{x'} V_{t+1}^\star(x').$$

The first is the myopic gap in favor of the myopic choice, the second measures how much an alternative signal changes the next-state distribution, and the third measures how much continuation values can vary next period.

Dynamic reversal therefore requires a near tie: an alternative signal can outperform the myopic choice only if continuation effects are large enough to overcome the local myopic gap. Let

$$\eta_t^{\mathrm{myo}}(x) := V_t^\star(x) - Q_t^\star\big(x, \pi_t^{\mathrm{myo}}(x)\big)$$

denote the local Bellman gap of the myopic choice at state-stage pair (x,t), where $V_t^\star$ and $Q_t^\star$ are the optimal value and action-value functions from the $\bar{R}$-space Bellman equation above, and $Q_t^{\lambda,\star}$ and V_t^λ are the corresponding G-space objects under the RD-penalized program.[16] The next proposition gives the sharp local screen; global and estimable operational versions are recorded in Appendix F.15.

[15] Throughout, $\mathrm{TV}(P,Q) := \sup_A |P(A) - Q(A)| = \frac{1}{2}\|P - Q\|_1$ denotes total-variation distance (the probabilist's convention).

[16] Throughout the formal results, $J_\nu(\pi) := \mathbb{E}_\nu^\pi[\sum_{t=1}^T G(X_t,\ell_t)]$ is the total expected G-space reward (relative to silence) under policy π and initial distribution ν, and $J_\nu^\lambda(\pi) := \mathbb{E}_\nu^\pi[\sum_{t=1}^T (G(X_t,\ell_t) - \lambda\,\Psi(X_t,\ell_t))]$ is its penalized counterpart. The RD value–cost frontier, Theorem 1, and Propositions D.1–D.2 are stated and proved in G-space. The total decision quality $\mathbb{E}[\sum_t \bar{R}(X_t,\ell_t)]$ differs from J_ν by the accumulated silence baseline $\mathbb{E}[\sum_t \bar{R}(X_t,\ell_0)]$, which is action-independent at each stage but policy-dependent through the state distribution; the computational benchmarks report $\bar{R}$-space rewards for the retention comparison.

PROPOSITION 4 **(Near-tie screen for the myopic benchmark)**. *At any state-stage pair (x,t), the myopic benchmark can be harmful only if the continuation distortion, scaled by the future-value spread, exceeds the local myopic gap:*

$$\eta_t^{\mathrm{myo}}(x) \le \max_{\ell\in\mathcal{L}} \big[\Phi_{t+1}\,\Delta_t^{\mathrm{TV}}(x,\ell) - \Gamma_t(x,\ell)\big]_+. \tag{5}$$

If $\Gamma_t(x,\ell) \ge \Phi_{t+1}\,\Delta_t^{\mathrm{TV}}(x,\ell)$ for all $\ell\in\mathcal{L}$, then the myopic benchmark is optimal at (x,t).

Proof. See Appendix F.15.

Proposition 4 identifies the state-stage pairs at which the full dynamic solution can depart from the myopic benchmark.[17]

3.5. Tractable Approximation of the Ordered-Menu Policy

The exact penalized dynamic program underlying Theorem 1 requires the continuation-value objects Q_t^λ. In many settings, however, a simpler one-step surrogate is attractive: at each state, select the signal level that maximizes the local gain–cost score $G_\ell(x) - \lambda\Psi_\ell(x)$.[18] The next result gives an explicit performance guarantee for that one-step rule relative to the exact penalized dynamic program.

PROPOSITION 5 **(Full-horizon performance guarantee for the one-step rule)**. *Fix $\lambda \ge 0$ and the silence baseline $\ell_0 = 0$. Let $\pi^{\lambda,\mathrm{app}}$ denote the one-step rule that selects*

$$\arg\max_{\ell\in\mathcal{L}} \{G_\ell(x) - \lambda\Psi_\ell(x)\}$$

at each (t,x). Suppose the exact penalized signal advantage relative to ℓ_0,

$$\Delta_{t,\ell}^{\lambda,\star}(x) := Q_t^{\lambda,\star}(x,\ell) - Q_t^{\lambda,\star}(x,\ell_0),$$

admits the decomposition

$$\Delta_{t,\ell}^{\lambda,\star}(x) = G_\ell(x) - \lambda\Psi_\ell(x) + \varepsilon_{t,\ell}^\lambda(x), \qquad |\varepsilon_{t,\ell}^\lambda(x)| \le b_{t,\ell}^\lambda(x).$$

Then the full-horizon loss of the one-step rule relative to the exact penalized dynamic program satisfies

$$J_\nu^\lambda(\pi^{\lambda,\star}) - J_\nu^\lambda(\pi^{\lambda,\mathrm{app}}) \le 2\sum_{t=1}^{T} \mathbb{E}_\nu^{\pi^{\lambda,\mathrm{app}}}\Big[\max_{\ell\in\mathcal{L}} b_{t,\ell}^\lambda(X_t)\Big]$$

[17] Global accumulation bounds and an estimable-primitives version are in Appendix F.15.

[18] Under the bounded-curvature linearization developed in Appendix B.2, this score approximates the exact penalized upgrade advantage up to a stagewise error bounded by the local curvature and the linearization-slope calibration error.

$$\leq 2\sum_{t=1}^{T}\left\|\max_{\ell\in\mathcal{L}} b_{t,\ell}^{\lambda}(\cdot)\right\|_{\infty}.$$

Under the bounded-curvature stagewise approximation recorded in Appendix B.2,

$$b_{t,\ell}^{\lambda}(x) = \frac{L_{t,x}}{2}\Big(\mathrm{Var}_{\ell}(\mathcal{U}\mid x) + \mathrm{Var}_{0}(\mathcal{U}\mid x) + \Psi_{\ell}(x)^{2}\Big) + |\lambda - \lambda_{t}^{\mathrm{loc}}(x)|\,|\Psi_{\ell}(x)|.$$

Here $\lambda_t^{\mathrm{loc}}(x)$ is the local linearization slope of the continuation-value function at the silence-baseline conditional mean, defined formally in Appendix B.2. The loss is therefore small whenever the stagewise approximation errors are small on the states the one-step rule actually visits.

Proof. See Appendix F.16.

Proposition 5 bounds the full-horizon loss of the one-step rule by twice the occupancy-weighted approximation error. In the computational studies (Section 4), the average bound-to-realized ratio is approximately 2.9×, indicating that the guarantee is conservative but still practically informative.

These results together suggest a simple workflow: first diagnose whether reduced-form design is inadequate and whether dynamic continuation effects are material; if so, construct an efficacy proxy and solve the appropriate design problem; then evaluate robustness under imperfect information. Section 4 validates this logic across the paper's benchmark environments.

4. Computational Illustration of the Design Methodology

This section evaluates the deployment logic developed in Section 3 focusing on three key questions. When is myopic signaling adequate? When does dynamic design matter, and under correct specification is SPD the correct benchmark? Under what named classes of misspecification does RD become the better design object? The answers yield a three-regime deployment rule: at low drift, myopic signaling is near-optimal; at material drift under correct specification, SPD is the natural benchmark and RD retains nearly all of its value; under over-restrictive burden misspecification or order-breaking observation noise, RD can regularize the design problem.

These illustrations are calibrated validations of that rule, not end-to-end structural estimation. The hospital benchmark uses a parametric restriction—monotone drift at rate β_d calibrated from Li and Piri 2025. The ambulatory benchmark uses the same form anchored at Ancker et al. 2017. The C-MAPSS benchmark uses a fully specified efficacy kernel. To make comparisons transparent, all cross-environment results are reported on a unified β_d grid with a common 8×5 state space, so that only drift varies; Appendix D records the construction.

Two implementation choices matter for interpretation. First, the computational RD family is generated by sweeping $\bar{R}(x,\ell) - \lambda_{\mathrm{RD}}\Psi(x,\ell)$ rather than the exact G-space program. Because the formal RD theory operates in G-space, and the two dynamic programs are not generally equivalent in a dynamic setting (Appendix D.4), the retention results should be read as characterizing

Setting	Environment class	Design verdict
Hospital pharmacy (evidence-anchored)	Li–Piri fatigue rate; other primitives specified	Dynamic optimum remains close to the myopic benchmark under this low-drift calibration
Ambulatory CDS (evidence-anchored)	Ancker log-IRR anchors drift	Larger overload moves the environment toward the design boundary (4.4–18.1% across band)
Fatigue sweep (counterfactual)	Hospital baseline with drift raised	Shows how a grounded healthcare benchmark crosses from modest to material dynamic effects
C-MAPSS (specified)	Turbofan dynamics; behavioral channel specified; structurally specified archetype sweep	Isolates the drift channel cleanly, quantifies design gains, and confirms the design logic survives across broader behavioral environments
Efficacy mismeasurement (specified / proxy)	Proxy-observed C-MAPSS grid	Quantifies value retention and budget distortion under a noisy efficacy proxy (Section 4.3)

Table 2 Computational environments used to validate the deployment rule. Evidence-anchored: one or more parameters anchored in published evidence; counterfactual: drift varied around a baseline; specified: fully structural.

a practical surrogate of the formal RD object. Second, the common target budget $\bar{B}$ is chosen from the RD family's realized cost range and SPD is then tuned to match, so the comparison is anchored to RD feasibility rather than to an exogenous neutral grid. These choices do not invalidate the exercises, but they do mean that the RD–SPD retention results compare practical implementations rather than the exact formal objects. Table 2 summarizes the benchmark environments; Appendix D records calibration details, auxiliary figures, and the full robustness grids.

4.1. Low Drift: When Myopic Signaling Is Sufficient

The low-drift case is the easiest. When efficacy drift is weak, dynamic signal design adds little, and simple myopic signal selection is effectively sufficient. The hospital-pharmacy benchmark illustrates this case. Using the fatigue estimate of Li and Piri (2025), we calibrate the per-signal drift coefficient to $\beta_d = 0.003394$ on the original hospital primitives. On the unified computational grid (8×5 states, $|\mathcal{L}|$=4, T=20), this corresponds to the low-drift point $\beta_d = 0.05$, which produces a comparably small dynamic-design gain on the common grid. Under this calibration, dynamic design improves little over the myopic benchmark, placing the hospital point in the near-equivalence region of Figure 3. The null-model placebo in Appendix D.2 removes signal dependence from K and reduces the RD gain to below 0.3%, consistent with the safe-harbor logic of Proposition 3. On the unified grid (Table D.2), direct SPD improves on the myopic benchmark by 1.95%, compared with 1.91% for RD at the common target budget. RD therefore retains 97.9% of the SPD design gain while disagreeing with SPD on 14.0% of state-stage pairs. In this low-drift region, neither SPD nor RD justifies the

added complexity: myopic signaling is already operationally adequate, and the deployment problem reduces to detecting when the environment exits this region.

4.2. Material Drift: When Dynamic Design Matters

Once drift is material, dynamic design delivers gains that myopic signal selection cannot capture. Two healthcare benchmarks and a structurally specified engineering benchmark illustrate this regime; full construction details are reported in Appendix D.

Ambulatory CDS. We use the encounter-level overload estimate in Ancker et al. (2017) as an empirical anchor for a higher-drift calibration. Because the Ancker evidence identifies the average drift magnitude but not the low-efficacy responsiveness parameter, we sweep $\mathsf{eff}_L \in [0.35, 0.70]$ as a sensitivity band. The low endpoint captures strong responsiveness deterioration consistent with the severe-alert-fatigue literature, while the high endpoint captures the mildest residual responsiveness the same evidence can support. Across this band, RD gains range from 4.42% to 18.13% over the myopic benchmark, moving the benchmark from near-equivalence toward a clearly design-relevant region as overload intensifies. The direct SPD benchmark remains extremely close throughout: SPD gains 4.49%, 10.89%, and 18.19% over the myopic benchmark at the high-, mid-, and low-eff_L anchors, with corresponding RD gains of 4.42%, 10.86%, and 18.13%. RD therefore retains 98.6%–99.7% of the SPD design gain across the full ambulatory band. Appendix D.4 reports calibration details, intermediate reference points, and the full unified-grid SPD–RD comparison table.

Drift sweep and structural benchmarks. To trace the drift transition systematically, we return to the hospital-pharmacy environment and vary only the fatigue parameter counterfactually, holding all remaining primitives fixed. Figure 3 plots RD gains on the unified grid across all calibration points: the evidence-anchored hospital point lies in the near-equivalence region, the ambulatory benchmarks span the design boundary, and the C-MAPSS points extend into the high-drift region where gains exceed 20%. At the boundary point $\beta_d = 0.30$, SPD gains 12.65% over the myopic benchmark and RD gains 12.61% (retention 99.6%). As drift rises further, absolute gains increase while the SPD–RD wedge remains negligible. Appendix D.4 reports the full drift-axis tables.

Oracle cost of RD relative to SPD. Structurally specified environments isolate the efficacy-drift channel more cleanly than the healthcare calibrations. In C-MAPSS, the efficacy kernel is specified and drift intensity varies directly, so the benchmark separates residual efficacy erosion from broader deployment frictions. As drift intensifies, dynamic-design gains rise from modest levels to above 20% (Table D.2)—not as field validation, but to show that strong efficacy drift makes myopic allocation materially suboptimal even when the physical dynamics are otherwise well behaved.

The direct SPD benchmark again remains extremely close to RD in value terms despite nontrivial state-stage policy disagreement. SPD gains 4.49%/15.68%/23.43% over the myopic benchmark

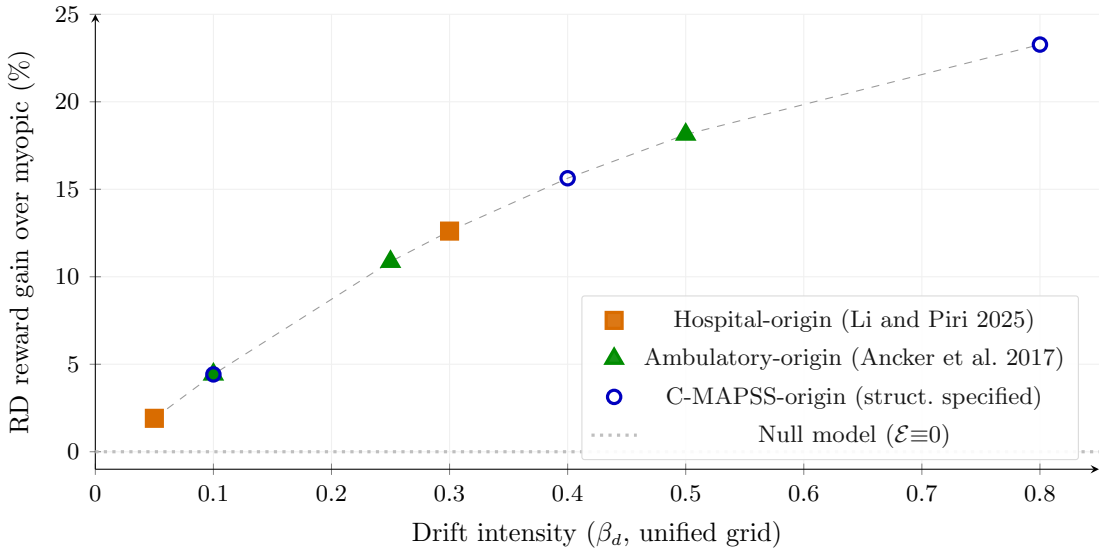

Figure 3 Dynamic-design gains rise monotonically with drift intensity on the unified computational grid. Each marker corresponds to one drift level; marker shapes indicate the empirical or structural calibration motivating that point (hospital-pharmacy, ambulatory CDS, or C-MAPSS). The ambulatory high-efficacy point and the C-MAPSS low-drift point coincide at the same grid location. At low drift, the gain is negligible and myopic signaling is effectively sufficient; at high drift, the gain exceeds 20%. Table D.2 reports the corresponding SPD–RD retention comparisons.

at the low-, boundary-, and high-drift configurations, against 4.42%/15.63%/23.27% for RD—retentions of 98.6%/99.6%/99.3%—even though the two policies disagree on 12.8%/6.8%/10.0% of state-stage pairs, respectively. The low-drift row is the merged Amb. CDS high-eff_L / C-MAPSS low-drift entry in Table D.2. Small value loss can therefore coexist with meaningful policy-level routing differences: RD is nearly costless in value terms not because it agrees with SPD at every state, but because the value wedge between their different choices is small. Across all unified-grid environments, RD retains between 97.9% and 99.7% of the SPD design gain (Appendix D.4).

Under correct full specification, SPD remains the natural oracle benchmark and should be used whenever the planner can credibly commit to a flexible cost object. The retention results above show that RD is nearly costless when that stronger object is available. The next subsection shows that RD can be preferable when it is not—precisely because the restriction that defines RD removes the degrees of freedom most vulnerable to misspecification. The point is not that RD is merely a convenient alternative to SPD; it is that RD is the correct design object in the weaker information regime, and that its near-zero oracle cost means the planner gives up almost nothing by adopting it when the stronger regime happens to hold.

4.3. Imperfect Information: When Misspecification Selects the Design Object

The preceding subsection showed that RD is nearly costless when the full cost object is credibly available. The question that determines whether RD is the right design object—rather than merely

a safe fallback—is whether it can also improve realized performance when the planner's model is imperfect. We study two misspecification channels: proxy-observation misspecification, in which the planner observes the efficacy state with error, and cost misspecification, in which the planner's estimated downstream-cost object deviates from the true burden.[19]

Proxy-observation misspecification. The preceding benchmarks assume that the planner observes the true efficacy class c_t. In practice, however, c_t is inferred from operational proxies and may be measured with error. We therefore ask how much oracle value survives when the planner uses a noisy proxy $\hat{c}_t$ rather than the true state. The test bed is a C-MAPSS–like grid with $|\mathcal{S}|$=8, $|\mathcal{C}|$=3 (low, mid, high), $|\mathcal{L}|$=4, and T=20. Two observation-noise models, parameterized by $\varepsilon \in [0, 0.30]$, govern the probability that the planner observes the wrong efficacy class. Model A (local ordered noise) sends errors only to adjacent classes, so the proxy preserves the efficacy ordering with high probability. Model B (order-breaking noise) sends errors to the opposite extreme, so even moderate noise can reverse the perceived ranking.

We compare five policies: oracle DP, proxy plug-in DP, proxy one-step, c-blind DP, and proxy myopic. The proxy-DP variants re-estimate the relevant transitions and rewards on the observed state space $(\mathcal{S}, \hat{\mathcal{C}})$ induced by the observation kernel and solve the corresponding proxy-state design problem. All policies are then evaluated exactly, by backward induction, in the true environment at three drift levels.

Interpretation. Under local ordered noise, proxy-based dynamic design remains close to the oracle even at high drift; under order-breaking noise, both value retention and budget control deteriorate much more sharply. The practical lesson concerns *state construction*: for proxy-based state-contingent design, preserving the efficacy ordering matters more than precise cardinal measurement. That is exactly the information regime in which RD is intended to operate. The planner need not observe the latent efficacy state perfectly; what matters is a planner-side proxy that preserves the design-relevant ordering well enough to support disciplined relative upgrade comparisons.

When misspecification selects RD over plug-in SPD. The preceding oracle comparison shows that RD gives up almost nothing when the model is correctly specified. The misspecification analysis identifies the conditions under which the same restriction can actively improve realized performance—making RD the selected design object, not merely a robust alternative. Two formal channels underlie that selection (Propositions D.1 and D.2): over-restrictive burden misspecification, in which estimated SPD wastes true-cost slack that RD does not need to estimate; and

[19] The C-MAPSS benchmark also illustrates the paper's supporting diagnostics: roughly 80% of the myopic-to-optimal gap concentrates in the top 10 state-stage pairs (Appendix F.15); the full-horizon performance guarantee of Proposition 5 is conservative by about 2.9× on average (Appendix F.16); and an archetype sweep confirms that design gains survive across broader behavioral environments (Appendix D).

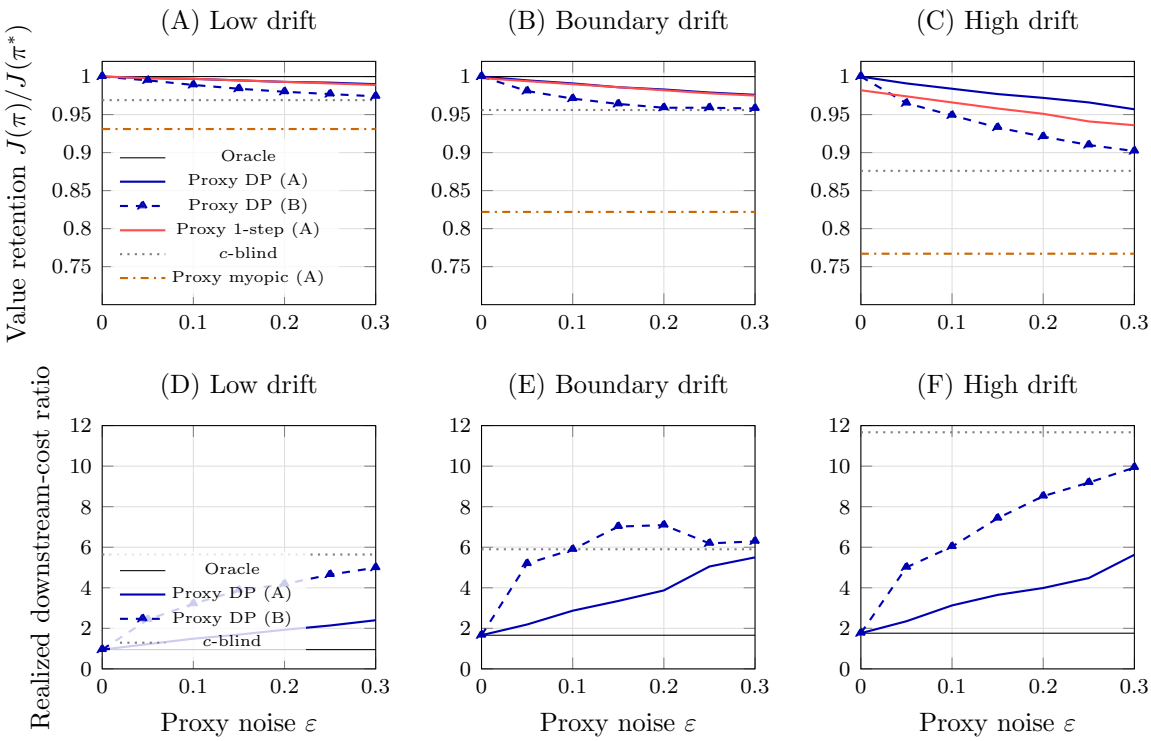

Figure 4 Robustness to efficacy-state mismeasurement on a three-class C-MAPSS grid with four signal levels and a twenty-period horizon. Top row: value retention relative to the oracle policy. Bottom row: realized downstream-cost ratio relative to the target budget; the proxy-myopic benchmark is omitted because its realized ratio exceeds the plotted range at all noise levels considered. Solid curves correspond to local ordered noise; dashed triangles correspond to order-breaking noise. Preserving the efficacy ordering yields substantially better value retention and budget control than order-breaking mismeasurement, especially as drift intensifies.

finite-sample estimation fragility, in which burden-estimation error scales with the dimensionality of the cost object while RD's restriction cost remains fixed. Empirically, the strongest RD gains arise under over-restrictive burden estimation and under order-breaking observation noise; the finite-sample channel provides theoretical grounding but does not produce a robust empirical advantage in the benchmark families we test (Appendix D.5).

Two caveats are important. First, the formal propositions are proved under $B \equiv \Psi$, whereas the computational exercises use a three-channel burden with $B \neq \Psi$, so the theory-to-empirics bridge is heuristic rather than exact. Second, in the burden-misspecification family the comparison is oracle-Ψ RD versus misspecified-B SPD. That informational asymmetry is deliberate: the Ψ-only restriction is, by construction, invariant to burden misspecification. Accordingly, the RD gains in that family reflect the asymmetry built into the exercise, not a matched-information benchmark.

The computations nonetheless confirm the selection pattern. Under *efficacy-state observation noise*, local ordered noise leaves estimated SPD slightly ahead by 0.01–0.07 reward units: when distortion is mild, the oracle flexibility of SPD still pays. But order-breaking noise reverses the

Regime	Recommended policy/object	Takeaway
Low drift & low misspecification	Myopic	Dynamic policy design adds little; the myopic benchmark is effectively sufficient and direct SPD nearly coincides with it in value terms.
Material drift & low misspecification	SPD if fully specified; otherwise RD	Dynamic design is relevant; RD retains nearly all SPD value while requiring less fragile cost specification.
Material drift & severe misspecification	RD	RD can outperform plug-in SPD by regularizing the design problem in the partial-specification regime.

Table 3 Deployment rule over design objects and policies across efficacy-drift and specification regimes. The recommendation depends not only on drift, but also on whether full cost specification is credible or only relative upgrade comparisons are credible.

ranking. Estimated RD outperforms estimated SPD by +0.19 to +1.02 reward units, with the largest gain in high-drift C-MAPSS, where the continuation channel is strongest. Under *cost misspecification*, the same pattern appears. Overweighting the exposure penalty makes estimated SPD overcautious, and RD outperforms by +0.74 to +0.86 reward units across all four environments. Underweighting or omitting that penalty restores a small SPD advantage, and the remaining misspecifications have negligible effects. The bound from Proposition D.1 correctly predicts the sign of the RD–SPD wedge across all 44 Family-B cells, and prior-averaged misspecification yields a positive mean RD advantage in all four headline environments. The formal results are therefore not only sufficient, but also quantitatively predictive. Full noise sweeps, perturbation grids, and per-environment tables are reported in Appendix D.5.

Table 3 summarizes the operational rule implied jointly by the theory and the evidence-anchored exercises.[20] The full cross-family numeric detail is reported in Tables D.3 to D.6 in Appendix D.5.

5. Discussion

In many human–AI systems, signals matter not only because they improve the current decision, but also because they shape the system's future capacity to improve decisions conditional on the realized action (Vaccaro et al. 2024, Li and Piri 2025). The relevant design question is therefore not only whether the system is accurate enough to signal now, but how signaling affects downstream performance over the deployment horizon (Chen et al. 2018). Once that channel is present, reduced-form continuation evidence is no longer sufficient for signal design, because it does not identify whether downstream consequences arise from action-mediated displacement or from efficacy drift conditional on the realized action. Our paper shows that these design-relevant components can instead be recovered through a factored-kernel representation that separates behavioral response, physical transitions, and efficacy drift.

[20] "Low misspecification" means that the object required by the chosen policy class—the full cost schedule for SPD, or the relative-cost index for RD—is credibly specified, not that the environment is globally well understood.

The paper's constructive contribution is to turn that diagnosis into a selection rule over design objects. Direct SPD is the correct object when the full constrained problem can be credibly specified. RD is the appropriate constructive object when marginal signal upgrades can be credibly assessed relative to silence but a fully flexible cardinal cost schedule cannot. Within that selected object, the paper's main theorem gives an exact characterization of the penalized ordered-menu policy through a single shadow price, while the supporting results identify safe cases, screen when myopic rules remain adequate, and bound the loss from tractable approximation. The framework is therefore not meant to advocate richer policies everywhere, but to distinguish the environments in which myopic signaling, direct SPD, and RD are each the economically credible recommendation.

Several scope conditions mark the boundary of our analysis. The efficacy state is intentionally parsimonious—rich enough to capture the intertemporal channel that matters for design, yet still interpretable and operationally estimable in settings where a more elaborate latent structure would be difficult to justify. The screening logic requires reliable action logging and meaningful variation in signal intensity, because those are precisely the ingredients needed to separate action-mediated displacement from efficacy drift. The within-cycle stationarity assumption abstracts from abrupt behavioral breaks and richer policy-dependent learning dynamics, but it allows diagnosis, policy construction, and deployment screening to be studied within a single tractable representation. The computational studies are calibrated rather than structurally estimated from a single operational deployment, and the paper does not claim universal RD dominance or full empirical identification in every environment. Its key results show instead that dynamic AI signal design is not a single optimization problem with a universally correct solution. When reduced-form evidence is decision-insufficient, the economically relevant design object—myopic signaling, direct SPD, or RD—must be selected by what the planner can credibly learn about the environment and what cost object the planner can credibly specify.

Acknowledgments

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Appendix A: Model Extensions, Assumptions, and Estimation

This appendix develops the modeling details underlying Section 2. It is organized around the recurring objects of the paper: the efficacy state c_t, the efficacy kernel K, the scope conditions under which the factorization is substantively meaningful, the observed-efficacy estimation path, and extensions that allow the planner to choose signal content or accommodate policy-dependent response. Unless otherwise noted, results are stated for a finite ordered signal menu $\mathcal{L} = \{\ell_0, \ldots, \ell_k\}$; binary specializations are noted only when they add content. The occupancy-LP formulation and deployment workflow are reported in Appendix C.

A.1. The efficacy state c_t: interpretation, observability, and construction

Throughout the paper, c_t is a planner-side summary statistic for the current capacity of the decision-support system to improve subsequent decisions. It is not intended as a literal psychological primitive, but as an information state constructed from routinely logged operational behavior. In the main-text benchmark this information state is an observed efficacy label c_t; more generally, the planner may work with a filtered score, a discretized proxy from logs, or a belief over efficacy classes. The screening results require only that the constructed state preserve the relevant ordering, so that higher values of c correspond to systematically greater future responsiveness or support quality. In practice, we discretize c_t into a small ordered set of efficacy classes, typically with $|\mathcal{C}| \in \{2, 3\}$.

Operational boundary. The framework admits three operational tiers. If the planner observes an efficacy label, or has a proxy that preserves the relevant ordering together with some source of identifying variation for the drift component of K, the full diagnose–screen–optimize workflow is available. If the planner has only a proxy with limited identifying variation, the framework still supports screening, calibration, and sensitivity analysis, but the efficacy-drift channel is necessarily more model-dependent. If neither a credible proxy nor meaningful variation in signal intensity is available, the framework remains useful as a design logic—in particular for clarifying what reduced-form continuation effects fail to reveal—but not yet as a fully operational policy-design procedure.

Worked example: constructing c from clinical decision-support override data. Consider a hospital deploying drug-interaction alerts. A practical implementation can construct c from routinely available EHR logs in three steps. *(1) Observable proxy:* for each clinician, compute an override-rate statistic $\widehat{p}^{\mathrm{ovr}}_{i,b}$ over a rolling window of W recent alerts (for example, W=50). *(2) Ordered discretization:* partition clinicians into a small number of efficacy groups using override-rate quantiles. With $|\mathcal{C}|$=3, one obtains c=high (bottom third, most compliant), c=mid, and c=low (top third). *(3) Transition estimation:* tabulate efficacy-class transitions across consecutive windows, stratified by severity s, realized action a, and signal level ℓ. This turns a continuous compliance trajectory into a discrete Markov information state that can be used directly in the design workflow; the panel in Ancker et al. (2017) indicates that such transition estimation is feasible at realistic data depth. The point is not that override rate is the unique proxy, but that the paper's efficacy state can be built from ordinary deployment logs rather than bespoke psychometric measurement. Section 4.3 quantifies how value retention and budget performance degrade when the planner works with a noisy efficacy proxy $\hat{c}_t$ rather than the true label.

A.2. The efficacy kernel K: mechanisms and empirical signatures

The kernel $K(c' \mid c, s, a, \ell)$ is reduced-form: it records whether today's signal intensity changes tomorrow's efficacy state after conditioning on the realized action. Two mechanism classes are especially relevant. The first is algorithm-side coverage drift, in which signaling shifts the system toward regions with different support quality or familiarity. The second is operator-side efficacy drift, in which repeated exposure changes future responsiveness through fatigue, effort substitution, or over-reliance. The key empirical question is therefore whether, after conditioning on the realized action, signal intensity still predicts future efficacy. If not, the environment satisfies the safe-harbor logic of Proposition 3. If so, residual efficacy drift is active and dynamic RD design may matter. Table B.1 summarizes the corresponding empirical signatures.

A.3. Scope conditions and model assumptions

When the factorization is defensible. The factorization is most natural when actions are discrete and well logged, signals are informational rather than physical, and physical evolution is governed by engineering or clinical processes that do not directly respond to the information channel once the realized action is fixed.

What breaks interpretation. If unobserved heterogeneity jointly affects physical outcomes and future responsiveness, or if the signal alters the environment through channels not captured by the logged action, then one should replace (F, K) with a joint reduced-form kernel $\tilde{P}(s', c' \mid s, c, a, \ell)$. The dynamic program and LP formulation remain valid under that reduced form, but the separation between displacement and efficacy drift is no longer behaviorally interpretable.

Why the fixed-μ benchmark can still be useful. Even when within-cycle response behavior is only approximately stationary, the main analysis remains informative over moderate deployment horizons. Proposition A.1 gives the corresponding robustness margin.

Proposition A.1 **(Value robustness to within-cycle behavioral drift)**. *If two response kernels μ and $\tilde{\mu}$ satisfy*

$$\sup_{(s,c),\ell} \|\mu(\cdot \mid s, c, \ell) - \tilde{\mu}(\cdot \mid s, c, \ell)\|_1 \leq \delta,$$

then for any fixed policy π:

$$|J_T(\pi; \mu) - J_T(\pi; \tilde{\mu})| \leq \frac{T(T+1)}{2} R_{\max} \delta,$$

and the regret from optimizing under the wrong kernel is bounded by $T(T{+}1)R_{\max}\delta$. The T^2 factor reflects compounding drift: with $R_{\max}{=}1$ and $\delta{=}0.01$, the value bound at $T{=}130$ is $T(T{+}1)\delta/2 = 85.1$ versus 2.1 at $T{=}20$, implying deployment cycles should rarely exceed $T{=}50$–100.

A.4. Observed efficacy, estimation path, and LP implementation

The observed-efficacy case is the benchmark operational formulation and the one most closely aligned with the deployment workflow in Section 2. Here the planner observes $c_t \in \mathcal{C}$ directly (Appendix A.1), so the planner state is (s_t, c_t) and the behavioral-response kernel $\mu(a \mid s, c, \ell)$ is exactly as in Section 2. The resulting signal-contingent transition kernel is

$$P_\ell(s', c' \mid s, c) = \sum_{a \in \mathcal{A}} \mu(a \mid s, c, \ell)\, F(s' \mid s, a)\, K(c' \mid c, s, a, \ell).$$

Assumption A.1 **(Conditional transition factorization (observed efficacy))**. *Conditional on* (s, c, a, ℓ),

$$\mathbb{P}(s', c' \mid s, c, a, \ell) = F(s' \mid s, a)\, K(c' \mid c, s, a, \ell).$$

Estimation path for F, K, and μ. The factorization is the estimation backbone of the deployment workflow in Appendix C: each kernel is recovered from a distinct slice of the deployment log, and the conditional independence above removes the need for cross-term identification. The physical kernel F is estimated from state–action–next-state transitions and is typically the easiest object, since physical or clinical transitions often respond to signals only through the realized action. The behavioral-response kernel $\mu(a \mid s, c, \ell)$ is estimated from signal–response tuples at the same decision epoch and is often recoverable from passive logs in deployed alerting and decision-support systems. The difficult object is the efficacy kernel $K(c' \mid c, s, a, \ell)$: passive continuation data cannot by themselves separate the residual ℓ-conditional drift channel from signal-induced selection in μ. Three operational routes address this difficulty: (1) designed variation (randomization or A/B testing of ℓ at fixed (s, a)); (2) institutional exogeneity (protocols or external shocks that vary signaling for reasons orthogonal to operator or case type); and (3) parametric restrictions (a low-dimensional drift specification combined with partial variation). Without at least one of these, the framework still clarifies what reduced-form continuation effects fail to identify and supports sensitivity analysis around plausible drift structures, but its screening interpretation should be treated as model-dependent rather than fully identified.

LP interpretation. The occupancy variable $x_t(s, c, \ell) = \mathbb{P}(S_t = s, C_t = c, \ell_t = \ell)$ represents state-time intervention mass. Because many operational constraints are linear in those flows, budgets, workload caps, and fairness restrictions can be imposed directly at the LP level. The same is true for switching frictions: augmenting the state with the previous signal Z_t and imposing

$$\sum_{t,s,c,z,\ell} x_t(s, c, z, \ell)\mathbf{1}\{\ell \neq z\} \leq B_{\text{switch}}$$

adds smoothness in intervention intensity without leaving the tractable occupancy-measure framework. Computational complexity and solver-scaling details are reported in Appendix C.

A.5. Choosing signal content as well as intensity

The baseline model fixes signal content within each intensity level. If the planner also controls content, three cases arise depending on whether content families can be ordered along the same prescriptiveness axis. When they can (Case 1), the ordered-menu theory extends directly after relabeling. When each family has its own intensity ladder (Case 2), one obtains family-specific indices but not a single scalar order. When content is genuinely multidimensional (Case 3), the scalar index and contiguous-zone results are no longer guaranteed, although the problem remains a finite CMDP solvable by LP. The Discussion notes these extensions as settings in which the scalar-index structure need not survive.

A.6. Behavioral loops and policy-dependent response

The baseline analysis treats $\mu(a \mid s, c, \ell)$ as fixed within a deployment cycle, which is the natural approximation when the planner's horizon is short relative to the speed at which operators adapt. If the deployed policy itself changes future responsiveness through habituation, effort substitution, or altered reliance, then the response

Mechanism class	Safe harbor: pure displacement ($\mathcal{E}=0$)	Drift-active: residual exposure ($\mathcal{E}\neq 0$)	Empirical implication
Algorithm-side coverage	Future support quality changes only through the action–state path.	Future support quality changes beyond the realized action (e.g., different information revealed, logged, or learned).	Conditional on a_t: does ℓ_t predict future support quality or OOD exposure?
Operator-side response	Actions change; future responsiveness does not, conditional on action.	Future responsiveness changes at fixed action (fatigue, effort substitution, skill erosion).	Conditional on a_t: does ℓ_t predict c_{t+1}, latency, overrides, or audit accuracy?

Table B.1 Mechanism classes behind signal-dependent efficacy. The key distinction is whether signal intensity still affects future efficacy after conditioning on the realized action.

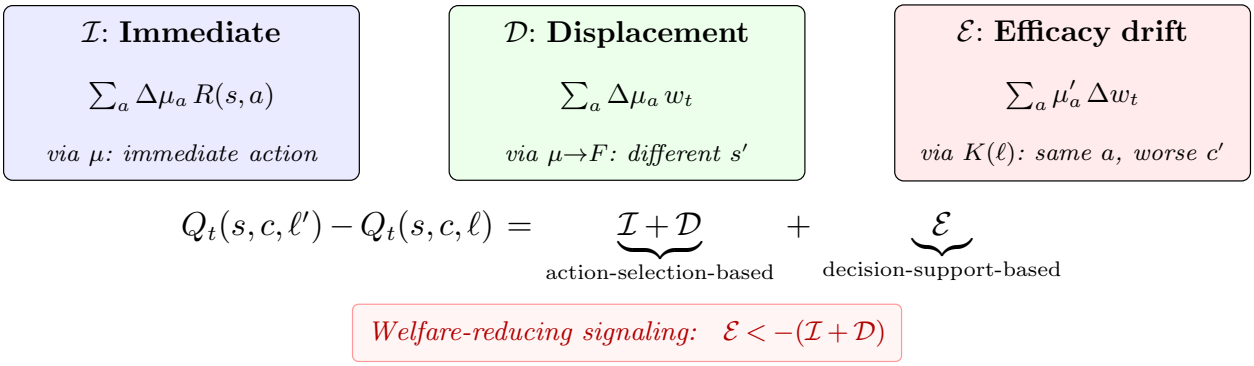

Figure B.1 IDE decomposition and harm condition. The three channels are immediate gain, action-mediated displacement, and efficacy drift. Welfare-reducing signaling occurs when efficacy drift is sufficiently negative to outweigh the combined immediate and displacement benefits. When future efficacy is signal-independent conditional on the realized action, the efficacy-drift channel vanishes.

kernel becomes policy-dependent: $\mu^\pi(a \mid s, c, \ell)$. The design problem then acquires an outer feedback loop whose equilibrium satisfies $\pi^\dagger = \mathcal{T}(\pi^\dagger)$, where $\mathcal{T}$ maps a candidate policy to the optimal response under the induced kernel. Under finite spaces, compactness and continuity of $\mathcal{T}$ guarantee existence, and contractivity yields uniqueness. Proposition A.1 then bounds the value loss from freezing μ at the $O(T^2 R_{\max}\delta)$ rate, so the fixed-μ benchmark remains the relevant first-order design object and the behavioral loop becomes a slower outer recalibration layer.

Appendix B: Supporting Structural Results

This appendix records the supporting objects used in the main text: the mechanism classes behind signal-dependent efficacy (Table B.1), the visual mnemonic for the IDE decomposition (Figure B.1), the proof of the IDE representation of the adjacent-upgrade index (Lemma 2), the one-step score approximation together with the downstream unfamiliarity index, and practical calibration of λ. Additional prevalence, regional-dominance, and lattice-monotonicity material is recorded in the online supplement available from the authors' website.

Figure B.1 provides a visual mnemonic for the IDE decomposition in Lemma 1.

B.1. Proof of the IDE representation of the adjacent-upgrade index

By Lemma 1, each adjacent-upgrade margin admits an IDE decomposition. For the comparison between levels $\ell+1$ and ℓ, let $\mathcal{I}_t^\ell(x;\lambda)$, $\mathcal{D}_t^\ell(x;\lambda)$, and $\mathcal{E}_t^\ell(x;\lambda)$ denote the immediate-gain, displacement, and efficacy-drift terms from Lemma 1 when the bounded continuation-value function there is taken to be V_{t+1}^λ, and define $\Delta\Psi^\ell(x) := \Psi(x,\ell{+}1) - \Psi(x,\ell)$. Then the adjacent-upgrade advantage can be written as

$$\Delta_t^\ell(x;\lambda) = \mathcal{I}_t^\ell(x;\lambda) + \mathcal{D}_t^\ell(x;\lambda) + \mathcal{E}_t^\ell(x;\lambda) - \lambda\,\Delta\Psi^\ell(x).$$

Hence, whenever $0 \le \theta_{t,\ell}(x) < +\infty$ and $\Delta\Psi^\ell(x) > 0$, continuity implies $\Delta_t^\ell\big(x;\theta_{t,\ell}(x)\big) = 0$, so

$$\theta_{t,\ell}(x) = \frac{\mathcal{I}_t^\ell\big(x;\theta_{t,\ell}(x)\big) + \mathcal{D}_t^\ell\big(x;\theta_{t,\ell}(x)\big) + \mathcal{E}_t^\ell\big(x;\theta_{t,\ell}(x)\big)}{\Delta\Psi^\ell(x)}.$$

Thus the finite adjacent-upgrade index is the shadow price at which the total IDE benefit of moving from ℓ to $\ell+1$ is exactly offset by the marginal downstream-cost increment. Under the one-step score approximation $Q_t^\lambda(x,\ell) \approx G(x,\ell) - \lambda\Psi(x,\ell)$ (recorded in Appendix B.2 and extended to ordered menus in Appendix E), the continuation terms are suppressed and the adjacent index reduces to the corresponding local gain–cost slope ratio.

B.2. One-Step Score Approximation and Calibration

We parameterize displacement effects using two familiarity maps $f_A : \mathcal{S}o[0,1]$ (algorithm familiarity) and $f_H : \mathcal{S}o[0,1]$ (human familiarity), and define downstream unfamiliarity $\mathcal{U} : \mathcal{S}imes\mathcal{C}o[0,1]$. A convenient specialization is the purely physical-state index

$$\mathcal{U}(s,c) \equiv (1 - f_A(s))(1 - f_H(s)),$$

but the ordered-menu results below allow unfamiliarity to depend on both the next physical state and the next efficacy state. The multiplicative form captures joint failure in the physical component: it is close to zero whenever either party has high familiarity.

For the linearization argument we specialize to the binary menu $\{0,1\}$ and write the recommendation-vs-silence objects as

$$G(s,c) := \mathbb{E}[R(s,a) \mid s,c,1] - \mathbb{E}[R(s,a) \mid s,c,0],$$

$$\Psi(s,c) := \mathbb{E}[\mathcal{U}(S_{t+1},C_{t+1}) \mid s,c,1] - \mathbb{E}[\mathcal{U}(S_{t+1},C_{t+1}) \mid s,c,0],$$

so that $G,\Psi : \mathcal{S}\times\mathcal{C} \to \mathbb{R}$. These are the binary specializations ($\ell{=}1$ vs. $\ell_0{=}0$) of the $G(x,\ell)$ and $\Psi(x,\ell)$ defined in Section 3.2; in the binary case the ℓ argument is redundant and we suppress it.

Assumption B.1 **(Local smooth continuation value for linearization)**. *At fixed (s,c), continuation value can be written as $V^\star(S_{t+1},C_{t+1}) = g(W)$ with $W = \mathcal{U}(S_{t+1},C_{t+1}) \in [0,1]$, where g is twice continuously differentiable and has bounded curvature on $[0,1]$.*

Under Assumption B.1, a first-order expansion around the baseline no-signal distribution gives

$$\Delta_\ell(s,c) \approx G_\ell(s,c) - \lambda\,\Psi_\ell(s,c),$$

which says that increases in expected downstream unfamiliarity are penalized at shadow price λ. Under the local linearization in Proposition B.1, this coefficient is $\lambda^{\mathrm{loc}} = -g'(m_0)$, which is nonnegative when $g'(m_0) \le 0$.

PROPOSITION B.1 **(Approximation error bound under bounded curvature)**. *Fix a state-efficacy pair (s,c) and an active signal level $\ell \in \mathcal{L} \setminus \{0\}$. Let $W := \mathcal{U}(S_{t+1}, C_{t+1}) \in [0,1]$, and suppose continuation value depends on next state through a twice differentiable map*

$$V^\star(S_{t+1}, C_{t+1}) = g(W),$$

with curvature bound $|g''(w)| \le L$ for all $w \in [0,1]$. Define

$$m_0 := \mathbb{E}[W \mid s, c, 0], \qquad \lambda^{\mathrm{loc}} := -g'(m_0),$$

and let $\mathrm{Var}_\ell(W)$ denote the variance of W under signal ℓ. Then

$$\Delta_\ell(s,c) = G_\ell(s,c) - \lambda^{\mathrm{loc}} \Psi_\ell(s,c) + \varepsilon_\ell(s,c),$$

with bound

$$\left|\varepsilon_\ell(s,c)\right| \le \frac{L}{2}\left(\mathrm{Var}_\ell(W) + \mathrm{Var}_0(W) + \Psi_\ell(s,c)^2\right).$$

Hence the one-step score approximation is exact when g is affine ($L=0$), and is accurate when downstream unfamiliarity dispersion is small. When a fixed design parameter λ is used instead of λ^{loc}, calibration error enters additively: $|\Delta_\ell(s,c) - (G_\ell - \lambda\Psi_\ell)| \le (L/2)(\mathrm{Var}_\ell(W) + \mathrm{Var}_0(W) + \Psi_\ell(s,c)^2) + |\lambda - \lambda^{\mathrm{loc}}|\,|\Psi_\ell(s,c)|$. If, in addition, $g'(m_0) \le 0$, then $\lambda^{\mathrm{loc}} \ge 0$, so the linear term admits the interpretation of a local cost price.

The proposition applies to any active signal level ℓ relative to the baseline $\ell = 0$. Proposition E.3 shows how the same level-by-level argument extends to ordered menus.

PROPOSITION B.2 **(Dual interpretation of λ)**. *Consider the constrained design problem* (Relative Design (RD))*. In the finite occupancy-LP setting, there is an optimal multiplier $\lambda^\star \ge 0$ such that any optimal policy of* (Relative Design (RD)) *also solves the unconstrained penalized problem*

$$\underset{\pi}{\textit{maximize}} \quad \mathbb{E}_\pi\left[\sum_{t=1}^{T}\left(G_{\ell_t}(X_t) - \lambda^\star \Psi_{\ell_t}(X_t)\right)\right].$$

Hence $\lambda^\star$ is the shadow price of tightening the relative downstream-cost budget $\bar{\Psi}$.

Interpretation and calibration of λ. By Proposition B.2, $\lambda^\star$ admits three complementary interpretations: (i) a dual multiplier from the occupancy LP; (ii) a Pigouvian price on signal-induced unfamiliarity; and (iii) a failure-cost rate $\lambda \approx C_{\mathrm{fail}} \cdot \partial p_{\mathrm{fail}} / \partial U$ near the baseline operating point. Once a calibration criterion is fixed, λ is estimated from data, primitives, and risk targets rather than chosen arbitrarily. The paragraph below lists the three approaches we use.

Approximation quality. Across twelve perturbed environments ($T \in \{6, 10, 14, 20\}$, $\lambda \in \{0.3, 1.5, 2.5\}$), the mean full-horizon value loss of the $G - \lambda\Psi$ one-step rule relative to the exact penalized dynamic program is 1.1–3.6%. This is small relative to the RD design gains of interest reported in Section 4. The formal full-horizon performance guarantee is Proposition 5 in the main text (Section 3.5) and is proved in Appendix F.16.

How to calibrate λ in practice. Three calibration approaches are available: (i) outcome-driven cross-validation, (ii) a risk-cost ratio $\lambda \approx C_{\mathrm{cat}} \cdot \Delta p_{\mathrm{cat}}$, and (iii) safety-standard mapping $\hat{\lambda}_{\mathrm{safety}} := \min\{\lambda : \widehat{p}_{\mathrm{incident}}(\pi^\star(\lambda)) \le \bar{p}\}$.

Appendix C: Diagnostics and Deployment Workflow

This appendix records the compressed deployment workflow together with the occupancy-LP implementation underlying the paper's design methodology.

Step 1: construct a planner-side efficacy state. Use routinely logged operational behavior to build an ordered efficacy proxy or observed efficacy label as in Appendix A.1.

Step 2: estimate the three primitives from separate data slices. Estimate F from state–action–next-state transitions, μ from signal–response tuples, and K from efficacy-state transitions conditional on (s, a, ℓ) using either designed variation, institutional exogeneity, or parametric restrictions as discussed in Appendix A.4.

Step 3: screen whether dynamic policy design is likely to matter. Compute the practical counterparts of immediate lift $\widehat{G}_\ell(x)$, transition distortion $\widehat{\Delta}^{\mathrm{TV}}(x)$, the comparison of $\hat{\lambda}\,\widehat{\Psi}_\ell(x)$ to $\widehat{G}_\ell(x)$, and within-cycle μ-drift. These quantities implement the near-tie and adequacy logic of Proposition 4.

Step 4: optimize within operational constraints. If design in a rich policy space is warranted, solve the occupancy-measure LP or its dynamic-programming counterpart under the desired workload, budget, safety, or switching constraints.

Occupancy LP and implementation. The signaling problem admits a standard occupancy-measure LP (Puterman 1994, Altman 1999). Any linear operational constraint—for example, prescriptive caps, workload budgets, fairness restrictions, or switching frictions—enters directly as a linear constraint on the occupancy measure, with λ appearing as the dual variable on the relative downstream-cost budget. The LP has $n{=}T|\mathcal{S}||\mathcal{C}||\mathcal{L}|$ variables; the benchmark discretizations used in the paper yield at most about 26,000 variables, so computation is modest for the state-space sizes considered here. In unconstrained settings, the same primitives can also be solved by standard backward induction. The LP formulation is especially useful because it accommodates operational constraints directly and yields the shadow-price interpretation of λ needed for the score-rule approximation.

Appendix D: Benchmark Supplement and Parameterization

This appendix records construction details for the computational illustration in Section 4. It is organized in five parts: (i) the C-MAPSS turbofan benchmark and its null-model placebo; (ii) the efficacy-state mismeasurement experiment; (iii) the perfect-information SPD benchmark anchoring the oracle RD-retention comparison; (iv) the imperfect-information robustness design comparing estimated RD and estimated SPD across three misspecification families; and (v) two stress tests for Proposition D.1, one checking whether its predicted sign and magnitude quantitatively track realized RD–SPD outcomes across the Family B perturbations and one characterizing the prior-averaged distribution of those outcomes. Additional sensitivity sweeps and witness-state diagnostics are reported in the online supplement. Throughout, ζ denotes the legacy C-MAPSS drift-rate parameter used in inherited sensitivity sweeps, and β_d denotes the normalized fatigue coefficient used in the evidence-anchored calibrations of Section 4.1.

D.1. C-MAPSS Turbofan Benchmark Details

The C-MAPSS environment combines physical dynamics calibrated from NASA sensor data (Saxena et al. 2008) with a structurally specified efficacy kernel, while the legacy drift rate ζ is swept as a sensitivity parameter. Detailed benchmark descriptions and certificate-tightness comparisons by drift intensity are reported in the online supplement.

D.2. Null-Model Placebo Test

The null-model placebo holds the calibrated physical dynamics fixed and removes the efficacy-drift channel by construction (K is signal-independent for all ℓ). Across $\zeta \in [0.5, 3.0]$, the gain over the myopic benchmark is below 0.3% (within SE). This is numerically consistent with the safe-harbor logic of Proposition 3 and, in this benchmark, indicates that once the efficacy-drift channel is removed, the remaining dynamic advantage is practically negligible.

D.3. Efficacy-State Mismeasurement Design

The mismeasurement exercise in Section 4.3 uses a $|\mathcal{C}|$=3 efficacy grid (low, mid, high) with two observation-noise models. Both models are parameterized by a single scalar $\varepsilon \in [0,1]$ governing the probability that the planner observes the wrong efficacy class.

Model A (local ordered noise). Errors go only to adjacent classes:

$$M_\varepsilon^A(\hat{c} \mid c) = \begin{cases} 1-\varepsilon & \text{if } \hat{c} = c, \\ \varepsilon & \text{if } |c-\hat{c}| = 1 \text{ and } c \in \{0,2\}, \\ \varepsilon/2 & \text{if } |c-\hat{c}| = 1 \text{ and } c = 1, \\ 0 & \text{otherwise.} \end{cases}$$

Model B (order-breaking noise). Errors go to the opposite extreme:

$$M_\varepsilon^B(\hat{c} \mid c) = \begin{cases} 1-\varepsilon & \text{if } \hat{c} = c, \\ \varepsilon & \text{if } (c,\hat{c}) \in \{(0,2),(2,0)\}, \\ \varepsilon/2 & \text{if } c = 1 \text{ and } \hat{c} \in \{0,2\}, \\ 0 & \text{otherwise.} \end{cases}$$

Proxy-state estimation. For each noise level ε, the proxy-observed model on $(\mathcal{S}, \hat{\mathcal{C}})$ is constructed by marginalizing the true primitives over the observation kernel: $\bar{R}_{\text{proxy}}(s,\hat{c},\ell) = \sum_c w(c \mid \hat{c})\, \bar{R}(s,c,\ell)$, with posterior weights $w(c \mid \hat{c}) \propto M_\varepsilon(\hat{c} \mid c) \cdot \text{prior}(c)$ and a uniform prior over $\mathcal{C}$. The proxy-state penalized DP and one-step rule are then solved on this proxy model. All policies are evaluated exactly via backward induction on the joint $(s,c,\hat{c})$ chain in the true environment. Additional noise grids, finite-sample re-estimation results, and a continuous latent-score extension are available in the online supplement.

Representative snapshot at ε=0.15. Table D.1 reports realized value retention and downstream-cost ratio at a single moderate noise level (ε=0.15, local ordered noise) across the three drift levels of Figure 4.

D.4. Direct SPD benchmark and RD-restriction comparison

This subsection records the construction used in Section 4 to compare the direct solution of the generic constrained signal-design problem (SPD) to the governable relative-design problem (RD) studied in the main theory. The perfect-information SPD reported here is an *oracle upper benchmark*: it quantifies how much design gain the generic constrained problem attains when the planner knows (μ, F, K, B) exactly, and therefore upper-bounds the value of any implementable design that uses the same budget.

	Low drift	Boundary drift	High drift
Value retention			
Proxy plug-in DP	0.995	0.986	0.977
Proxy one-step	0.995	0.986	0.958
c-blind DP	0.969	0.956	0.876
Proxy myopic	0.931	0.822	0.767
Realized downstream-cost ratio			
Oracle DP	0.95	1.66	1.76
Proxy plug-in DP	1.69	3.35	3.65
c-blind DP	5.64	5.90	11.67
Proxy myopic	15.70	43.18	27.10

Table D.1 Performance at proxy noise level 0.15 (local ordered noise, Model A). Value retention is the ratio of realized reward to oracle reward. Realized downstream-cost ratio is cumulative downstream cost divided by the target budget.

Generic burden function. All environments are solved on the unified C-MAPSS–like grid with $|\mathcal{S}|$=8, $|\mathcal{C}|$=5, $|\mathcal{L}|$=4, and T=20. On this grid we define a three-channel burden function

$$B(x,\ell) \;=\; a_1\,\Psi(x,\ell) \;+\; a_2\,\phi(\ell) \;+\; a_3\,\Pr\big(C_{t+1} \le c_{\text{low}} \mid x,\ell\big),$$

where $\Psi(x,\ell)$ is the relative downstream unfamiliarity term used in (RD), $\phi(\ell) \in [0,1]$ is a direct monotone exposure penalty in signal intensity, and $\Pr(C_{t+1} \le c_{\text{low}} \mid x,\ell)$ captures next-period low-efficacy risk. We use coefficient weights $(a_1, a_2, a_3) = (1.0,\, 0.5,\, 0.5)$ in the main-text table and verify robustness below. The low-efficacy threshold is $c_{\text{low}} = 1$ on the $|\mathcal{C}|$=5 grid, and the second channel uses $\phi(\ell_0) = 0$ and the third is computed as $\Pr(C_{t+1} \le c_{\text{low}} \mid x,\ell) - \Pr(C_{t+1} \le c_{\text{low}} \mid x,\ell_0)$, so that all three channels vanish at ℓ_0.

Common target budget. For each environment, the target budget $\bar{B}$ is chosen as the 40% point of the RD policy family's realized downstream-cost range on the true model, i.e., $\bar{B} = B^{\text{RD}}_{\min} + 0.40\,(B^{\text{RD}}_{\max} - B^{\text{RD}}_{\min})$, where $B^{\text{RD}}_{\min}$ and $B^{\text{RD}}_{\max}$ are the minimum and maximum realized B-costs across the RD policy family swept over $\lambda_{\text{RD}} \in [0, 60]$. This choice ensures that the RD family contains at least one feasible policy and places the target well inside the feasible cost range; in all environments tested, the SPD constraint also binds at this target.

SPD solution. SPD is solved by penalized dynamic programming on the true burden $B(x,\ell)$. We binary-search λ_{SPD} over $[0, 30]$ for 32 iterations to hit $\bar{B}$ to within numerical tolerance. The resulting deterministic Markov policy π_{SPD} is then evaluated exactly via backward induction on the true environment.

RD budget matching. The RD policy family is the set of penalized-DP solutions with reward $\bar{R}(x,\ell) - \lambda_{\text{RD}}\Psi(x,\ell)$ across the λ_{RD} grid, where $\bar{R}(x,\ell)$ is the expected immediate decision quality of Section 2; the formal RD theory operates in G-space, but the $\bar{R}$-penalized and G-penalized dynamic programs are not generally equivalent because the accumulated silence baseline $\bar{R}(x,\ell_0)$ enters the continuation and is policy-dependent through the state distribution. The $\bar{R}$-based sweep here is therefore a practical surrogate for the G-space program: it generates a family of candidate policies, each of which is then evaluated exactly via backward induction on the true environment in both $\bar{R}$-space (for the retention comparison) and under the true burden B. To the extent that the silence baseline varies slowly relative to the signal-dependent gains

Environment	RD gain over myopic	SPD gain over myopic	RD ret. of SPD	Policy disagree.
Hospital (Li–Piri)	1.91%	1.95%	97.9%	14.0%
Amb. CDS (high eff_L) / C-MAPSS (low drift)	4.42%	4.49%	98.6%	12.8%
Amb. CDS (mid eff_L)	10.86%	10.89%	99.7%	5.4%
Amb. CDS (low eff_L)	18.13%	18.19%	99.7%	6.1%
Hospital sweep (boundary)	12.61%	12.65%	99.6%	6.6%
C-MAPSS (boundary drift)	15.63%	15.68%	99.6%	6.8%
C-MAPSS (high drift)	23.27%	23.43%	99.3%	10.0%

Table D.2 SPD vs. RD comparison across the computational environments. Gains are reported relative to the myopic benchmark under a common three-channel burden at a matched target budget. Retention is the fraction of SPD's design gain that RD captures. Policy disagreement is the fraction of state-stage pairs on which the two policies select different signal levels. The Amb. CDS (high efficacy) and C-MAPSS (low drift) entries are merged because they share identical primitives on the unified computational grid.

and costs, the $\bar{R}$-sweep and G-sweep families will be close; in the unified-grid benchmarks tested, SPD and RD policies differ on at most 14% of state-stage pairs and the value wedge is negligible, so any surrogate-gap effect is absorbed within the reported retention margins. Each family member is evaluated under the true burden $B(x, \ell)$, and the benchmark RD policy π_{RD} is the member with $\mathbb{E}\big[\sum_t B(X_t, \ell_t)\big] \leq \bar{B}$ that attains the highest true reward. This ex-post selection over the RD family is analogous to the λ_{SPD} binary search for SPD: both procedures aim to find the best feasible policy within their respective policy classes, using exact backward-induction evaluation on the true environment. All policies—the myopic benchmark (myopic $\arg\max_\ell \bar{R}(x, \ell)$), SPD, and RD—are evaluated exactly via backward induction.

Environments. Eight calibration points from Section 4 are mapped onto the unified grid by varying only β_d: Hospital (Li–Piri) at β_d=0.05; Ambulatory CDS at $\beta_d \in \{0.10, 0.25, 0.50\}$ for high, mid, and low eff_L; Hospital counterfactual sweep (boundary) at β_d=0.30; and C-MAPSS at $\beta_d \in \{0.10, 0.40, 0.80\}$ for low, boundary, and high drift. Because the unified grid is parameterized only by β_d, the Ambulatory-high-eff_L and C-MAPSS-low-drift calibrations collapse onto the same β_d=0.10 point and yield identical results; they are merged into a single row in Table D.2, giving seven displayed rows for eight calibration points. The two calibrations correspond to different managerial settings reached from opposite directions—one evidence-anchored healthcare, one structurally specified engineering—and constitute a single numerical instance shared by two substantive environments.

Robustness to burden weights. We re-run the comparison with two alternative coefficient settings to test sensitivity to the weights (a_1, a_2, a_3): an exposure-heavy weighting (1.0, 1.0, 0.5) and a low-efficacy-heavy weighting (1.0, 0.5, 1.0). Under both alternatives the qualitative conclusion is unchanged: SPD improves over the myopic benchmark monotonically with drift, and RD retention of SPD remains between 97% and 100% across all eight environments, with the largest retention gap (about three percentage points) occurring in the lowest-drift environment. The full robustness tables are available in the online supplement. The simulation script `scripts/spd_vs_rd_benchmark.py` reproduces the results with deterministic seed 2026.

Family	Condition	Best RD	Median
Oracle (perfect info)	–	−0.006	−0.023
Family A (obs noise)	Local, ε=0.30	−0.013	−0.035
	Order-break, ε=0.10	+0.741	+0.366
	Order-break, ε=0.20	+1.017	+0.504
	Order-break, ε=0.30	+0.334	+0.304
Family B (burden misspec.)	Overweight ϕ ($\hat{a}_2$=1.0)	+0.862	+0.782
	Underweight ϕ ($\hat{a}_2$=0.25)	−0.137	−0.138
	Omit ϕ	−0.137	−0.138
	Spurious interaction	−0.019	−0.030
Family C (finite sample)	N=500, best env.	+0.122	−0.060
	N=5000, best env.	−0.046	−0.117

Table D.3 Estimated RD versus estimated SPD under imperfect information. The reported quantity is the realized reward gap (RD minus SPD) evaluated in the true environment under the true burden at a common target budget. Positive values indicate that RD outperforms estimated SPD. Per-family details are in Tables D.4 to D.6.

D.5. Estimated SPD vs. estimated RD under imperfect information

This subsection is intended as a mechanism illustration, not as a rigorous empirical test of Propositions D.1 and D.2 and not as a claim of uniform RD dominance. Three caveats apply throughout. First, Propositions D.1 and D.2 are proved under the maintained identification $B \equiv \Psi$, whereas the empirical exercises below use a three-channel burden with $Beq\Psi$, so the mapping from propositions to benchmark is heuristic rather than exact. Second, each perturbation family uses a different information structure for RD by design: Family A uses a proxy-observed $\hat{\Psi}_\varepsilon$; Family B uses the true Ψ (because the Ψ-only restriction is, by construction, invariant to SPD's burden misspecification); and Family C uses a plug-in $\widehat{\Psi}$ estimated from finite samples. These are different objects, so the cross-family comparison in Table D.3 should be read as illustrating distinct mechanisms rather than testing a single condition. Third, the oracle benchmark of Appendix D.4 is unchanged. All evaluations are exact backward-induction averages on the *true* environment under the *true* three-channel burden at the common target $\bar{B}$. Simulation script: extttscripts/spd_vs_rd_imperfect.py, seed 2026.

Environments. We use the four boundary/high-drift environments from Table D.2: C-MAPSS boundary (β_d=0.40), C-MAPSS high (β_d=0.80), Ambulatory CDS low-eff_L (β_d=0.50), and the hospital sweep boundary (β_d=0.30). Low-drift and safe-harbor cases are excluded because the design gain there is small for any policy.

Family A: efficacy-state observation noise. The planner observes a noisy class $\hat{c} \sim M_\varepsilon(\cdot \mid c)$ in one of two forms from Appendix D.3: Model A (local ordered noise, in which $\hat{c}$ is adjacent to c with probability ε) or Model B (order-breaking noise, in which $\hat{c}$ is sent to the opposite extreme with probability ε). The proxy-observed kernel, reward, and burden on $(\mathcal{S}, \hat{\mathcal{C}})$ are obtained by posterior averaging over the true efficacy class under a uniform prior. Estimated SPD solves the penalized DP on the proxy model with the budget binary-searched to $\bar{B}$; estimated RD sweeps $\lambda_{\mathrm{RD}} \in [0, 40]$ on the proxy $\hat{\Psi}_\varepsilon$ and selects the family member whose realized true B-cost is at most $\bar{B}$ and whose realized true reward is highest. Both are evaluated exactly by backward induction on the joint $(s, c, \hat{c})$ chain. The grid is $\varepsilon \in \{0, 0.10, 0.20, 0.30\}$; Figure D.1 plots the corresponding reward-gap sweep for both noise models.

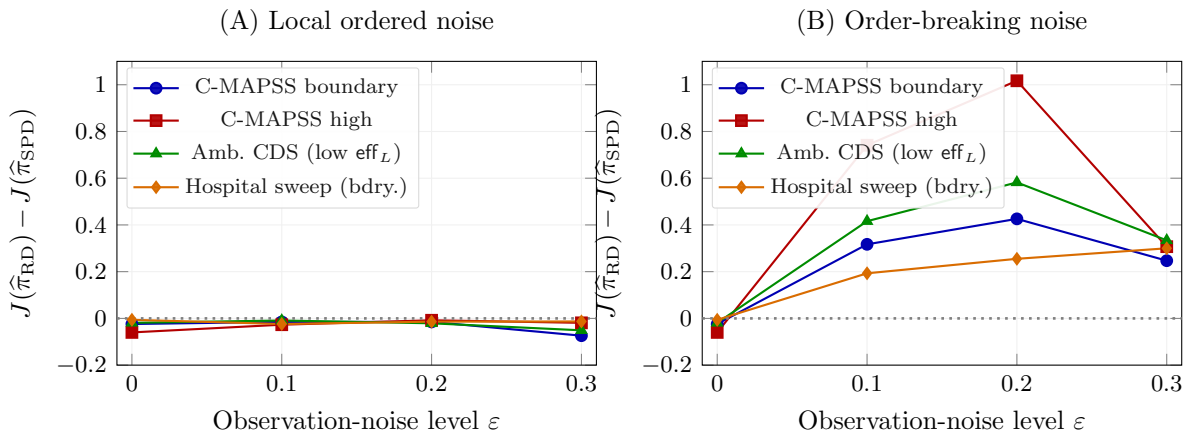

Figure D.1 Estimated RD vs. estimated SPD under observation-noise perturbation across four boundary/high-drift environments. Each curve plots the realized reward gap (RD minus SPD) as a function of observation-noise level. Panel (A): local ordered noise, where RD slightly loses. Panel (B): order-breaking noise, where RD wins substantially. Cross-family summary in Table D.3.

Family A summary. The two panels of Figure D.1 exhibit a clear sign flip between Models A and B. Under local ordered noise (Panel (A)), estimated RD loses only marginally to estimated SPD across all four environments, with the reward gap never exceeding about 0.07 in absolute value. Under order-breaking noise (Panel (B)), the proxy-observed kernel becomes severely misspecified and estimated SPD overfits; estimated RD then wins substantially, with reward gaps between +0.2 and +1.0 reward units at $\varepsilon \in \{0.10, 0.20\}$.

Family B: burden misspecification. The planner keeps the true transition and reward primitives but solves SPD with a misspecified burden $\hat{B}(x,\ell) = \hat{a}_1\,\Psi(x,\ell) + \hat{a}_2\,\phi(\ell) + \hat{a}_3\,\Pr(C_{t+1} \le \hat{c}_{\mathrm{low}} \mid x,\ell) + \gamma \mathbb{1}\{c \text{ high}\}\mathbb{1}\{\ell \text{ high}\}$. We test twelve conditions: weight perturbations of $(\hat{a}_1, \hat{a}_2, \hat{a}_3)$, dropped channels ($\Psi$, ϕ, $\Pr_{\mathrm{low}}$), wrong thresholds $\hat{c}_{\mathrm{low}} \in \{0, 2\}$, and spurious interactions $\gamma \in \{0.25, 0.50\}$. By design, this family creates an information asymmetry: RD uses its Ψ-only restriction on the true Ψ (because the restriction is invariant to burden misspecification), while SPD is solved with the misspecified $\hat{B}$. The comparison is therefore oracle-Ψ RD versus misspecified-B SPD, not a matched-information benchmark. SPD's budget is matched to $\bar{B}$ under the belief $\hat{B}$, and evaluation is always under the true B. The results are asymmetric. When the planner overweights the direct intensity penalty ($\hat{a}_2$=1.0 vs 0.5), SPD becomes overcautious and RD beats it by +0.74 to +0.86 reward units across all four environments. When ϕ is underweighted or omitted, SPD becomes less restrictive and catches slightly more reward, so RD loses marginally. Wrong thresholds, dropped Ψ or low-efficacy channels, and spurious interactions have effects below 0.1 in absolute magnitude.

Theoretical explanation. The asymmetry admits a one-sided explanation. Throughout Proposition D.1 and Proposition D.2, we maintain the identification $B(x,\ell) \equiv \Psi(x,\ell)$. Under this identification, the RD and SPD constrained problems share the same feasible set. The two propositions below and the reward comparisons J_ν are stated and proved in G-space (relative to silence), with the RD value–cost frontier $V^\star(\bar{\Psi})$ of Proposition 2 as the reference frontier. Misspecification is modeled by letting the planner solve SPD with an estimated burden $\hat{B} \neq B$. The two propositions below characterize two distinct imperfect-information

channels: Proposition D.1 identifies a named over-restrictive misspecification class under which RD dominates plug-in SPD exactly, and Proposition D.2 identifies a finite-sample estimation regime in which the same RD structure dominates flexible plug-in SPD because of estimation fragility. Extensions to $B \neq \Psi$ require a separately defined SPD frontier $V^{\star}_{\mathrm{SPD}}(\bar{\Psi}; B)$ and are deferred to the online supplement.

PROPOSITION D.1 **(RD dominance under over-restrictive burden misspecification)**. *Assume $B \equiv \Psi$ on the nominal model. Let $V^{\star}(\bar{\Psi})$ denote the RD value–cost frontier from Proposition 2, and let $\lambda^{\star} \in \partial V^{\star}(\bar{\Psi})$ be a dual-optimal multiplier at $\bar{\Psi}$. Let $\hat{B}(x, \ell)$ be the planner's estimated burden, and let $\pi_{\mathrm{SPD}}(\hat{B})$ be a deterministic Markov policy induced by the penalized DP on $\hat{B}$ at some $\hat{\lambda} \geq 0$, chosen so that its believed burden cost satisfies*

$$\mathbb{E}^{\pi_{\mathrm{SPD}}}\Big[\sum_{t=1}^{T} \hat{B}(X_t, \ell_t)\Big] \leq \bar{\Psi}.$$

When the believed budget binds, this inequality holds with equality. Let π_{RD} be the best RD-family member satisfying $\mathbb{E}^{\pi}[\sum_t B(X_t, \ell_t)] \leq \bar{\Psi}$—that is, the RD policy with the highest true reward among those that are budget-feasible—and let $\eta := V^{\star}(\bar{\Psi}) - J_{\nu}(\pi_{\mathrm{RD}}) \geq 0$ denote the oracle restriction cost of the RD formulation at $\bar{\Psi}$. Define the true-burden slack of the estimated SPD policy,

$$\Delta\bar{\Psi}_{\mathrm{SPD}} := \bar{\Psi} - \mathbb{E}^{\pi_{\mathrm{SPD}}}\Big[\sum_{t=1}^{T} B(X_t, \ell_t)\Big].$$

Suppose that the misspecification is over-restrictive *in the sense that $\hat{B}(x, \ell) \geq B(x, \ell)$ for every $(x, \ell) \in \mathcal{X} \times \mathcal{L}$. Then:*

(i) $\Delta\bar{\Psi}_{\mathrm{SPD}} \geq 0$;

(ii) $J_{\nu}(\pi_{\mathrm{SPD}}) \leq V^{\star}(\bar{\Psi}) - \lambda^{\star}\,\Delta\bar{\Psi}_{\mathrm{SPD}}$;

(iii) whenever $\lambda^{\star}\,\Delta\bar{\Psi}_{\mathrm{SPD}} > \eta$, the true-environment realized reward satisfies $J_{\nu}(\pi_{\mathrm{RD}}) > J_{\nu}(\pi_{\mathrm{SPD}})$.

Proof. See Appendix F.17.

Mechanism. Proposition D.1 identifies a named misspecification class under which RD can dominate plug-in SPD. When the planner's estimated burden upper-bounds the true burden pointwise, the SPD policy, chosen to satisfy the budget under the inflated belief, is over-restrictive relative to the true constraint: its realized true burden cost falls below $\bar{\Psi}$ by $\Delta\bar{\Psi}_{\mathrm{SPD}} \geq 0$, and by concavity of the frontier this slack represents unrealized reward of size at least $\lambda^{\star}\,\Delta\bar{\Psi}_{\mathrm{SPD}}$. If the believed budget binds, this reduces to exact believed-budget saturation as a special case. If the induced SPD-side reward loss exceeds the oracle RD-restriction cost η, RD strictly dominates. Table D.2 shows that η is small in absolute terms across the paper's benchmark environments (at most 0.16 reward units on the unified grid, or at most 2.1% of the SPD design gain—the maximum at the lowest-drift hospital row), so even moderate values of $\lambda^{\star}\,\Delta\bar{\Psi}_{\mathrm{SPD}}$ are sufficient to flip the ranking. The clean single-condition RD-win observed above—overweighted intensity penalty—is the natural three-channel analogue of the over-restrictive case of Proposition D.1. The proposition is stated under the maintained identification $B \equiv \Psi$; the empirical Family B setup allows a three-channel $B \neq \Psi$, so the correspondence is not a verbatim application of the proposition but a reading in which the same over-restrictive mechanism—an inflated believed burden forces the estimated SPD policy to leave true-burden slack on the table, and that slack is converted to reward loss at rate $\lambda^{\star}$—operates pointwise when

overweighting ϕ through $\hat{a}_2$, which induces $\hat{B}(x,\ell) \geq B(x,\ell)$ uniformly. The empirical magnitudes ($\Delta\bar{\Psi}_{\mathrm{SPD}} \approx$ 1.1–1.3 burden units combined with dual-optimal $\lambda^\star$ on the order of 0.5–0.9; compare the SPD realized B-cost to the target $\bar{\Psi}$ in Table D.5) reproduce the observed +0.74 to +0.86 reward-unit gap under this reading.

Sample-complexity perspective. Proposition D.1 identifies a class-specific misspecification channel through which RD can dominate plug-in SPD. A second, complementary channel arises from finite-sample estimation fragility when the planner fits a flexible cost model from limited data. Suppose the planner's burden lies in a d-dimensional class $\mathcal{B}_d = \{B_a : a \in \mathbb{R}^d\}$ with bounded basis functions and is estimated by $\hat{B} = B_{\hat{a}}$ with an unbiased estimator $\hat{a}$. If the plug-in SPD regret is locally quadratically sensitive to burden-estimation error, then the expected true-environment regret of plug-in SPD scales with the estimation error in $\hat{a}$. The RD policy, by contrast, does not depend on a, so its expected regret remains bounded by the oracle restriction cost η. The next proposition records this second imperfect-information channel.

Proposition D.2 **(RD dominance under burden-estimation fragility)**. *Assume $B \equiv \Psi$ on the nominal model, so that $V^\star(\bar{\Psi})$ from Proposition 2 is the RD value–cost frontier and all comparisons below are in G-space. Let $\mathcal{B}_d = \{B_a : a \in \mathbb{R}^d\}$ be a d-dimensional linear burden class with true burden $B^\star = B_{a^\star}$, and let $\hat{a}$ be an unbiased estimator of $a^\star$ such that*

$$\Pr(\hat{a} \in \mathcal{N}) = 1 \qquad \textit{and} \qquad \mathbb{E}\|\hat{a} - a^\star\|^2 = \frac{\sigma^2 d}{N\kappa_{\min}}$$

for some sample size N, some constant $\kappa_{\min} > 0$, and some neighborhood $\mathcal{N} \subseteq \mathbb{R}^d$ of $a^\star$.

Suppose the plug-in SPD regret is locally quadratically identified on $\mathcal{N}$ in the sense that there exist constants

$$0 < \underline{L} \leq \overline{L} < \infty$$

such that, for every $a \in \mathcal{N}$,

$$\underline{L}\,\|a - a^\star\|^2 \;\leq\; V^\star(\bar{\Psi}) - J_\nu(\pi_{\mathrm{SPD}}(B_a)) \;\leq\; \overline{L}\,\|a - a^\star\|^2.$$

Then

$$\frac{\underline{L}\,\sigma^2 d}{N\kappa_{\min}} \;\leq\; \mathbb{E}\Big[V^\star(\bar{\Psi}) - J_\nu\big(\pi_{\mathrm{SPD}}(B_{\hat{a}})\big)\Big] \;\leq\; \frac{\overline{L}\,\sigma^2 d}{N\kappa_{\min}}.$$

Moreover,

$$V^\star(\bar{\Psi}) - J_\nu(\pi_{\mathrm{RD}}) \leq \eta$$

for the best budget-feasible RD-family policy at $\bar{\Psi}$. In particular, whenever

$$N < N^\star(d, \sigma, \eta, \kappa_{\min}, \underline{L}) := \frac{\underline{L}\,\sigma^2 d}{\eta\,\kappa_{\min}},$$

the RD policy has strictly lower expected true-environment regret than the corresponding plug-in SPD policy estimated from the same data.

Proof. See Appendix F.18.

Reading the threshold. Proposition D.2 identifies a second imperfect-information channel through which RD can dominate plug-in SPD. As the burden parameter space grows (d large) or the sample size shrinks (N

Environment	Local noise (worst ε)	Order-breaking noise (peak ε)
C-MAPSS boundary	-0.074 (at ε=0.30)	$+0.426$ (at ε=0.20)
C-MAPSS high drift	-0.027 (at ε=0.10)	$+1.017$ (at ε=0.20)
Amb. CDS (low eff_L)	-0.051 (at ε=0.30)	$+0.582$ (at ε=0.20)
Hospital sweep (boundary)	-0.024 (at ε=0.10)	$+0.300$ (at ε=0.30)

Table D.4 Family A (observation noise) worst-case summary. One row per environment and noise model, reporting the realized reward gap (RD minus SPD) at the most adverse noise level. Negative entries indicate a small SPD advantage under local noise; positive entries indicate an RD advantage under order-breaking noise.

small), the expected true-environment regret of the flexible plug-in SPD policy increases because burden-estimation error is amplified by the local quadratic sensitivity of the plug-in map. RD, by contrast, pays only the fixed oracle restriction cost η. The threshold $N^\star$ is increasing in d and in $\underline{L}\sigma^2/\kappa_{\min}$ and decreasing in η. Together, Proposition D.1 and Proposition D.2 identify two complementary channels through which RD can dominate plug-in SPD: one arising from over-restrictive burden misspecification and one arising from finite-sample estimation fragility.

Family C: finite-sample estimation. The planner draws N training episodes from the true environment under a uniform exploratory policy. From these episodes we compute maximum-likelihood estimates $(\widehat{F}, \widehat{K}, \widehat{\mu})$ with a Dirichlet prior of 0.5 for smoothing, then build the plug-in reward $\widehat{R}$ and plug-in burden $\widehat{B}$. SPD and RD are solved on the plug-in model and evaluated on the true environment. The grid is $N \in \{500, 1000, 2500, 5000\}$ with eight Monte Carlo replications per configuration.

Reported quantities. For each cell we record the realized true reward of SPD and RD, the realized true B-cost of each, the RD$-$SPD realized-reward gap, and for Family C the RD-win probability across replications. Figure D.1 above plots the Family A reward gap against ε for both noise models, showing the clean separation between local-noise (RD loses marginally) and order-breaking (RD wins substantially).

Full numeric results. Table D.3 above summarizes the cross-family pattern. The three tables below give the per-family detail for Family A (observation noise), Family B (burden misspecification), and Family C (finite-sample estimation). Entries are realized true rewards averaged over the uniform initial distribution; positive values in "RD$-$SPD" indicate that estimated RD beats estimated SPD on the true environment.

The finite-sample grid is therefore interpreted as an illustration of the estimation-fragility channel formalized in Proposition D.2, rather than as the paper's main quantitative benchmark for RD dominance.

Interpretation and theory alignment. The oracle and imperfect-information comparisons together describe a bias–variance trade-off: under accurate models, SPD's additional flexibility wins marginally; under severe model error, RD's Ψ-only restriction regularizes the design. The RD advantage under order-breaking noise scales with drift intensity in the same direction as the SPD–myopic gap—the largest wedge (+1.02) occurs in C-MAPSS high drift at ε=0.20, exactly where the dynamic continuation channel is strongest and where Proposition 4's near-tie logic implies that small perturbations to the continuation kernel can most easily overturn the penalized-DP ranking. The RD wins are not universal: finite-sample estimation error does not deliver a robust RD advantage, and under mild local observation noise RD still loses marginally. The story is therefore disciplined rather than universal: where the planner's model is severely misspecified, the RD restriction that yields the ordered-menu characterization also provides a regularization benefit; where the model is nearly correct, the same restriction costs a small amount of realized reward.

Condition	C-MAPSS bdry	C-MAPSS high	Amb low eff_L	Hosp sweep
wts 1.0/0.5/0.5 (oracle)	−0.022	−0.063	−0.039	−0.019
wts 1.0/1.0/0.5	+0.822	+0.742	+0.781	+0.862
wts 1.0/0.25/0.5	−0.137	−0.152	−0.140	−0.138
wts 1.0/0.5/1.0	−0.015	−0.056	−0.033	−0.012
wts 1.0/0.5/0.25	−0.025	−0.066	−0.042	−0.023
omit ϕ	−0.137	−0.152	−0.140	−0.138
omit $\mathrm{Pr}_{\mathrm{low}}$	−0.028	−0.069	−0.045	−0.025
omit Ψ	−0.034	−0.076	−0.050	−0.028
threshold $\hat{c}_{\mathrm{low}}$=0	−0.016	−0.049	−0.033	−0.008
threshold $\hat{c}_{\mathrm{low}}$=2	−0.025	−0.067	−0.043	−0.024
spurious γ=0.25	−0.022	−0.063	−0.039	−0.019
spurious γ=0.50	−0.022	−0.063	−0.039	−0.019

Table D.5 Family B (burden misspecification): realized rewards across twelve misspecification conditions for the four boundary/high-drift environments. The relative-design policy uses its downstream-cost-only restriction throughout. The clean RD-win region is the overweighted exposure-penalty condition.

Environment	Best cell (mean / win prob.)	Overall mean	Overall win prob.
C-MAPSS boundary	N=500: +0.007 / 0.38	−0.087	0.22
C-MAPSS high drift	N=500: +0.122 / 0.75	+0.004	0.44
Amb. CDS (low eff_L)	N=1000: −0.053 / 0.38	−0.101	0.31
Hospital sweep (boundary)	N=1000: −0.037 / 0.50	−0.153	0.16

Table D.6 Family C (finite-sample estimation) aggregated summary. "Best cell" gives the sample-size configuration with the largest mean realized-reward gap (RD minus SPD) across Monte Carlo draws. Only one cell exhibits a positive mean with a win probability above one-half; the finite-sample regime does not support a robust RD advantage claim.

D.6. RD-vs-SPD stress tests: quantitative bound check and prior-averaged perturbations

This subsection records two stress tests that go beyond the hand-picked perturbations of Appendix D.5: a quantitative predictive check for the bound in Proposition D.1, and a prior-averaged distribution of RD-vs-SPD outcomes under random burden misspecification. Simulation script: `scripts/spd_vs_rd_stress.py`, seed 2026.

Prior-averaged misspecification (S2). We next draw burden misspecifications from a prior rather than picking them adversarially. For each environment we sample 200 specifications: multiplicative log-normal noise (σ=0.55) on the three baseline weights, independent channel dropout at rate 0.10, a perturbed low-efficacy threshold, and a spurious interaction term added with probability 0.15. Each draw is solved by plug-in SPD on the perturbed burden, evaluated on the true environment under the true burden, and compared to the RD frontier. Figure D.2 shows the resulting distribution.

The distribution is asymmetric. RD−SPD has a positive mean (+0.10 to +0.21) but a negative median (−0.07 to −0.12): RD wins outright in 36%–42% of draws, and when it wins it wins by substantially more than SPD wins when SPD wins. The 90th-percentile RD advantage (+0.63 to +1.05) dwarfs the 10th-percentile SPD advantage (−0.14 to −0.15). This is the bias–variance pattern Proposition D.1 and Proposition D.2 predict: RD's regret is capped while SPD's has a heavy right tail from over-restrictive realizations, so RD's expected advantage is positive even though SPD wins more individual draws.

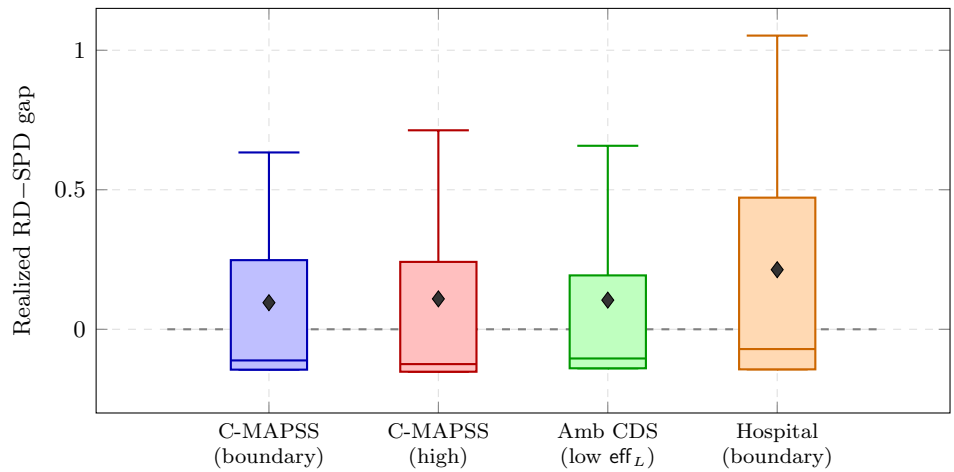

Figure D.2 Prior-averaged distribution of RD minus SPD realized reward across 200 random burden misspecifications, by environment. Boxes span the 25th–75th percentiles, whiskers the 10th–90th. The dashed line at zero separates RD wins (above) from SPD wins (below). The mean is positive across all four environments, driven by an asymmetric right tail.

Together, S2 and S3 sharpen the paper's RD-vs-SPD message. Proposition D.1 identifies the one-sided sufficient condition under which RD must dominate, and Proposition D.2 extends the logic to finite-sample estimation via $N < N^\star$. S3 shows that the predicted advantage $\lambda^\star \Delta\bar{\Psi}_{\mathrm{SPD}} - \eta$ is quantitatively—not just directionally—predictive. S2 shows that under symmetric prior-averaged perturbations, RD's expected advantage remains positive across benchmark environments because of the asymmetric right tail. The conditions under which RD dominates plug-in SPD are therefore identified both theoretically and empirically.

Appendix E: Multi-Level Signal Design

This appendix records supporting material for the ordered-menu extension: the multi-level formulation, the score-line geometry, the pruning logic, and numerical illustrations on the C-MAPSS grid.

E.1. Ordered Signal Menus

The main text adopts the integer-indexed convention $\mathcal{L} = \{0, 1, \ldots, k\}$, with $\ell = 0$ denoting silence and adjacent pairs written as $\ell{+}1$ and $\ell{-}1$. We retain that convention for the design objects (G_ℓ, Ψ_ℓ, score-line geometry, and pruning logic), using the equivalent generic-label form $\{\ell_0, \ell_1, \ldots, \ell_k\}$ with $\ell_0 = 0$ only when the exposition benefits from naming non-adjacent levels explicitly. Higher levels correspond to more prescriptive forms of assistance (for example, silence < alert < partial information < guided suggestion < full prescription). As the menu grows, the state space, factored kernel, dynamic program, and occupancy-measure LP are unchanged; only the geometry of the feasible signal choices changes.

For each state $x = (s, c)$ and level $\ell \in \mathcal{L}$, define the *gain* and *relative downstream cost* relative to silence:

$$G_\ell(x) := \mathbb{E}\big[R(s, A) \mid x, \ell\big] - \mathbb{E}\big[R(s, A) \mid x, 0\big], \tag{6}$$

$$\Psi_\ell(x) := \mathbb{E}\big[\mathcal{U}(S_{t+1}, C_{t+1}) \mid x, \ell\big] - \mathbb{E}\big[\mathcal{U}(S_{t+1}, C_{t+1}) \mid x, 0\big], \tag{7}$$

with $G_0(x) = \Psi_0(x) = 0$ by construction, and the *score*

$$S_\ell(x; \lambda) \;:=\; G_\ell(x) - \lambda\, \Psi_\ell(x), \qquad \lambda \geq 0. \tag{8}$$

The one-step ordered-menu rule (the $G-\lambda\Psi$ approximate signal policy with $|\mathcal{L}|>2$) chooses

$$\pi^{\lambda}(x) \in \operatorname*{arg\,max}_{\ell\in\mathcal{L}} S_{\ell}(x;\lambda),$$

defaulting to silence ($\ell=0$) when the score maximum is nonpositive. The binary case recovers: $\mathcal{L}=\{0,1\}$, $G_1=G$, $\Psi_1=\Psi$, and the rule reduces to "signal iff $G-\lambda\Psi\geq 0$."

E.2. Displacement Ordering from the Factored Kernel

The threshold geometry below is most useful when higher-intensity signals carry weakly larger downstream costs. The following lemma derives that ordering from the paper's primitive channels rather than imposing it as an auxiliary assumption.

Lemma E.1 **(Monotonicity of Ψ_ℓ in signal intensity)**. *Let $0=\ell_0<\ell_1<\cdots<\ell_k$. Suppose:*

(a) Action–unfamiliarity alignment. *For every state (s,c), define the action-conditional expected next-period unfamiliarity at level ℓ,*

$$W(a,\ell) := \sum_{s',c'} F(s'\mid s,a)\,K(c'\mid c,s,a,\ell)\,\mathcal{U}(s',c').$$

Then for every pair $\ell'>\ell$, $\sum_a \mu(a\mid s,c,\ell')\,W(a,\ell)\geq\sum_a \mu(a\mid s,c,\ell)\,W(a,\ell)$; that is, increasing signal intensity shifts the action distribution toward actions that induce weakly greater downstream unfamiliarity.

(b) Monotone efficacy drift. *For every (s,c,a), the transition probability $K(\mathsf{low}\mid c,s,a,\ell)$ into the low-efficacy state is weakly increasing in ℓ.*

(c) High-low unfamiliarity ordering. *For every (x,ℓ,a), next-period unfamiliarity is weakly higher when the next efficacy state is* low *than when it is* high*:*

$$\mathbb{E}[\mathcal{U}(S_{t+1},C_{t+1})\mid x,\ell,a,C_{t+1}=\mathsf{low}]\geq\mathbb{E}[\mathcal{U}(S_{t+1},C_{t+1})\mid x,\ell,a,C_{t+1}=\mathsf{high}]\,.$$

Then $\Psi_\ell(x)$ is weakly increasing in ℓ for every state x.

Proof. See Appendix F.20.

Remark E.1 (Ordered-menu monotonicity conditions). Conditions (a)–(c) are the ordered-menu analogues of the mechanism-side comparative statics discussed in Appendix A.2. On the operator side, a natural interpretation is skill atrophy or effort substitution: stronger signals such as full solutions can reduce future unaided capability more than softer signals such as hints, so residual efficacy drift and hence $\Psi_\ell(x)$ rise with prescriptiveness (Budzyn et al. 2025, Bastani et al. 2025, Poulidis et al. 2025a). Conditions (a) and (c) supply the ordered-state structure needed for efficacy drift to interact monotonically with signal intensity. The lemma thus closes the gap between the factored-kernel primitives and the cost ordering used in the multi-level propositions below.

E.3. Ordered-Menu Threshold Structure

Each score line $\lambda \mapsto S_\ell(x;\lambda) = G_\ell(x) - \lambda\,\Psi_\ell(x)$ is affine in λ, with intercept $G_\ell(x)$ and slope $-\Psi_\ell(x)$. The optimal level is therefore determined by the upper envelope of these lines.

PROPOSITION E.1 **(Threshold structure of the one-step score rule on an ordered menu)**. *Fix an augmented state $x = (s,c) \in \mathcal{X}$. Assume $\Psi_0(x) \le \Psi_1(x) \le \cdots \le \Psi_k(x)$ (the cost ordering of Lemma E.1), and pre-prune any levels with identical cost coordinates by keeping only the one with the highest gain. Then the ordered-menu policy $\pi^\lambda(x) \in \arg\max_{\ell\in\mathcal{L}} S_\ell(x;\lambda)$ satisfies:*

(i) Both the maximal and minimal selections from $\arg\max_{\ell\in\mathcal{L}} S_\ell(x;\lambda)$ are weakly decreasing in λ: a higher shadow price selects weakly lower signal intensity.

(ii) The λ-axis is partitioned into at most $k+1$ contiguous intervals, each associated with an active score-maximizing level, with ties possible only at boundary values.

(iii) Transitions between adjacent active levels occur at finitely many thresholds $\bar{\lambda}_j(x) = (G_j(x) - G_{j'}(x))/(\Psi_j(x) - \Psi_{j'}(x))$, where j and j' are consecutive efficient levels (see Proposition E.2).

Proof. See Appendix F.21.

The binary case is the special case $k = 1$: the single threshold $\bar{\lambda}(x) = G_1(x)/\Psi_1(x)$ reproduces the rule "signal iff $\lambda \le \bar{\lambda}(x)$."

E.4. Efficient Levels and Menu Pruning

Not every level in the menu is worth offering. The next proposition characterizes which levels are *efficient*—score-maximizing for some λ—and which are *dominated*—never score-maximizing.

PROPOSITION E.2 **(Efficient levels under the one-step score rule)**. *Fix an augmented state $x = (s,c) \in \mathcal{X}$. Assume $\Psi_0(x) \le \Psi_1(x) \le \cdots \le \Psi_k(x)$, and pre-prune any levels with identical cost coordinates by keeping only the one with the highest gain. Consider the menu points $\{(\Psi_\ell(x), G_\ell(x)) : \ell \in \mathcal{L}\}$ in the (Ψ, G)-plane.*

(i) A level ℓ is score-maximizing under $S_\ell(x;\lambda)$ for some $\lambda \ge 0$ if and only if the point $(\Psi_\ell(x), G_\ell(x))$ lies on the portion of the upper concave envelope *of the menu points that is exposed by a supporting line of slope $\lambda \ge 0$.*

(ii) Any level strictly below the upper concave envelope is never score-maximizing for any $\lambda \ge 0$. Such a level can be removed from the menu at state x without changing the score-maximizing level under $S_\ell(x;\lambda)$ for any λ.

(iii) Let $\ell^{(0)} < \ell^{(1)} < \cdots < \ell^{(m)}$ be the efficient levels (those on the portion of the upper concave envelope exposed by nonnegative slopes), with $\ell^{(0)} = 0$. Then their adjacent slopes

$$r_j(x) := \frac{G_{\ell^{(j)}}(x) - G_{\ell^{(j-1)}}(x)}{\Psi_{\ell^{(j)}}(x) - \Psi_{\ell^{(j-1)}}(x)}, \qquad j = 1, \ldots, m,$$

are weakly decreasing and nonnegative:

$$r_1(x) \ge r_2(x) \ge \cdots \ge r_m(x) \ge 0.$$

Proof. See Appendix F.22.

Remark E.2 (Genericity of full menu activation). After pre-pruning levels with identical cost coordinates, all levels on the menu are efficient if and only if the adjacent slopes $r_j(x)$ of Proposition E.2 are weakly decreasing and nonnegative across all levels; the menu points then lie on the upper concave envelope in the (Ψ, G)-plane. If the adjacent slopes are *strictly* decreasing and strictly positive, every level occupies a nonempty open λ-interval and the menu admits no pruning in the stronger interior-activation sense. Ties among adjacent slopes correspond to collinear menu points; negative slopes correspond to points on the envelope that are not exposed by any supporting line of nonnegative slope; and zero slopes correspond to points exposed only at the boundary price $\lambda = 0$. Generically these degeneracies are absent, and every level occupies a nonempty open λ-interval.

Corollary E.1 **(Interior efficient levels can be locally score-maximizing)**. *Fix an augmented state $x = (s, c) \in \mathcal{X}$, and let $\ell^{(0)} < \ell^{(1)} < \cdots < \ell^{(m)}$ be the efficient levels from Proposition E.2, with $\ell^{(0)} = 0$. For any interior level $\ell^{(j)}$ with $0 < j < m$, if the adjacent slopes satisfy $r_{j+1}(x) < r_j(x)$, then there exists a nonempty open interval $(\underline{\lambda}_j, \bar{\lambda}_j)$ such that $\ell^{(j)}$ is the unique score-maximizing signal level under the one-step rule for all $\lambda \in (\underline{\lambda}_j, \bar{\lambda}_j)$. In particular, if $r_1(x) > r_2(x) > \cdots > r_m(x) > 0$, then every interior efficient level is uniquely score-maximizing on a nonempty open interval.*

Proof. See Appendix F.23.

This result is not driven by noise or uncertainty. It is a structural consequence of the concave gain–cost frontier: at the relevant shadow prices, intermediate levels can offer a better gain-per-unit-cost trade-off than either silence or full prescription. In the benchmark environments, 57.5% of states have at least one efficient interior level.

Proposition E.3 **(Multi-level approximation bound)**. *Under the bounded-curvature assumption of Proposition B.1, for each level $\ell \in \mathcal{L}$ the net signal value decomposes as*

$$\Delta_\ell(x) \;=\; G_\ell(x) - \lambda^{\mathrm{loc}}\,\Psi_\ell(x) + \varepsilon_\ell(x),$$

with per-level error bound $|\varepsilon_\ell(x)| \le (L/2)\big(\mathrm{Var}_\ell(W) + \mathrm{Var}_0(W) + \Psi_\ell(x)^2\big)$. The per-state regret of the one-step rule $\hat{\ell}(x) := \arg\max_{\ell\in\mathcal{L}}\big\{G_\ell(x) - \lambda^{\mathrm{loc}}\,\Psi_\ell(x)\big\}$ relative to the exact optimal level $\ell^\star(x) := \arg\max_{\ell\in\mathcal{L}} \Delta_\ell(x)$ satisfies $\Delta_{\ell^\star}(x) - \Delta_{\hat{\ell}}(x) \le 2\max_{\ell\in\mathcal{L}} |\varepsilon_\ell(x)|$.

Proof. See Appendix F.25.

E.5. Numerical Illustration

We illustrate the ordered-menu theory on the 8×5 C-MAPSS state grid (T=20, ζ=1.4, structurally specified K) with a five-level menu $\mathcal{L} = \{0, 0.25, 0.50, 0.75, 1.0\}$ (silence, alert, partial information, guided suggestion, full prescription).

Score-line geometry and upper envelope. Figure E.2 provides the ordered-menu geometry in fuller detail. At state $x_1 = (h$=*fair*, c=VH), all five levels are active and the λ-axis is partitioned into five intervals. At state $x_2 = (h$=*very good*, c=VL), the two intermediate levels ℓ=0.50, 0.75 are dominated and therefore pruned.

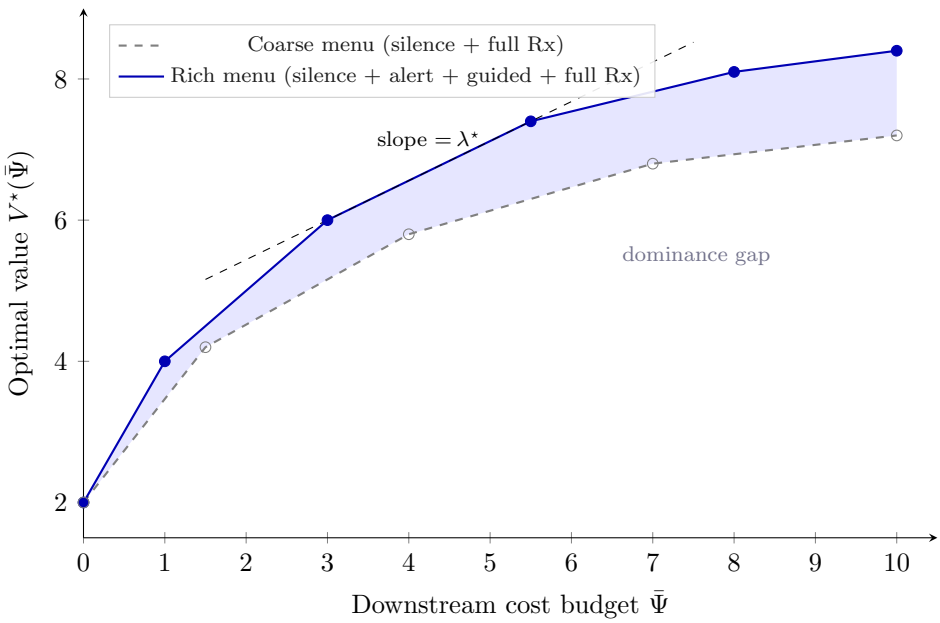

Figure E.1 **Value–cost frontier and menu richness. The rich-menu frontier weakly dominates the coarse-menu frontier at every cost budget (shaded gap). The tangent line at the rich frontier has slope equal to the shadow price of downstream cost. Stylized illustration; exact frontiers are piecewise affine with finitely many kinks.**

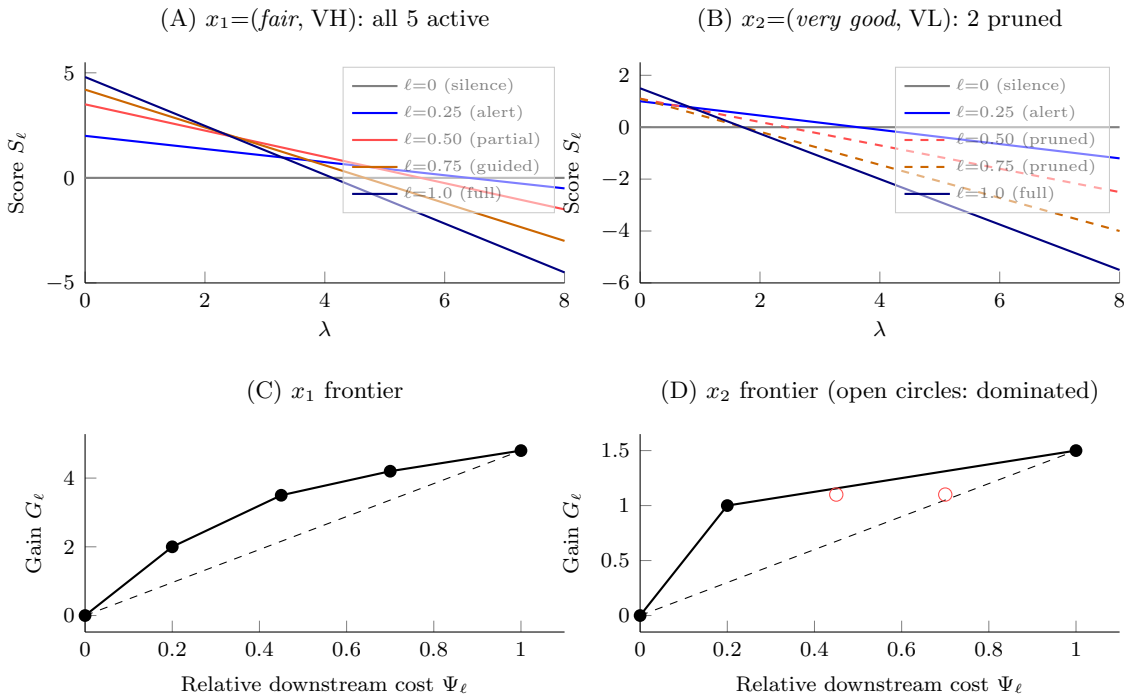

Figure E.2 **Score-line geometry. Top: score versus shadow price at two representative states; the upper envelope determines the score-maximizing level under the one-step score rule. At one state all five levels are active; at the other, two are pruned (dashed). Bottom: gain–cost frontier; dominated levels (open circles) lie below the hull.**

Appendix F: Proofs

This appendix contains proofs for the propositions, lemmas, theorems, and corollaries stated in the main paper and appendices.

F.1. Proof of Proposition 1 (Reduced-form decision insufficiency)

Proof. We prove the proposition in two parts.

Part (i): two observationally equivalent factorizations with opposite deployment implications. Fix a planner state $x=(s,c)$. Let $\mathcal{A}=\{a_0,a_1\}$, $\mathcal{C}=\{\mathsf{high},\mathsf{low}\}$, and $\mathcal{L}=\{0,1\}$. For each signal level $m\in\{0,1\}$, let

$$\bar{P}_m(\cdot\mid x):=\sum_{a\in\mathcal{A}}\mu(a\mid x,m)\,K(\cdot\mid c,s,a,m)$$

denote the reduced-form efficacy transition. By assumption, $\bar{P}_0(\cdot\mid x)\neq\bar{P}_1(\cdot\mid x)$.

We first construct a factorization satisfying signal-independence conditional on action. Define

$$\mu^S(a_0\mid x,0)=1,\quad \mu^S(a_1\mid x,0)=0,\quad \mu^S(a_0\mid x,1)=0,\quad \mu^S(a_1\mid x,1)=1,$$

and set

$$K^S(\cdot\mid c,s,a_0,m)=\bar{P}_0(\cdot\mid x),\qquad K^S(\cdot\mid c,s,a_1,m)=\bar{P}_1(\cdot\mid x),\qquad m\in\{0,1\}.$$

Then K^S is signal-independent conditional on action at the focal state x. For each $a\in\mathcal{A}$, we have $K^S(\cdot\mid c,s,a,0)=K^S(\cdot\mid c,s,a,1)$. Hence this factorization satisfies, at x, the exclusion restriction that eliminates residual efficacy drift. For each $m\in\{0,1\}$, we also have $\sum_{a\in\mathcal{A}}\mu^S(a\mid x,m)\,K^S(\cdot\mid c,s,a,m)=\bar{P}_m(\cdot\mid x)$.

We next construct a *drift-only* factorization. Let $\mu^D(a_0\mid x,0)=\mu^D(a_0\mid x,1)=\frac{1}{2}$, $\mu^D(a_1\mid x,0)=\mu^D(a_1\mid x,1)=\frac{1}{2}$, so that $\|\Delta\mu^D\|_{\mathrm{TV}}=0$. Set $K^D(\cdot\mid c,s,a,m)=\bar{P}_m(\cdot\mid x)$ for every $a\in\mathcal{A}$, $m\in\{0,1\}$. Then signal dependence is carried by K^D. Since $\bar{P}_0(\cdot\mid x)\neq\bar{P}_1(\cdot\mid x)$, this factorization has residual efficacy drift. We have

$$\sum_{a\in\mathcal{A}}\mu^D(a\mid x,m)\,K^D(\cdot\mid c,s,a,m)=\bar{P}_m(\cdot\mid x).$$

The two factorizations produce the same reduced-form pair but have opposite structural properties: one satisfies signal-independence conditional on action, while the other loads all signal dependence into K^D.

Part (ii): explicit common completion with opposite local rankings. We now show that the two factorizations from Part (i) can be embedded in a *common* two-period environment in which they induce opposite local signal rankings at the same planner state. Because $\bar{P}_0(\cdot\mid x)\neq\bar{P}_1(\cdot\mid x)$, there exists a measurable set $B\subseteq\mathcal{C}$ such that

$$\Delta_B:=\bar{P}_1(B\mid x)-\bar{P}_0(B\mid x)>0.$$

Consider a two-period problem with initial state $x=(s,c)$ at the current stage, zero current reward,

$$R(s,a)=0\qquad\text{for all } a\in\mathcal{A},$$

and physical state space enlarged to include two successor states s^-,s^+. Let the physical transition kernel satisfy

$$F(s^-\mid s,a_0)=1,\qquad F(s^+\mid s,a_1)=1.$$

Choose any constant $M > \Delta_B$, and define the next-period continuation value by

$$V_{t+1}(s', c') := M\,\mathbf{1}\{s' = s^+\} - \mathbf{1}\{c' \in B\}.$$

We first evaluate the *safe* factorization (μ^S, K^S). Under μ^S, signal 0 induces action a_0 and signal 1 induces action a_1. Hence,

$$Q_t^S(x,0) = \sum_{c' \in \mathcal{C}} \bar{P}_0(c' \mid x)\, V_{t+1}(s^-, c') = -\bar{P}_0(B \mid x),$$

whereas

$$Q_t^S(x,1) = \sum_{c' \in \mathcal{C}} \bar{P}_1(c' \mid x)\, V_{t+1}(s^+, c') = M - \bar{P}_1(B \mid x).$$

Therefore,

$$Q_t^S(x,1) - Q_t^S(x,0) = M - \big(\bar{P}_1(B \mid x) - \bar{P}_0(B \mid x)\big) = M - \Delta_B > 0.$$

So in the safe factorization the higher signal is strictly beneficial at x.

Now evaluate the *drift-only* factorization (μ^D, K^D). Under μ^D, the action distribution is signal-invariant:

$$\mu^D(a_0 \mid x, 0) = \mu^D(a_0 \mid x, 1) = \tfrac{1}{2}, \qquad \mu^D(a_1 \mid x, 0) = \mu^D(a_1 \mid x, 1) = \tfrac{1}{2}.$$

Thus the expected physical-state contribution $M\,\mathbf{1}\{s' = s^+\}$ is the same under both signals and cancels in the difference. More explicitly,

$$Q_t^D(x,m) = \tfrac{1}{2}M - \bar{P}_m(B \mid x), \qquad m \in \{0,1\},$$

so

$$Q_t^D(x,1) - Q_t^D(x,0) = -\big(\bar{P}_1(B \mid x) - \bar{P}_0(B \mid x)\big) = -\Delta_B < 0.$$

So in the drift-only factorization the higher signal is strictly harmful at x.

Finally, the two environments have the same reduced-form efficacy transitions at x by Part (i), and they also have the same immediate signal-level payoffs at x because $R(s,a) = 0$ for every action, so

$$\bar{R}(x,0) = \bar{R}(x,1) = 0$$

in both environments. Hence the same reduced-form objects are consistent with opposite local signal rankings:

$$Q_t^S(x,1) > Q_t^S(x,0), \qquad Q_t^D(x,1) < Q_t^D(x,0).$$

This proves the claimed opposite deployment implications.

Parts (i)–(ii) establish the claim. □

F.2. Proof of Corollary 1 (Decision-insufficiency of reduced-form evidence)

Proof. Consider the two-period completion constructed in Part (ii) of the proof of Proposition 1, and let the initial distribution ν put unit mass on the focal state x. Denote the two environments by

$$\mathcal{E}^S = (\mu^S, K^S, F, R, V_{t+1}) \qquad \text{and} \qquad \mathcal{E}^D = (\mu^D, K^D, F, R, V_{t+1}).$$

By construction, these two environments induce the same reduced-form efficacy transitions at x:

$$\bar{P}_m^S(\cdot \mid x) = \bar{P}_m^D(\cdot \mid x), \qquad m \in \{0,1\},$$

and the same immediate signal-level payoffs,

$$\bar{R}^S(x,0) = \bar{R}^D(x,0), \qquad \bar{R}^S(x,1) = \bar{R}^D(x,1).$$

Let ρ be any signal-design rule that conditions only on these reduced-form efficacy transitions and immediate signal-level payoffs, and not on the factorization into μ, F, and K. Since those observables are identical in $\mathcal{E}^S$ and $\mathcal{E}^D$ at the focal state x, the rule ρ must choose the same signal level at x in both environments.

But Proposition 1 shows that the exact local rankings are opposite:

$$Q_t^S(x,1) > Q_t^S(x,0), \qquad Q_t^D(x,1) < Q_t^D(x,0).$$

Hence the optimal signal at x is 1 in $\mathcal{E}^S$ and 0 in $\mathcal{E}^D$. Because ρ takes the same action in both environments, it must disagree with the optimal local choice in at least one of them. Therefore any rule based only on these reduced-form objects is suboptimal in at least one observationally equivalent environment.

For the lifecycle lower bound, fix the environment in which ρ is suboptimal, and let π^ρ denote the policy induced by ρ there. Suppose there exists a state-stage pair $(\bar{x}, \bar{t})$ such that

$$d_{\bar{t}}^{\pi^\rho}(\bar{x};\nu) \geq p \qquad \text{and} \qquad \delta_{\bar{t}}^{\pi^\rho}(\bar{x}) \geq \eta,$$

where

$$\delta_t^\pi(x) := V_t^\star(x) - Q_t^\star(x, \pi_t(x))$$

is the local Bellman gap. The standard occupancy decomposition for finite-horizon MDPs (Puterman 1994, Sec. 4.5) yields

$$J_\nu(\pi^\star) - J_\nu(\pi^\rho) = \sum_{t=1}^{T} \sum_{x \in \mathcal{X}} d_t^{\pi^\rho}(x;\nu)\, \delta_t^{\pi^\rho}(x).$$

Every term on the right-hand side is nonnegative, so

$$J_\nu(\pi^\star) - J_\nu(\pi^\rho) \geq d_{\bar{t}}^{\pi^\rho}(\bar{x};\nu)\, \delta_{\bar{t}}^{\pi^\rho}(\bar{x}) \geq p\eta.$$

Thus the induced lifecycle loss is at least $p\eta$. □

F.3. Proof of Lemma 1 (IDE decomposition)

Proof. Fix a stage t, a planner state (s,c), and any two signal levels $\ell', \ell \in \mathcal{L}$. For convenience, introduce the shorthand

$$\mu_a := \mu(a \mid s,c,\ell), \qquad \mu'_a := \mu(a \mid s,c,\ell'), \qquad \Delta\mu_a := \mu'_a - \mu_a.$$

Also define the action-contingent continuation value

$$w_t(a,\ell) := w_t(s,c,a,\ell) = \sum_{s',c'} F(s' \mid s,a)\, K(c' \mid c,s,a,\ell)\, V_{t+1}(s',c').$$

We now expand the two action values:

$$Q_t(s,c,\ell') = \sum_a \mu'_a \big(R(s,a) + w_t(a,\ell')\big), \tag{9}$$

$$Q_t(s,c,\ell) = \sum_a \mu_a \big(R(s,a) + w_t(a,\ell)\big). \tag{10}$$

Subtract (10) from (9):

$$Q_t(s,c,\ell') - Q_t(s,c,\ell) = \sum_a \mu'_a R(s,a) - \sum_a \mu_a R(s,a) + \sum_a \mu'_a w_t(a,\ell') - \sum_a \mu_a w_t(a,\ell). \tag{11}$$

The reward term simplifies to $\sum_a \Delta\mu_a\, R(s,a)$. For the continuation term, add and subtract $\sum_a \mu'_a w_t(a,\ell)$:

$$\sum_a \mu'_a w_t(a,\ell') - \sum_a \mu_a w_t(a,\ell) = \sum_a \mu'_a \big(w_t(a,\ell') - w_t(a,\ell)\big) + \sum_a \Delta\mu_a\, w_t(a,\ell). \tag{12}$$

Substituting (12) back into (11) yields

$$Q_t(s,c,\ell') - Q_t(s,c,\ell) = \underbrace{\sum_a \Delta\mu_a\, R(s,a)}_{\mathcal{I}} + \underbrace{\sum_a \Delta\mu_a\, w_t(a,\ell)}_{\mathcal{D}} + \underbrace{\sum_a \mu'_a \big(w_t(a,\ell') - w_t(a,\ell)\big)}_{\mathcal{E}},$$

which is exactly (1), completing the proof.

Note that nowhere in the derivation did we use optimality of V_{t+1}. We only used that the expectations defining $w_t(a,\ell)$ are well defined. Hence the same identity holds for any bounded continuation-value function V_{t+1}. □

F.4. Proof of Proposition B.2 (Dual interpretation)

Proof. We prove the result by writing the constrained problem in standard occupancy-LP form, invoking finite-dimensional LP duality, and then showing that the optimal multiplier induces an equivalent penalized problem over the occupancy polytope.

Let x collect the finite-horizon occupancy measures, and write the standard occupancy-flow polytope as $\mathcal{P} = \{x \geq 0 : Ax = b\}$, where A and b encode the usual flow-conservation constraints and initial-condition terms. Then the constrained design problem can be written as

$$\begin{aligned} \underset{x}{\text{maximize}} \quad & \langle g, x\rangle \\ \text{subject to} \quad & Ax = b, \\ & \langle \psi, x\rangle \leq \bar{\Psi}, \\ & x \geq 0. \end{aligned} \tag{Primal}$$

From the stated feasibility assumption we have that this optimization problem is feasible. Moreover, because the horizon, state space, and signal menu are finite, $\mathcal{P}$ is a nonempty bounded polytope. Hence the primal problem has a finite attained optimum.

We consider a free dual vector y for the flow constraints $Ax = b$ and a nonnegative multiplier λ for the unfamiliarity-budget constraint $\langle \psi, x\rangle \leq \bar{\Psi}$. The dual LP is

$$\begin{aligned} \underset{y,\ \lambda \geq 0}{\text{minimize}} \quad & \langle b, y\rangle + \lambda\bar{\Psi} \\ \text{subject to} \quad & A^\top y + \lambda\psi \geq g. \end{aligned} \tag{Dual}$$

Because the primal and the dual are finite-dimensional linear programs, and the primal is feasible with finite optimal value, standard LP strong duality applies. Let $x^\star$ be any primal-optimal solution and let $(y^\star, \lambda^\star)$ be any dual-optimal solution. Then

$$\langle g, x^\star\rangle = \langle b, y^\star\rangle + \lambda^\star\bar{\Psi}, \tag{13}$$

and complementary slackness gives

$$\lambda^\star\big(\langle \psi, x^\star\rangle - \bar{\Psi}\big) = 0. \tag{14}$$

Combining (13) and (14),

$$\langle g - \lambda^\star \psi,\ x^\star\rangle = \langle b, y^\star\rangle. \tag{15}$$

Now fix any $x \in \mathcal{P}$. Dual feasibility in the dual implies

$$g - \lambda^\star \psi \le A^\top y^\star \qquad \text{componentwise.}$$

Since $x \ge 0$, it follows that

$$\langle g - \lambda^\star \psi,\ x\rangle \le \langle A^\top y^\star,\ x\rangle = \langle y^\star,\ Ax\rangle = \langle y^\star,\ b\rangle.$$

By (15), the right-hand side equals $\langle g - \lambda^\star \psi,\ x^\star\rangle$. Therefore

$$\langle g - \lambda^\star \psi,\ x\rangle \le \langle g - \lambda^\star \psi,\ x^\star\rangle \qquad \text{for all } x \in \mathcal{P},$$

so

$$x^\star \in \arg\max_{x \in \mathcal{P}} \langle g - \lambda^\star \psi,\ x\rangle. \tag{16}$$

Define the penalized one-period reward

$$r^{\lambda^\star}(x, \ell) := G_\ell(x) - \lambda^\star \Psi_\ell(x).$$

Then (16) is exactly the penalized occupancy LP associated with this reward. By the standard equivalence between feasible occupancy measures and randomized Markov policies in finite-horizon MDPs, (16) is equivalent to

$$\pi^\star \in \arg\max_{\pi} \mathbb{E}_\pi\left[\sum_{t=1}^{T}\big(G_{\ell_t}(X_t) - \lambda^\star \Psi_{\ell_t}(X_t)\big)\right].$$

Thus any primal-optimal policy solves the penalized problem at the dual-optimal multiplier $\lambda^\star$; if the budget binds, $\lambda^\star$ is the shadow price of relaxing that budget. □

F.5. Markov implementation of the constrained optimum

Proposition F.1 **(Markov implementation)**. *The constrained design problem* (Relative Design (RD)) *admits an optimal Markov policy that randomizes at at most one state-stage pair. Moreover, whenever the deterministic penalized policy induced by* (4) *is primal feasible and satisfies complementary slackness at a dual-optimal multiplier $\lambda^\star$, it attains the constrained optimum.*

Proof. Introduce the time-augmented state $\tilde{x} = (t, x)$. Then the finite-horizon problem is equivalent to a finite-state CMDP with a single linear resource constraint. By Altman (1999, Theorem 3.3), there exists an optimal randomized Markov policy for the constrained problem. Because there is only one linear resource constraint, Altman (1999, Theorem 3.4) further implies that one may choose an optimal randomized Markov policy that randomizes at at most one augmented state. Translating back to the original finite-horizon formulation, one may therefore choose an optimal Markov policy that randomizes at at most one state-stage pair.

For the deterministic claim, let $\lambda^\star$ be any dual-optimal multiplier for the constrained occupancy LP, and let $\pi^{\lambda^\star}$ denote the deterministic Markov policy induced by the penalized dynamic program at price $\lambda^\star$,

equivalently by the index rule in Theorem 1 Part (ii). Let $x^{\lambda^\star}$ be its occupancy measure. Because $\pi^{\lambda^\star}$ is optimal for the penalized problem with one-period reward

$$r^{\lambda^\star}(x,\ell) = G(x,\ell) - \lambda^\star \Psi(x,\ell),$$

its occupancy measure solves

$$x^{\lambda^\star} \in \arg\max_{x\in\mathcal{P}} \langle g - \lambda^\star \psi,\, x\rangle,$$

where $\mathcal{P} = \{x \geq 0 : Ax = b\}$ is the finite-horizon occupancy polytope.

Assume now that $x^{\lambda^\star}$ is feasible for the original budget constraint and satisfies complementary slackness:

$$\langle \psi, x^{\lambda^\star}\rangle \leq \bar{\Psi}, \qquad \lambda^\star\big(\langle \psi, x^{\lambda^\star}\rangle - \bar{\Psi}\big) = 0.$$

By dual optimality of $\lambda^\star$,

$$V^\star(\bar{\Psi}) = \min_{\lambda\geq 0}\Big\{h(\lambda) + \lambda\bar{\Psi}\Big\} = h(\lambda^\star) + \lambda^\star\bar{\Psi},$$

where $h(\lambda) := \max_{x\in\mathcal{P}}\langle g - \lambda\psi,\, x\rangle$. Since $x^{\lambda^\star}$ maximizes the penalized objective at $\lambda^\star$, $h(\lambda^\star) = \langle g - \lambda^\star\psi,\, x^{\lambda^\star}\rangle$. Therefore

$$\begin{aligned} V^\star(\bar{\Psi}) &= \langle g - \lambda^\star\psi,\, x^{\lambda^\star}\rangle + \lambda^\star\bar{\Psi} \\ &= \langle g,\, x^{\lambda^\star}\rangle - \lambda^\star\langle \psi, x^{\lambda^\star}\rangle + \lambda^\star\bar{\Psi} \\ &= \langle g,\, x^{\lambda^\star}\rangle, \end{aligned}$$

where the last step uses complementary slackness. Thus $x^{\lambda^\star}$ attains the constrained primal optimum. Hence the constrained optimum is attained by the deterministic Markov policy $\pi^{\lambda^\star}$. □

F.6. Proof of Theorem 1 (Index-threshold characterization of the ordered-menu policy)

Proof. We prove the two parts of the theorem under regularity conditions (R1)–(R3). The selector-cell representation then follows directly from the selector formula.

Part (i): indexability. Fix a stage t, a state $x = (s,c)$, and an adjacent pair $\ell, \ell+1$. By condition (R1), the map $\lambda \mapsto \Delta_t^\ell(x;\lambda)$ is weakly decreasing. By condition (R3), it is continuous. Therefore the upper level set

$$A_t^\ell(x) := \{\lambda \geq 0 : \Delta_t^\ell(x;\lambda) \geq 0\}$$

is either empty, or an interval of the form $[0,\bar{\lambda}]$ for some finite $\bar{\lambda} \geq 0$, or the half-line $[0,\infty)$.

Define $\theta_{t,\ell}(x) := \sup A_t^\ell(x)$, with the convention $\sup\varnothing = -\infty$. Then $\theta_{t,\ell}(x) \in [-\infty,+\infty]$.

If $A_t^\ell(x) = \varnothing$, then $\Delta_t^\ell(x;\lambda) < 0$ for every $\lambda \geq 0$, so $\theta_{t,\ell}(x) = -\infty$. If $A_t^\ell(x) = [0,\infty)$, then $\theta_{t,\ell}(x) = +\infty$, and $\Delta_t^\ell(x;\lambda) \geq 0$ for every $\lambda \geq 0$. Otherwise $A_t^\ell(x) = [0,\bar{\lambda}]$ with finite $\bar{\lambda}$, and then

$$\Delta_t^\ell(x;\lambda) \geq 0 \iff \lambda \leq \bar{\lambda} = \theta_{t,\ell}(x).$$

This proves the threshold property in the extended-real sense.

Part (ii): ordered indices and penalized-policy characterization. By condition (R2), for every $\lambda \geq 0$,

$$\Delta_t^\ell(x;\lambda) \geq \Delta_t^{\ell+1}(x;\lambda).$$

Hence

$$\{\lambda : \Delta_t^{\ell+1}(x;\lambda) \geq 0\} \subseteq \{\lambda : \Delta_t^{\ell}(x;\lambda) \geq 0\},$$

and taking suprema yields

$$\theta_{t,\ell}(x) \geq \theta_{t,\ell+1}(x).$$

Therefore

$$\theta_{t,0}(x) \geq \theta_{t,1}(x) \geq \cdots \geq \theta_{t,k-1}(x).$$

Now fix $\lambda \geq 0$. Since the menu is $\mathcal{L} = \{0,\ldots,k\}$, the adjacent difference along the menu is

$$Q_t^{\lambda}(x,m) - Q_t^{\lambda}(x,m-1) = \Delta_t^{m-1}(x;\lambda), \qquad m = 1,\ldots,k.$$

Because the adjacent increments are weakly decreasing in the menu index, once one adjacent increment becomes negative, all subsequent adjacent increments are also weakly negative. Thus the sequence $m \mapsto Q_t^{\lambda}(x,m)$ is weakly increasing up to some point and weakly decreasing thereafter.

Define

$$m^{\lambda}(x) := \max\Big\{m \in \{0,\ldots,k\} : \Delta_t^{j}(x;\lambda) \geq 0 \text{ for all } j = 0,\ldots,m-1\Big\},$$

with the convention that the condition is vacuous when $m = 0$. We claim that $m^{\lambda}(x)$ is the maximal maximizer of $Q_t^{\lambda}(x,\cdot)$.

First, if $m \leq m^{\lambda}(x)$, then

$$Q_t^{\lambda}(x,m) - Q_t^{\lambda}(x,0) = \sum_{j=0}^{m-1} \Delta_t^{j}(x;\lambda) \geq 0,$$

so all levels up to $m^{\lambda}(x)$ weakly improve upon silence.

Second, if $m > m^{\lambda}(x)$, then by definition $\Delta_t^{m^{\lambda}(x)}(x;\lambda) < 0$, and by condition (R2),

$$\Delta_t^{j}(x;\lambda) \leq 0 \qquad \text{for all } j \geq m^{\lambda}(x).$$

Therefore

$$Q_t^{\lambda}(x,m) - Q_t^{\lambda}(x,m^{\lambda}(x)) = \sum_{j=m^{\lambda}(x)}^{m-1} \Delta_t^{j}(x;\lambda) \leq 0.$$

Hence no level above $m^{\lambda}(x)$ can strictly improve upon it, so $m^{\lambda}(x)$ is the maximal maximizer.

Finally, by Part (i),

$$\Delta_t^{j}(x;\lambda) \geq 0 \iff \lambda \leq \theta_{t,j}(x).$$

Thus

$$m^{\lambda}(x) = \max\{m \in \{0,\ldots,k\} : \theta_{t,m-1}(x) \geq \lambda\},$$

with the convention $\theta_{t,-1}(x) := +\infty$. Therefore

$$\pi_t^{\lambda,\max}(x) = \max\{\ell \in \mathcal{L} : \theta_{t,\ell-1}(x) \geq \lambda\},$$

which is exactly (4). The implementation claim at a dual-optimal multiplier $\lambda^{\star}$ then follows from Proposition F.1.

Selector-cell representation. Fix (s,t,λ). For each $m=0,\dots,k$, define

$$\mathcal{Z}_m(s,t;\lambda) := \{c\in\mathcal{C} : \pi_t^{\lambda,\max}(s,c)=m\}.$$

By the selector formula above,

$$\mathcal{Z}_m(s,t;\lambda) = \{c\in\mathcal{C} : \theta_{t,m-1}(s,c)\ge\lambda>\theta_{t,m}(s,c)\}, \qquad m=0,\dots,k,$$

with the conventions $\theta_{t,-1}(s,c):=+\infty$ and $\theta_{t,k}(s,c):=-\infty$. These sets are pairwise disjoint and their union is all of $\mathcal{C}$, because the selector chooses exactly one menu level at each c. Hence $\pi_t^{\lambda,\max}(s,\cdot)$ partitions $\mathcal{C}$ into at most $|\mathcal{L}|$ nonempty selector cells. □

F.7. Proof of Proposition 2 (Value–cost frontier)

Proof. We prove the two parts in order.

Part (i): monotonicity, concavity, and piecewise affineness. Let $\mathcal{P}:=\{x\ge 0 : Ax=b\}$ denote the finite-horizon occupancy polytope, and write the constrained design problem as $V^\star(\bar\Psi)=\max\{\langle g,x\rangle : x\in\mathcal{P},\ \langle\psi,x\rangle\le\bar\Psi\}$. If $\bar\Psi'\ge\bar\Psi$, the feasible set at budget $\bar\Psi$ is contained in the feasible set at budget $\bar\Psi'$, so $V^\star(\bar\Psi')\ge V^\star(\bar\Psi)$ and $V^\star$ is weakly increasing.

By Proposition B.2,

$$V^\star(\bar\Psi)=\min_{\lambda\ge 0}\big\{h(\lambda)+\lambda\bar\Psi\big\}, \qquad h(\lambda):=\max_{x\in\mathcal{P}}\langle g-\lambda\psi,x\rangle.$$

For each fixed λ, the map $\bar\Psi\mapsto h(\lambda)+\lambda\bar\Psi$ is affine. Therefore $V^\star$ is the pointwise infimum of affine functions of $\bar\Psi$, and is hence concave. Because the underlying constrained problem is a finite-dimensional parametric LP with a single right-hand-side parameter, $V^\star$ is piecewise affine.

Part (ii): supergradient interpretation. Fix a budget $\bar\Psi$, and let $\lambda^\star(\bar\Psi)$ be any dual-optimal multiplier: $V^\star(\bar\Psi)=h(\lambda^\star(\bar\Psi))+\lambda^\star(\bar\Psi)\,\bar\Psi$. For any other budget $\bar\Psi'$, dual feasibility of the same multiplier yields

$$V^\star(\bar\Psi')\le h(\lambda^\star(\bar\Psi))+\lambda^\star(\bar\Psi)\,\bar\Psi' = V^\star(\bar\Psi)+\lambda^\star(\bar\Psi)(\bar\Psi'-\bar\Psi).$$

Thus $\lambda^\star(\bar\Psi)$ is a supergradient of the concave function $V^\star$ at $\bar\Psi$. If $V^\star$ is differentiable at $\bar\Psi$, the supergradient is unique, so $dV^\star/d\bar\Psi=\lambda^\star$. □

F.8. Budget implementation through a fixed index system

Proposition F.2 **(Budget implementation through a fixed index system)**. *Under the theorem-local regularity conditions (R1)–(R3) of Theorem 1:*

(i) the dynamic signaling indices $\theta_{t,\ell}(s,c)$ are invariant to $\bar\Psi$;

(ii) different budgets operate by selecting different dual-optimal shadow prices $\lambda^\star(\bar\Psi)$ applied to the same index system through the selector (4)*;*

(iii) between kink budgets of the frontier $V^\star$, the dual-optimal shadow price is constant, and under nondegeneracy of the penalized dynamic program the induced deterministic penalized policy is constant on the interval.

Proof. We prove the three parts in order, under the theorem-local regularity conditions (R1)–(R3) of Theorem 1.

Part (i): indices are invariant to the budget. The dynamic signaling indices $\theta_{t,\ell}(s,c) = \sup\{\lambda \geq 0 : \Delta_t^\ell(s,c;\lambda) \geq 0\}$ are defined entirely from the family of penalized DPs parameterized by λ. They depend only on the primitives of that penalized family and are therefore invariant to $\bar{\Psi}$.

Part (ii): budgets act through the shadow price on the same index system. Let $\Lambda^\star(\bar{\Psi}) := \arg\min_{\lambda \geq 0}\{h(\lambda) + \lambda\bar{\Psi}\}$ denote the set of dual-optimal multipliers at budget $\bar{\Psi}$, where $h(\lambda) := \max_{x \in \mathcal{P}} \langle g - \lambda\psi, x\rangle$ as in the proof of Proposition 2. By Proposition B.2, any policy optimal for the constrained problem is also optimal for the penalized problem at any $\lambda^\star \in \Lambda^\star(\bar{\Psi})$. Conversely, by Proposition F.1, any penalized-optimal policy at such a multiplier that is primal feasible and satisfies complementary slackness is constrained-optimal. In particular, for any $\lambda^\star \in \Lambda^\star(\bar{\Psi})$, the maximal penalized selector takes the index-threshold form

$$\pi_t^{\lambda^\star,\max}(s,c) = \max\{\ell \in \mathcal{L} : \theta_{t,\ell-1}(s,c) \geq \lambda^\star\}$$

from Theorem 1 Part (ii). Thus budgets act through the shadow price $\lambda^\star$, while the index system itself remains fixed.

Part (iii): constant shadow price and constant policy between kink budgets. By Proposition 2 Part (i), $V^\star(\bar{\Psi})$ is piecewise affine with finitely many kink budgets $\bar{\Psi}_1 < \cdots < \bar{\Psi}_m$. Fix an open interval between consecutive kinks. Because $V^\star$ is affine on that interval, its superdifferential is a singleton and is constant there. Thus there exists a constant multiplier λ_j such that $\partial V^\star(\bar{\Psi}) = \{\lambda_j\}$ for every $\bar{\Psi}$ in that interval; equivalently, the same dual-optimal shadow price applies throughout the interval.

By Part (i) of this proposition, the adjacent-upgrade indices $\theta_{t,\ell}(s,c)$ do not depend on the budget level $\bar{\Psi}$; only the dual threshold changes with the budget. Therefore, throughout the present interval, the associated penalized-optimal threshold family is generated by thresholding the same index system at the same constant shadow price λ_j.

Now suppose additionally that the penalized dynamic program at λ_j is nondegenerate, in the sense that for every state-stage pair (x,t) there is a unique maximizing signal level in $\arg\max_{\ell \in \mathcal{L}} Q_t^{\lambda_j}(x,\ell)$. Then the penalized-optimal policy is deterministic and uniquely determined by λ_j, and hence is constant throughout the interval. □

F.9. Kink-interval implementation and kink behavior of the constrained optimum

This subsection records the full interval-wise implementation statement underlying Proposition F.2, including the complementary-slackness condition under which the constant deterministic penalized policy attains the constrained optimum on an interior interval, and the edge behavior of $V^\star$ at a kink budget.

Within any open interval between consecutive kink budgets on which the penalized dynamic program at the associated constant multiplier λ_j is nondegenerate, the induced deterministic penalized policy is constant throughout the interval by Proposition F.2 Part (iii). Whenever this deterministic penalized policy is also primal feasible for the budget constraint and satisfies complementary slackness, Proposition F.1 implies that

it attains the constrained optimum throughout the same interval. This is the sense in which an interior interval of the budget path supports a single deterministic Markov implementation of the constrained optimum.

At a kink budget, by contrast, $V^\star$ is not differentiable, so $\partial V^\star(\bar{\Psi})$ is not singleton. Hence more than one dual-optimal multiplier may be active at the kink. Equivalently, the shadow price need not be unique at the kink, and different penalized-optimal selectors may be associated with different multipliers in the superdifferential. This is exactly the sense in which the budget path can change regime at a kink.

F.10. Frontier dominance of richer menus

PROPOSITION F.3 **(Frontier dominance of richer menus)**. *Let $\mathcal{L}_{\text{coarse}} \subset \mathcal{L}_{\text{rich}}$, and let $V^\star_{\mathcal{L}}(\bar{\Psi})$ denote the RD frontier under menu $\mathcal{L}$. Then*

$$V^\star_{\mathcal{L}_{\text{rich}}}(\bar{\Psi}) \geq V^\star_{\mathcal{L}_{\text{coarse}}}(\bar{\Psi}) \qquad \textit{for every } \bar{\Psi}.$$

Proof. Let $\mathcal{L}_{\text{coarse}} \subset \mathcal{L}_{\text{rich}}$. Every policy feasible under $\mathcal{L}_{\text{coarse}}$ is also feasible under $\mathcal{L}_{\text{rich}}$, since the rich menu simply enlarges the set of admissible signal choices while leaving the state dynamics, objective, and cost constraint otherwise unchanged. Hence the feasible policy set under $\mathcal{L}_{\text{coarse}}$ is a subset of the feasible policy set under $\mathcal{L}_{\text{rich}}$. Taking suprema over the two feasible sets yields

$$V^\star_{\mathcal{L}_{\text{rich}}}(\bar{\Psi}) \geq V^\star_{\mathcal{L}_{\text{coarse}}}(\bar{\Psi}) \qquad \text{for every } \bar{\Psi}.$$

This proves frontier dominance. □

F.11. Primitive conditions for regime direction

The following two propositions provide explicit sufficient conditions for each canonical regime direction referenced in the main text, making the bidirectional characterization symmetric in primitives. Throughout, for any two signal levels $\ell' > \ell$, we use the IDE channel notation $\mathcal{I}(s, c, \ell', \ell)$, $\mathcal{D}(s, c, \ell', \ell)$, $\mathcal{E}(s, c, \ell', \ell)$ from Lemma 1.

PROPOSITION F.4 **(Primitive conditions for the conservation regime)**. *At fixed stage t and physical state s, suppose:*

(a) $\mathcal{I}(s, c, \ell', \ell)$ is weakly decreasing in c for all $\ell' > \ell$: immediate gains from stronger signals are larger at low efficacy (more room for improvement when already degraded);

(b) $\mathcal{E}(s, c, \ell', \ell)$ is weakly decreasing in c for all $\ell' > \ell$: the drift penalty is at least as severe at high efficacy as at low efficacy (more to lose from a high baseline);

(c) $\mathcal{D}(s, c, \ell', \ell)$ is weakly decreasing in c for all $\ell' > \ell$: displacement is less valuable at high efficacy (trajectory already near-optimal).

Then the dynamic net-benefit difference

$$\big[\bar{R}(s, c, \ell') + \mathbb{E}[V^\lambda_{t+1}(X_{t+1}) \mid s, c, \ell']\big] - \big[\bar{R}(s, c, \ell) + \mathbb{E}[V^\lambda_{t+1}(X_{t+1}) \mid s, c, \ell]\big]$$

is weakly decreasing in c for every $\ell' > \ell$. Equivalently, the pre-penalty dynamic objective has decreasing differences in (c, ℓ). Thus, absent the explicit downstream-cost penalty, higher-efficacy states weakly favor less prescriptive signals.

Proof. Fix $c^H \succ c^L$ and $\ell' > \ell$. By Lemma 1 with continuation value V^λ_{t+1}, the net-benefit difference equals $\mathcal{I}+\mathcal{D}+\mathcal{E}$. The cross-difference between c^H and c^L is therefore $[\mathcal{I}(c^H)-\mathcal{I}(c^L)]+[\mathcal{D}(c^H)-\mathcal{D}(c^L)]+[\mathcal{E}(c^H)-\mathcal{E}(c^L)]$. Under conditions (a)–(c), each bracket is weakly nonpositive, giving decreasing differences in (c,ℓ) for the pre-penalty dynamic objective. □

Proposition F.5 **(Primitive conditions for the exploitation regime)**. *At fixed stage t and physical state s, suppose the reversal of (a)–(c) in the preceding proposition:*

(a′) $\mathcal{I}(s,c,\ell',\ell)$ is weakly increasing in c for all $\ell' > \ell$: immediate gains from stronger signals are larger at high efficacy (signals are more effective when the system is well calibrated);

(b′) $\mathcal{E}(s,c,\ell',\ell)$ is weakly increasing in c for all $\ell' > \ell$: the drift penalty is weakly smaller at high efficacy (degradation is less consequential from a high baseline);

(c′) $\mathcal{D}(s,c,\ell',\ell)$ is weakly increasing in c for all $\ell' > \ell$: displacement is more valuable at high efficacy (well-calibrated signals push actions in the right direction).

Then the dynamic net-benefit difference is weakly increasing in c for every $\ell' > \ell$. Equivalently, the pre-penalty dynamic objective has increasing differences in (c,ℓ). Thus, absent the explicit downstream-cost penalty, higher-efficacy states weakly favor more prescriptive signals.

Proof. Fix $c^H \succ c^L$ and $\ell' > \ell$. By Lemma 1 with continuation value V^λ_{t+1}, the net-benefit difference equals $\mathcal{I}+\mathcal{D}+\mathcal{E}$. The cross-difference between c^H and c^L is therefore

$$[\mathcal{I}(c^H)-\mathcal{I}(c^L)]+[\mathcal{D}(c^H)-\mathcal{D}(c^L)]+[\mathcal{E}(c^H)-\mathcal{E}(c^L)].$$

Under conditions (a′)–(c′), each bracket is weakly nonnegative. Hence the full cross-difference is weakly nonnegative, so the dynamic net-benefit difference is weakly increasing in c for every $\ell' > \ell$. Equivalently, the pre-penalty dynamic objective has increasing differences in (c,ℓ). Thus, absent the explicit downstream-cost penalty, higher-efficacy states weakly favor more prescriptive signals. □

The two propositions determine the direction of the pre-penalty monotonicity in efficacy. This yields the same zonal-ordering direction for the penalized selector only under additional conditions ensuring that the explicit downstream-cost term does not overturn that monotonicity. Under decreasing differences in the pre-penalty objective, higher-efficacy states weakly favor less prescriptive signals; under increasing differences, they weakly favor more prescriptive signals. Appendix F.6 gives a stronger lattice-monotonicity formulation under additional single-crossing structure.

F.12. Signal-invariant harm and local estimable certificate

Remark F.1 (When the efficacy-drift channel vanishes). If $K(\cdot \mid c,s,a,\ell') = K(\cdot \mid c,s,a,\ell)$ for all a,c,s,ℓ,ℓ', then $\mathcal{E} \equiv 0$. Thus signal-independent efficacy is sufficient for eliminating the efficacy-drift channel.

Corollary F.1 **(Welfare-reducing signaling at signal-invariant states)**. *If there exists an action $a^\star \in \mathcal{A}$ such that $\mu(a^\star \mid s,c,\ell) = 1$ for every signal level ℓ, then $\Delta\mu_a = 0$ for all a, so $\mathcal{I} = \mathcal{D} = 0$ and $Q_t(s,c,\ell') - Q_t(s,c,\ell) = \mathcal{E}$. Welfare-reducing signaling occurs at a signal-invariant state if and only if $\mathcal{E} < 0$.*

Proof. Fix (s,c,t) and two signal levels $\ell' \geq \ell$. By assumption, $\mu(\cdot \mid s,c,\ell)$ places full mass on the same action for every signal level at (s,c). Hence $\Delta\mu_a := \mu(a \mid s,c,\ell') - \mu(a \mid s,c,\ell) = 0$ for all $a \in \mathcal{A}$. Applying Lemma 1, $\mathcal{I} = \sum_a \Delta\mu_a R(s,a) = 0$ and $\mathcal{D} = \sum_a \Delta\mu_a w_t(s,c,a,\ell) = 0$. Therefore $Q_t(s,c,\ell') - Q_t(s,c,\ell) = \mathcal{E}$, and welfare-reducing signaling occurs if and only if $\mathcal{E} < 0$. □

F.13. Proof of Proposition 3 (Safe harbor)

Proof. Fix a state-stage pair (s,c,t) and an ordered pair of signal levels $\ell' \geq \ell$ such that locally beneficial signaling holds for this comparison, i.e., $\mathcal{I} + \mathcal{D} \geq 0$.

Under the proposition's hypothesis, efficacy is signal-independent conditional on the realized action. Thus, for every action a,

$$K(\cdot \mid c,s,a,\ell') = K(\cdot \mid c,s,a,\ell).$$

By the definition of the action-contingent continuation value in Lemma 1, we have

$$\begin{aligned} w_t(s,c,a,\ell') &= \sum_{s',c'} F(s' \mid s,a) K(c' \mid c,s,a,\ell') V^\star_{t+1}(s',c') \\ &= \sum_{s',c'} F(s' \mid s,a) K(c' \mid c,s,a,\ell) V^\star_{t+1}(s',c') \\ &= w_t(s,c,a,\ell). \end{aligned}$$

Therefore, the efficacy-drift term in Lemma 1 equals zero:

$$\begin{aligned} \mathcal{E} &= \sum_a \mu(a \mid s,c,\ell') \Big(w_t(s,c,a,\ell') - w_t(s,c,a,\ell) \Big) \\ &= 0. \end{aligned}$$

Applying Lemma 1 then gives

$$Q_t(s,c,\ell') - Q_t(s,c,\ell) = \mathcal{I} + \mathcal{D} \geq 0,$$

where the inequality is exactly the local beneficial-signaling condition for this state-stage pair and ordered signal pair. Hence harmful signaling is impossible for this comparison. Because the argument is pointwise, the same conclusion holds at any state-stage pair and ordered signal pair satisfying the stated condition. □

Remark. Every direct-control MDP satisfies signal-independent efficacy as a degenerate special case. The contrapositive of the theorem implies that harmful signaling under locally beneficial signaling requires $\mathcal{E} \neq 0$.

F.14. Constructive Example: Harmful Signaling at a Signal-Invariant State

We construct a two-period instance ($T = 2$) with probabilistic efficacy dynamics and an immediate-gain vs. future-cost trade-off. Let $\mathcal{S} = \{s_1, s_2\}$, $\mathcal{A} = \{a_1, a_2\}$, $\mathcal{C} = \{\mathsf{high}, \mathsf{low}\}$, $\mathcal{L} = \{0,1\}$. Physical transitions are $F(s_1 \mid s_1, a) = 0.7$, $F(s_2 \mid s_1, a) = 0.3$ for all a (action-independent). Rewards: $R(s_1,a_1) = 1$, $R(s_1,a_2) = 0.2$, $R(s_2,a_1) = 0.8$, $R(s_2,a_2) = 0$.

At (s_1, high), the operator already takes the optimal action regardless of the signal: $\mu(a_1 \mid s_1, \mathsf{high}, \ell) = 1$ for $\ell \in \{0,1\}$. This is the signal-invariant condition. At (s_2, high), the signal improves the action distribution: $\mu(a_1 \mid s_2, \mathsf{high}, 0) = 0.5$, $\mu(a_1 \mid s_2, \mathsf{high}, 1) = 0.9$. Under low efficacy, the operator is less responsive: $\mu(a_1 \mid s, \mathsf{low}, 0) = 0.5$, $\mu(a_1 \mid s, \mathsf{low}, 1) = 0.6$ for both states.

We also have $K(\mathsf{high} \mid \mathsf{high}, s, a, 0) = 0.95, K(\mathsf{high} \mid \mathsf{high}, s, a, 1) = 0.7,$ for all s, a. Signaling erodes efficacy since each prescriptive signal reduces the probability of remaining in high from 0.95 to 0.70, regardless of the action taken. This is the action-conditioned residual dependence: $K(\cdot \mid \mathsf{high}, s, a, 1) \neq K(\cdot \mid \mathsf{high}, s, a, 0)$ even at fixed a. Under low efficacy: $K(\mathsf{high} \mid \mathsf{low}, s, a, \ell) = 0.3$ for all ℓ (partial recovery, signal-independent).

We now show that harmful signaling exists at a signal-invariant state. At $(s_1, \mathsf{high}, t{=}1)$, $\mu(\cdot \mid s_1, \mathsf{high}, \ell)$ places full mass on a_1 for all ℓ, so $\Delta\mu_a = 0$ for all a. Therefore $\mathcal{I} = \mathcal{D} = 0$, and the value comparison reduces to $\mathcal{E}$. The continuation-value difference between $\ell = 0$ and $\ell = 1$ arises entirely through K:

$$\mathcal{E} \;=\; \mu(a_1 \mid s_1, \mathsf{high}, 1)\big[w(s_1, \mathsf{high}, a_1, 1) - w(s_1, \mathsf{high}, a_1, 0)\big],$$

where $w(s_1, \mathsf{high}, a_1, \ell) = \sum_{s', c'} F(s' \mid s_1, a_1)\, K(c' \mid \mathsf{high}, s_1, a_1, \ell)\, V_2^\star(s', c')$.

At the terminal stage $t{=}2$, there is no continuation, so $V_2^\star(s, c) = \max_\ell \sum_a \mu(a \mid s, c, \ell)\, R(s, a)$. Evaluating: $V_2^\star(s_1, \mathsf{high}) = 1.0$ (full mass on a_1), $V_2^\star(s_1, \mathsf{low}) = \max\{0.6, 0.68\} = 0.68$ (best at $\ell{=}1$), $V_2^\star(s_2, \mathsf{high}) = \max\{0.40, 0.72\} = 0.72$ (best at $\ell{=}1$), $V_2^\star(s_2, \mathsf{low}) = \max\{0.40, 0.48\} = 0.48$. Hence $V_2^\star(s', \mathsf{high}) > V_2^\star(s', \mathsf{low})$ for both physical states. Because $K(\mathsf{high} \mid \mathsf{high}, s_1, a_1, 1) = 0.70 < 0.95 = K(\mathsf{high} \mid \mathsf{high}, s_1, a_1, 0)$, the continuation value under $\ell{=}1$ is strictly lower than under $\ell{=}0$. Hence $\mathcal{E} < 0$ and $Q_1(s_1, \mathsf{high}, 1) < Q_1(s_1, \mathsf{high}, 0)$: the signal is harmful despite the operator already taking the optimal action.

At (s_2, high), by contrast, the signal improves the action distribution ($\Delta\mu_{a_1} = 0.4$), so unlike at (s_1, high) it delivers a genuine current decision benefit. This construction therefore isolates the mechanism behind Corollary F.1: signaling can be strictly harmful at a signal-invariant state because the current action is unchanged while the residual efficacy cost persists. In such states, $\mathcal{I} = \mathcal{D} = 0$, so the sign of the comparison is determined entirely by $\mathcal{E}$. □

F.15. Proof of Proposition 4 (Near-tie screen)

Proof. We first derive the bound in (5). We then verify the stated sufficient condition for local optimality of the myopic choice. Fix a stage t and a state x, and let $\bar{\ell} := \pi_t^{\mathrm{myo}}(x)$ under the fixed tie-breaking rule above. By construction, $\bar{\ell} \in \arg\max_{\ell \in \mathcal{L}} M_\ell(x)$, and so for every $\ell \in \mathcal{L}$,

$$\Gamma_t(x, \ell) := M_{\bar{\ell}}(x) - M_\ell(x) \geq 0.$$

We now compare the exact action value of an arbitrary level ℓ to that of the myopically selected level $\bar{\ell}$. Write

$$Q_t^\star(x, \ell) = M_\ell(x) + C_\ell(x),$$

where

$$C_\ell(x) := \mathbb{E}\big[V_{t+1}^\star(X_{t+1}) \mid X_t = x, \ell_t = \ell\big] = \int V_{t+1}^\star(x')\, P_\ell(dx' \mid x).$$

Then

$$\begin{aligned} Q_t^\star(x, \ell) - Q_t^\star(x, \bar{\ell}) &= \Big(M_\ell(x) - M_{\bar{\ell}}(x)\Big) + \Big(C_\ell(x) - C_{\bar{\ell}}(x)\Big) \\ &= -\Gamma_t(x, \ell) + \Big(C_\ell(x) - C_{\bar{\ell}}(x)\Big). \end{aligned} \tag{17}$$

We next bound the continuation-value difference. Define

$$m_{t+1} := \inf_{x'} V^\star_{t+1}(x'), \qquad M_{t+1} := \sup_{x'} V^\star_{t+1}(x'), \qquad \Phi_{t+1} := M_{t+1} - m_{t+1}.$$

Then the shifted function

$$f(x') := V^\star_{t+1}(x') - m_{t+1}$$

satisfies

$$0 \le f(x') \le \Phi_{t+1}.$$

Therefore,

$$\begin{aligned} C_\ell(x) - C_{\bar\ell}(x) &= \int V^\star_{t+1}(x') \big(P_\ell - P_{\bar\ell}\big)(dx' \mid x) \\ &= \int f(x') \big(P_\ell - P_{\bar\ell}\big)(dx' \mid x). \end{aligned} \tag{18}$$

Now apply the standard total-variation inequality for bounded functions:

$$\left| \int f \, d(P_\ell - P_{\bar\ell}) \right| \le \Phi_{t+1} \, \mathrm{TV}\big(P_\ell(\cdot \mid x), P_{\bar\ell}(\cdot \mid x)\big).$$

Hence

$$C_\ell(x) - C_{\bar\ell}(x) \le \Phi_{t+1} \, \Delta^{\mathrm{TV}}_t(x, \ell). \tag{19}$$

Substituting (19) into (17) gives, for every $\ell \in \mathcal{L}$,

$$Q^\star_t(x, \ell) - Q^\star_t(x, \bar\ell) \le \Phi_{t+1} \Delta^{\mathrm{TV}}_t(x, \ell) - \Gamma_t(x, \ell).$$

Now maximize over $\ell \in \mathcal{L}$. By definition,

$$\eta^{\mathrm{myo}}_t(x) = V^\star_t(x) - Q^\star_t(x, \bar\ell) = \max_{\ell \in \mathcal{L}} \Big(Q^\star_t(x, \ell) - Q^\star_t(x, \bar\ell) \Big).$$

Therefore,

$$\eta^{\mathrm{myo}}_t(x) \le \max_{\ell \in \mathcal{L}} \Big(\Phi_{t+1} \Delta^{\mathrm{TV}}_t(x, \ell) - \Gamma_t(x, \ell) \Big).$$

Since the choice $\ell = \bar\ell$ is feasible and gives value zero inside the maximum, we may write the same bound as

$$\eta^{\mathrm{myo}}_t(x) \le \max_{\ell \in \mathcal{L}} \Big[\Phi_{t+1} \Delta^{\mathrm{TV}}_t(x, \ell) - \Gamma_t(x, \ell) \Big]_+,$$

which is exactly (5). For the second claim, suppose

$$\Gamma_t(x, \ell) \ge \Phi_{t+1} \Delta^{\mathrm{TV}}_t(x, \ell) \qquad \text{for all } \ell \in \mathcal{L}.$$

Then every bracketed term on the right-hand side is nonpositive, and hence

$$\eta^{\mathrm{myo}}_t(x) = 0.$$

Therefore the myopically selected level $\bar\ell$ is optimal at (x, t). □

Auxiliary consequences. The next two corollaries record immediate consequences used later. The conservative global certificate and the T^2 tightness construction are moved to the online supplement.

Corollary F.2 **(Global accumulation bound)**. *Summing the local screen across periods yields*

$$\|V_1^\star - V_1^{\pi^{\mathrm{myo}}}\|_\infty \le \sum_{t=1}^{T} \|\max_\ell \big[\Phi_{t+1}\Delta_t^{\mathrm{TV}}(\cdot,\ell) - \Gamma_t(\cdot,\ell)\big]_+\|_\infty. \tag{20}$$

If $\Gamma_t(x,\ell) \ge \Phi_{t+1}\Delta_t^{\mathrm{TV}}(x,\ell)$ *for all* t, x, ℓ*, then* π^{myo} *is globally optimal.*

Proof of Corollary F.2. We combine the local bound of Proposition 4 with the standard Bellman-recursion argument.

Define

$$R_t(x) := V_t^\star(x) - V_t^{\pi^{\mathrm{myo}}}(x).$$

By the Bellman recursion under policy π^{myo},

$$R_t(x) = \delta_t^{\mathrm{myo}}(x) + \mathbb{E}^{\pi^{\mathrm{myo}}}\left[R_{t+1}(X_{t+1}) \mid X_t = x\right], \tag{21}$$

where

$$\delta_t^{\mathrm{myo}}(x) = V_t^\star(x) - Q_t^\star(x, \pi_t^{\mathrm{myo}}(x)) = \eta_t^{\mathrm{myo}}(x).$$

Taking suprema over x in (21) gives

$$\|R_t\|_\infty \le \|\eta_t^{\mathrm{myo}}\|_\infty + \|R_{t+1}\|_\infty.$$

Iterating from $t = 1$ to T and using $R_{T+1} \equiv 0$, we obtain

$$\|V_1^\star - V_1^{\pi^{\mathrm{myo}}}\|_\infty = \|R_1\|_\infty \le \sum_{t=1}^{T} \|\eta_t^{\mathrm{myo}}\|_\infty. \tag{22}$$

Now apply Proposition 4 pointwise:

$$\eta_t^{\mathrm{myo}}(x) \le \max_{\ell\in\mathcal{L}} \Big[\Phi_{t+1}\Delta_t^{\mathrm{TV}}(x,\ell) - \Gamma_t(x,\ell)\Big]_+.$$

Taking suprema over x and substituting into (22) yields

$$\|V_1^\star - V_1^{\pi^{\mathrm{myo}}}\|_\infty \le \sum_{t=1}^{T} \left\| \max_{\ell\in\mathcal{L}} \Big[\Phi_{t+1}\Delta_t^{\mathrm{TV}}(\cdot,\ell) - \Gamma_t(\cdot,\ell)\Big]_+ \right\|_\infty,$$

which is exactly (20).

For the second claim, suppose $\Gamma_t(x,\ell) \ge \Phi_{t+1}\Delta_t^{\mathrm{TV}}(x,\ell)$, for all t, x, ℓ. Then the term inside the norm is identically zero for every t, so the right-hand side above is zero, and we have

$$J_\nu(\pi^{\mathrm{myo}}) = J_\nu(\pi^\star),$$

so the myopic benchmark attains the optimal value and is globally optimal. □

Corollary F.3 **(Local estimable certificate)**. *Fix* t*,* x*, and let* $\ell_0 = \ell_t^{\mathrm{myo}}(x)$*. For any* $\ell \in \mathcal{L}$*, define* $\Delta_t^\mu(x,\ell) := \mathrm{TV}\big(\mu(\cdot \mid x,\ell),\, \mu(\cdot \mid x,\ell_0)\big)$ *and* $\bar{\mathcal{E}}_t(x,\ell) := \sum_a \mu(a \mid x,\ell_0)\, \mathrm{TV}\big(K(\cdot \mid c,s,a,\ell),\, K(\cdot \mid c,s,a,\ell_0)\big)$*. Then* $\Delta_t^{\mathrm{TV}}(x,\ell) \le \tilde{\rho}(F,K)\, \Delta_t^\mu(x,\ell) + \bar{\mathcal{E}}_t(x,\ell)$*.*

Proof of Corollary F.3. Fix a stage t, a state $x=(s,c)$, and let $\ell_0 := \pi_t^{\mathrm{myo}}(x)$. We will bound the transition distortion

$$\Delta_t^{\mathrm{TV}}(x,\ell) = \mathrm{TV}\big(P_\ell(\cdot \mid x), P_{\ell_0}(\cdot \mid x)\big)$$

in terms of primitive quantities. We write the factored kernel as

$$P_\ell(\cdot \mid x) = \sum_a \mu(a \mid x,\ell)\, H_a^\ell(\cdot), \quad \text{where} \quad H_a^\ell(\cdot) := F(\cdot \mid s,a)\, K(\cdot \mid c,s,a,\ell).$$

Similarly, we have

$$P_{\ell_0}(\cdot \mid x) = \sum_a \mu(a \mid x,\ell_0)\, H_a^{\ell_0}(\cdot).$$

Adding and subtracting the intermediate term $\sum_a \mu(a \mid x,\ell_0)\, H_a^\ell(\cdot)$, we have

$$\begin{aligned} P_\ell(\cdot \mid x) - P_{\ell_0}(\cdot \mid x) = &\sum_a \Big(\mu(a \mid x,\ell) - \mu(a \mid x,\ell_0)\Big) H_a^\ell(\cdot) \\ &+ \sum_a \mu(a \mid x,\ell_0)\Big(H_a^\ell(\cdot) - H_a^{\ell_0}(\cdot)\Big). \end{aligned} \tag{23}$$

Taking total variation and applying the triangle inequality, we have

$$\Delta_t^{\mathrm{TV}}(x,\ell) \le \frac{1}{2}\left\| \sum_a \big(\mu(a \mid x,\ell) - \mu(a \mid x,\ell_0)\big) H_a^\ell(\cdot) \right\|_1 + \frac{1}{2}\left\| \sum_a \mu(a \mid x,\ell_0)\big(H_a^\ell(\cdot) - H_a^{\ell_0}(\cdot)\big) \right\|_1. \tag{24}$$

We bound the two pieces separately. For the response-shift term, let

$$\Delta\mu_a := \mu(a \mid x,\ell) - \mu(a \mid x,\ell_0), \qquad \tau := \mathrm{TV}\big(\mu(\cdot \mid x,\ell), \mu(\cdot \mid x,\ell_0)\big).$$

Since $\sum_a \Delta\mu_a = 0$, there exist probability vectors p^+, p^- such that

$$\Delta\mu_a = \tau\big(p_a^+ - p_a^-\big) \qquad \text{for all } a \in \mathcal{A}.$$

Define $M^+ := \sum_a p_a^+ H_a^\ell(\cdot), M^- := \sum_a p_a^- H_a^\ell(\cdot)$. Then M^+ and M^- are probability measures, and

$$\sum_a \Delta\mu_a H_a^\ell(\cdot) = \tau\big(M^+ - M^-\big).$$

Therefore

$$\tfrac{1}{2}\left\| \sum_a \Delta\mu_a H_a^\ell(\cdot) \right\|_1 = \tau\, \mathrm{TV}(M^+, M^-).$$

We now use convexity of total variation in each argument:

$$\begin{aligned} \mathrm{TV}(M^+, M^-) &= \mathrm{TV}\left(\sum_a p_a^+ H_a^\ell, \sum_{a'} p_{a'}^- H_{a'}^\ell \right) \\ &\le \sum_{a,a'} p_a^+ p_{a'}^- \,\mathrm{TV}\big(H_a^\ell, H_{a'}^\ell\big) \\ &\le \tilde{\rho}(F,K), \end{aligned}$$

because $\tilde{\rho}(F,K)$ is the maximum of $\mathrm{TV}(H_a^\ell, H_{a'}^\ell)$ over all (s,c,ℓ,a,a'). Hence

$$\begin{aligned} \tfrac{1}{2}\left\| \sum_a \big(\mu(a \mid x,\ell) - \mu(a \mid x,\ell_0)\big) H_a^\ell(\cdot) \right\|_1 &\le \tilde{\rho}(F,K)\, \mathrm{TV}\big(\mu(\cdot \mid x,\ell), \mu(\cdot \mid x,\ell_0)\big) \\ &= \tilde{\rho}(F,K)\, \Delta_t^\mu(x,\ell). \end{aligned} \tag{25}$$

For the efficacy-drift term, note that $H_a^\ell(\cdot) = F(\cdot \mid s,a) \otimes K(\cdot \mid c,s,a,\ell), H_a^{\ell_0}(\cdot) = F(\cdot \mid s,a) \otimes K(\cdot \mid c,s,a,\ell_0)$. Because the physical kernel is the same in both products,

$$\mathrm{TV}\big(H_a^\ell(\cdot), H_a^{\ell_0}(\cdot)\big) = \mathrm{TV}\big(K(\cdot \mid c,s,a,\ell), K(\cdot \mid c,s,a,\ell_0)\big).$$

Therefore,

$$\begin{aligned}\tfrac{1}{2}\left\|\sum_a \mu(a \mid x,\ell_0)\big(H_a^\ell(\cdot) - H_a^{\ell_0}(\cdot)\big)\right\|_1 &\le \sum_a \mu(a \mid x,\ell_0)\,\mathrm{TV}\big(H_a^\ell(\cdot), H_a^{\ell_0}(\cdot)\big) \\ &= \sum_a \mu(a \mid x,\ell_0)\,\mathrm{TV}\big(K(\cdot \mid c,s,a,\ell), K(\cdot \mid c,s,a,\ell_0)\big) \\ &= \bar{\mathcal{E}}_t(x,\ell). \end{aligned} \tag{26}$$

Combining (24), (25), and (26), we get

$$\Delta_t^{\mathrm{TV}}(x,\ell) \le \tilde{\rho}(F,K)\,\Delta_t^\mu(x,\ell) + \bar{\mathcal{E}}_t(x,\ell),$$

and from Proposition 4

$$\eta_t^{\mathrm{myo}}(x) \le \max_{\ell \in \mathcal{L}} \Big[\Phi_{t+1}\Delta_t^{\mathrm{TV}}(x,\ell) - \Gamma_t(x,\ell)\Big]_+.$$

Substituting the bound above, we have

$$\eta_t^{\mathrm{myo}}(x) \le \max_{\ell \in \mathcal{L}} \Big[\Phi_{t+1}\big(\tilde{\rho}(F,K)\Delta_t^\mu(x,\ell) + \bar{\mathcal{E}}_t(x,\ell)\big) - \Gamma_t(x,\ell)\Big]_+.$$

This is the local estimable certificate. To obtain the stage-summed bound under the standing assumption $|R(s,a)| \le R_{\max}$, note that for every stage $t+1$ and state x',

$$-(T-t)R_{\max} \le V_{t+1}^\star(x') \le (T-t)R_{\max}.$$

Therefore,

$$\Phi_{t+1} = \sup_{x'} V_{t+1}^\star(x') - \inf_{x'} V_{t+1}^\star(x') \le 2(T-t)R_{\max}.$$

Substituting this bound into the local estimable certificate gives the stated trajectory-level inequality. □

We will now show the global refinement to worst-case primitives, but before doing so, we state and prove the following useful lemma.

Lemma F.1 **(Conservative myopic regret bound)**. *Let*

$$\Delta_t^{\mathrm{TV}}(x) := \max_{\ell \in \mathcal{L}} \Delta_t^{\mathrm{TV}}(x,\ell), \qquad \bar{\Delta}^{\mathrm{TV}} := \max_{t,x} \Delta_t^{\mathrm{TV}}(x), \qquad \bar{\Delta}_t^{\mathrm{TV}} := \sup_x \Delta_t^{\mathrm{TV}}(x).$$

Then the per-state myopic loss satisfies $0 \le \eta_t^{\mathrm{myo}}(x) \le \Phi_{t+1}\,\Delta_t^{\mathrm{TV}}(x)$*, and the global finite-horizon myopic regret is bounded by*

$$\|V_1^\star - V_1^{\pi^{\mathrm{myo}}}\|_\infty \le \sum_{t=1}^{T-1} \Phi_{t+1}\,\bar{\Delta}_t^{\mathrm{TV}} \le T(T-1)R_{\max}\,\bar{\Delta}^{\mathrm{TV}}.$$

Proof. Fix stage t and state $x = (s,c)$. Let $\ell^\star \in \arg\max_\ell\{M_\ell(x) + C_\ell(x)\}$ denote the optimal signal, and let $\ell^{\mathrm{myo}} \in \arg\max_\ell M_\ell(x)$ denote the myopic signal.

Decompose the per-state myopic loss into myopic and continuation components:

$$\eta_t^{\mathrm{myo}}(x) = \underbrace{\big(M_{\ell^\star} - M_{\ell^{\mathrm{myo}}}\big)}_{\le 0 \text{ (myopic maximizes } M)} + \big(C_{\ell^\star} - C_{\ell^{\mathrm{myo}}}\big).$$

Because ℓ^{myo} maximizes the myopic score, the first term is nonpositive. Thus,

$$\eta_t^{\mathrm{myo}}(x) \le C_{\ell^\star}(x) - C_{\ell^{\mathrm{myo}}}(x).$$

Recall that the continuation value is $C_\ell(x) = \mathbb{E}_{P_\ell(\cdot|x)}[V_{t+1}^\star(X_{t+1})]$.

By the standard total-variation coupling inequality—for any bounded function f and distributions P, Q:

$$|\mathbb{E}_P f - \mathbb{E}_Q f| \le \mathrm{sp}(f) \cdot \mathrm{TV}(P, Q).$$

Applying this with $f = V_{t+1}^\star$, we have

$$C_{\ell^\star}(x) - C_{\ell^{\mathrm{myo}}}(x) \le \Phi_{t+1} \cdot \mathrm{TV}\big(P_{\ell^\star}(\cdot \mid x),\, P_{\ell^{\mathrm{myo}}}(\cdot \mid x)\big) \le \Phi_{t+1} \cdot \Delta_t^{\mathrm{TV}}(x).$$

Nonnegativity ($\eta_t^{\mathrm{myo}}(x) \ge 0$) follows from the optimality of $\ell^\star$. This establishes the local bound $0 \le \eta_t^{\mathrm{myo}}(x) \le \Phi_{t+1}\, \Delta_t^{\mathrm{TV}}(x)$.

For the global bound, define the regret function $R_t(x) := V_t^\star(x) - V_t^{\pi^{\mathrm{myo}}}(x)$ and the per-state gap $\delta_t(x) := V_t^\star(x) - Q_t^\star\big(x, \pi_t^{\mathrm{myo}}(x)\big)$. By the Bellman recursion, $R_t(x) \le \delta_t(x) + \|R_{t+1}\|_\infty$. Taking the supremum over x and telescoping from $t = 1$ to $T - 1$ with $R_{T+1} \equiv 0$:

$$\|R_1\|_\infty \le \sum_{t=1}^{T-1} \|\delta_t\|_\infty \le \sum_{t=1}^{T-1} \Phi_{t+1}\, \bar{\Delta}_t^{\mathrm{TV}}.$$

Under the standing assumption $|R(s,a)| \le R_{\max}$, we have $\Phi_{t+1} \le 2(T-t)R_{\max}$. Substituting:

$$\|V_1^\star - V_1^{\pi^{\mathrm{myo}}}\|_\infty \le 2R_{\max} \sum_{t=1}^{T-1} (T-t)\, \bar{\Delta}_t^{\mathrm{TV}} \le T(T-1)R_{\max}\, \bar{\Delta}^{\mathrm{TV}}.$$

□

F.16. Proof of Proposition 5 (Full-horizon performance guarantee)

Proof. We prove the result in two steps. First, we derive the local Bellman-gap bound for the approximate rule. We then aggregate those local losses across the trajectory using the occupancy decomposition.

Let $Q_t^{\lambda,\star}(x, \ell)$ denote the exact penalized action value at stage t, state x, and level ℓ, and let

$$V_t^{\lambda,\star}(x) := \max_{\ell \in \mathcal{L}} Q_t^{\lambda,\star}(x, \ell).$$

Using the baseline level ℓ_0 fixed in the statement, define the exact penalized signal advantage by

$$\Delta_{t,\ell}^{\lambda,\star}(x) := Q_t^{\lambda,\star}(x, \ell) - Q_t^{\lambda,\star}(x, \ell_0).$$

Because the baseline term $Q_t^{\lambda,\star}(x, \ell_0)$ does not depend on ℓ, maximizing $\Delta_{t,\ell}^{\lambda,\star}(x)$ over ℓ is equivalent to maximizing $Q_t^{\lambda,\star}(x, \ell)$ over ℓ.

Let $\ell_t^\star(x) \in \arg\max_{\ell \in \mathcal{L}} \Delta_{t,\ell}^{\lambda,\star}(x)$ be an exact penalized-optimal level, and let $\hat{\ell}_t(x) := \pi_t^{\lambda,\mathrm{app}}(x) \in \arg\max_{\ell \in \mathcal{L}} \{G_\ell(x) - \lambda \Psi_\ell(x)\}$ be the approximate one-step choice. Define the local Bellman gap of the approximate rule by

$$\delta_t^{\lambda,\mathrm{app}}(x) := V_t^{\lambda,\star}(x) - Q_t^{\lambda,\star}(x, \hat{\ell}_t(x)).$$

By the argmax equivalence just established,

$$\delta_t^{\lambda,\mathrm{app}}(x) = \Delta_{t,\ell_t^\star(x)}^{\lambda,\star}(x) - \Delta_{t,\hat{\ell}_t(x)}^{\lambda,\star}(x).$$

Now use the assumed approximation decomposition $\Delta_{t,\ell}^{\lambda,\star}(x) = G_\ell(x) - \lambda\Psi_\ell(x) + \varepsilon_{t,\ell}^{\lambda}(x)$ with $|\varepsilon_{t,\ell}^{\lambda}(x)| \le b_{t,\ell}^{\lambda}(x)$. Substituting,

$$\delta_t^{\lambda,\mathrm{app}}(x) = \Big(G_{\ell_t^\star}(x) - \lambda\Psi_{\ell_t^\star}(x)\Big) - \Big(G_{\hat{\ell}_t}(x) - \lambda\Psi_{\hat{\ell}_t}(x)\Big) + \varepsilon_{t,\ell_t^\star}^{\lambda}(x) - \varepsilon_{t,\hat{\ell}_t}^{\lambda}(x). \tag{27}$$

By optimality of $\hat{\ell}_t(x)$ for the approximate score $G_\ell(x) - \lambda\Psi_\ell(x)$, the score-difference term on the right-hand side is nonpositive. Therefore

$$\delta_t^{\lambda,\mathrm{app}}(x) \le \big|\varepsilon_{t,\ell_t^\star}^{\lambda}(x)\big| + \big|\varepsilon_{t,\hat{\ell}_t}^{\lambda}(x)\big| \le 2\max_{\ell\in\mathcal{L}} b_{t,\ell}^{\lambda}(x).$$

Applying the standard occupancy decomposition for finite-horizon MDPs (Puterman 1994, Sec. 4.5) to the penalized dynamic program, let $d_t^{\pi^{\lambda,\mathrm{app}}}(x;\nu)$ denote the stage-t occupancy measure under the approximate policy from initial distribution ν. Then

$$J_\nu^\lambda(\pi^{\lambda,\star}) - J_\nu^\lambda(\pi^{\lambda,\mathrm{app}}) = \sum_{t=1}^{T}\sum_{x\in\mathcal{X}} d_t^{\pi^{\lambda,\mathrm{app}}}(x;\nu)\,\delta_t^{\lambda,\mathrm{app}}(x).$$

Substituting the local bound,

$$\begin{aligned} J_\nu^\lambda(\pi^{\lambda,\star}) - J_\nu^\lambda(\pi^{\lambda,\mathrm{app}}) &\le 2\sum_{t=1}^{T}\sum_{x\in\mathcal{X}} d_t^{\pi^{\lambda,\mathrm{app}}}(x;\nu)\,\max_{\ell\in\mathcal{L}} b_{t,\ell}^{\lambda}(x) \\ &= 2\sum_{t=1}^{T}\mathbb{E}_\nu^{\pi^{\lambda,\mathrm{app}}}\Big[\max_{\ell\in\mathcal{L}} b_{t,\ell}^{\lambda}(X_t)\Big]. \end{aligned} \tag{28}$$

This is the first displayed bound in the proposition. Finally, since $\max_{\ell\in\mathcal{L}} b_{t,\ell}^{\lambda}(X_t) \le \big\|\max_{\ell\in\mathcal{L}} b_{t,\ell}^{\lambda}(\cdot)\big\|_\infty$, we obtain

$$J_\nu^\lambda(\pi^{\lambda,\star}) - J_\nu^\lambda(\pi^{\lambda,\mathrm{app}}) \le 2\sum_{t=1}^{T}\big\|\max_{\ell\in\mathcal{L}} b_{t,\ell}^{\lambda}(\cdot)\big\|_\infty,$$

which is the second displayed bound. □

F.17. Proof of Proposition D.1 (Regularization under over-restrictive burden misspecification)

Proof. Under over-restrictive misspecification, $\hat{B} \ge B$ pointwise, so for any policy π,

$$\mathbb{E}^\pi\Big[\textstyle\sum_{t=1}^{T} B(X_t,\ell_t)\Big] \le \mathbb{E}^\pi\Big[\textstyle\sum_{t=1}^{T} \hat{B}(X_t,\ell_t)\Big].$$

Applying this to π_{SPD} and using the believed-feasibility condition

$$\mathbb{E}^{\pi_{\mathrm{SPD}}}\Big[\textstyle\sum_{t=1}^{T} \hat{B}(X_t,\ell_t)\Big] \le \bar{\Psi},$$

we obtain

$$\mathbb{E}^{\pi_{\mathrm{SPD}}}\Big[\textstyle\sum_{t=1}^{T} B(X_t,\ell_t)\Big] \le \bar{\Psi}.$$

Hence

$$\Delta\bar{\Psi}_{\mathrm{SPD}} = \bar{\Psi} - \mathbb{E}^{\pi_{\mathrm{SPD}}}\Big[\textstyle\sum_{t=1}^{T} B(X_t,\ell_t)\Big] \ge 0,$$

which proves (i).

For (ii), by the definition of $\Delta\bar{\Psi}_{\mathrm{SPD}}$,

$$\mathbb{E}^{\pi_{\mathrm{SPD}}}\Big[\textstyle\sum_{t=1}^{T} B(X_t,\ell_t)\Big] = \bar{\Psi} - \Delta\bar{\Psi}_{\mathrm{SPD}}.$$

Thus π_{SPD} is feasible for the constrained design problem with budget $\bar{\Psi} - \Delta\bar{\Psi}_{\mathrm{SPD}}$, and therefore

$$J_\nu(\pi_{\mathrm{SPD}}) \le V^\star\big(\bar{\Psi} - \Delta\bar{\Psi}_{\mathrm{SPD}}\big).$$

By Proposition 2, $V^\star$ is concave in $\bar{\Psi}$ and $\lambda^\star \in \partial V^\star(\bar{\Psi})$, so the supergradient inequality gives

$$V^\star\big(\bar{\Psi} - \Delta\bar{\Psi}_{\mathrm{SPD}}\big) \le V^\star(\bar{\Psi}) - \lambda^\star\,\Delta\bar{\Psi}_{\mathrm{SPD}}.$$

Combining the two displays yields (ii).

For (iii), by the definition of η,

$$J_\nu(\pi_{\mathrm{RD}}) = V^\star(\bar{\Psi}) - \eta.$$

Subtracting the bound in (ii) gives

$$J_\nu(\pi_{\mathrm{RD}}) - J_\nu(\pi_{\mathrm{SPD}}) \ge \lambda^\star\,\Delta\bar{\Psi}_{\mathrm{SPD}} - \eta,$$

which is strictly positive whenever

$$\lambda^\star\,\Delta\bar{\Psi}_{\mathrm{SPD}} > \eta.$$

□

F.18. Proof of Proposition D.2 (Finite-sample dominance of RD over plug-in SPD under burden-estimation error)

Proof. Define the plug-in SPD regret random variable

$$\mathcal{R}_{\mathrm{SPD}}(\hat{a}) := V^\star(\bar{\Psi}) - J_\nu\big(\pi_{\mathrm{SPD}}(B_{\hat{a}})\big).$$

By assumption, $\Pr(\hat{a} \in \mathcal{N}) = 1$, and on $\mathcal{N}$ the plug-in SPD regret satisfies the quadratic sandwich

$$\underline{L}\,\|\hat{a} - a^\star\|^2 \le \mathcal{R}_{\mathrm{SPD}}(\hat{a}) \le \overline{L}\,\|\hat{a} - a^\star\|^2 \qquad \text{a.s.}$$

Taking expectations yields

$$\underline{L}\,\mathbb{E}\|\hat{a} - a^\star\|^2 \le \mathbb{E}[\mathcal{R}_{\mathrm{SPD}}(\hat{a})] \le \overline{L}\,\mathbb{E}\|\hat{a} - a^\star\|^2.$$

Using the assumed Euclidean mean-square error identity,

$$\mathbb{E}\|\hat{a} - a^\star\|^2 = \frac{\sigma^2 d}{N\kappa_{\min}},$$

we obtain

$$\frac{\underline{L}\,\sigma^2 d}{N\kappa_{\min}} \le \mathbb{E}\Big[V^\star(\bar{\Psi}) - J_\nu\big(\pi_{\mathrm{SPD}}(B_{\hat{a}})\big)\Big] \le \frac{\overline{L}\,\sigma^2 d}{N\kappa_{\min}}.$$

For RD, by the definition of the oracle restriction cost η,

$$V^\star(\bar{\Psi}) - J_\nu(\pi_{\mathrm{RD}}) \le \eta$$

for the best budget-feasible RD-family policy at $\bar{\Psi}$. Because π_{RD} does not depend on $\hat{a}$, this bound is deterministic with respect to burden-estimation noise.

Therefore, if

$$\frac{\underline{L}\,\sigma^2 d}{N\kappa_{\min}} > \eta,$$

then

$$\mathbb{E}\Big[V^\star(\bar{\Psi}) - J_\nu\big(\pi_{\mathrm{SPD}}(B_{\hat{a}})\big)\Big] > \eta \ge V^\star(\bar{\Psi}) - J_\nu(\pi_{\mathrm{RD}}),$$

so the RD policy has strictly lower expected true-environment regret than the corresponding plug-in SPD policy. Rearranging gives

$$N < \frac{\underline{L}\,\sigma^2 d}{\eta\,\kappa_{\min}} = N^\star(d, \sigma, \eta, \kappa_{\min}, \underline{L}).$$

□

F.19. Structural Interpretability of the Factored Kernel

Remark F.2 (Structural interpretability of the factored kernel). The factored-kernel representation $P_\ell = \mu \cdot F \cdot K$ is policy-equivalent to the unfactored augmented-state MDP. Its value is structural: it yields the $\mathcal{I}$–$\mathcal{D}$–$\mathcal{E}$ decomposition, exposes modular estimation objects (μ, F, K), certifies the disappearance of the harmful channel when K is ℓ-independent conditional on action, and makes clear why the reduced form cannot separate efficacy drift from action-mediated displacement.

Justification. Policy-equivalence is immediate from

$$P_\ell(s', c' \mid s, c) = \sum_{a \in \mathcal{A}} \mu(a \mid s, c, \ell)\, F(s' \mid s, a)\, K(c' \mid c, s, a, \ell),$$

since the Bellman equation depends on the transition law only through these entries. The decomposition claim follows from Lemma 1; the certification claim from Proposition 3; and the non-separability claim from Proposition 1. □

F.20. Proof of Lemma E.1 (Monotonicity of Ψ_ℓ)

Proof. Fix a state $x = (s, c)$ and two signal levels satisfying $\ell' > \ell$. Recall that $\Psi_\ell(x)$ is defined relative to the baseline level ℓ_0 through expected downstream unfamiliarity. To prove monotonicity of Ψ_ℓ, it is enough to prove monotonicity of expected unfamiliarity itself. Let $U_\ell(x) := \mathbb{E}[\mathcal{U}(S_{t+1}, C_{t+1}) \mid x, \ell]$. Then

$$\Psi_\ell(x) = U_\ell(x) - U_{\ell_0}(x).$$

Since the baseline term $U_{\ell_0}(x)$ does not depend on ℓ, it suffices to show that

$$U_{\ell'}(x) \geq U_\ell(x).$$

We start by conditioning on the realized action. By the law of iterated expectations,

$$U_\ell(x) = \sum_a \mu(a \mid x, \ell)\, W(a, \ell),$$

where $W(a, \ell) := \mathbb{E}[\mathcal{U}(S_{t+1}, C_{t+1}) \mid x, \ell, a]$.

We now compare $W(a, \ell')$ and $W(a, \ell)$ for a fixed action a. By condition (b), increasing signal intensity weakly raises the probability of transitioning into the low-efficacy state:

$$K(\mathsf{low} \mid c, s, a, \ell') \geq K(\mathsf{low} \mid c, s, a, \ell).$$

By condition (c), next-period unfamiliarity is weakly larger when the next efficacy state is low than when it is high. Therefore,

$$W(a, \ell') \geq W(a, \ell) \qquad \text{for every action } a.$$

Using the expectation under ℓ', we have

$$\begin{aligned} U_{\ell'}(x) &= \sum_a \mu(a \mid x, \ell')\, W(a, \ell') \\ &\geq \sum_a \mu(a \mid x, \ell')\, W(a, \ell), \end{aligned} \tag{29}$$

where the inequality uses the pointwise bound $W(a, \ell') \geq W(a, \ell)$.

Next invoke condition (a), which states that increasing signal intensity shifts the action distribution toward actions associated with weakly higher unfamiliarity weights. Applying that condition to the weights $W(a,\ell)$ gives

$$\begin{aligned}\sum_a \mu(a \mid x,\ell')\, W(a,\ell) &\geq \sum_a \mu(a \mid x,\ell)\, W(a,\ell) \\ &= U_\ell(x). \end{aligned} \tag{30}$$

Combining (29) and (30), we get

$$U_{\ell'}(x) \geq U_\ell(x),$$

and thus

$$\Psi_{\ell'}(x) = U_{\ell'}(x) - U_{\ell_0}(x) \geq U_\ell(x) - U_{\ell_0}(x) = \Psi_\ell(x).$$

□

F.21. Proof of Proposition E.1 (Ordered-menu threshold structure)

Proof. Fix an augmented state $x = (s,c) \in \mathcal{X}$. For each signal level $\ell \in \mathcal{L}$, recall that the score is

$$S_\ell(x;\lambda) := G_\ell(x) - \lambda \Psi_\ell(x).$$

We prove the three claims in order.

Part (i): monotonicity of the optimal level in λ. Take two levels $\ell' < \ell$. Then

$$S_\ell(x;\lambda) - S_{\ell'}(x;\lambda) = \big(G_\ell(x) - G_{\ell'}(x)\big) - \lambda\big(\Psi_\ell(x) - \Psi_{\ell'}(x)\big). \tag{31}$$

By hypothesis, $\Psi_0(x) \leq \Psi_1(x) \leq \cdots \leq \Psi_k(x)$, and so we have $\Psi_\ell(x) - \Psi_{\ell'}(x) \geq 0$. From (31), we have that the difference $S_\ell(x;\lambda) - S_{\ell'}(x;\lambda)$ is weakly decreasing in λ. This is exactly the decreasing-differences condition in the pair (ℓ,λ). Since the signal menu $\mathcal{L}$ is a finite ordered lattice, Topkis's monotonicity theorem implies that both the maximal and minimal selections from $\arg\max_{\ell\in\mathcal{L}} S_\ell(x;\lambda)$ are weakly decreasing in λ; in particular, any unique selector is weakly decreasing.

Part (ii): interval structure of the active regions. Fix a level $\ell \in \mathcal{L}$. Its optimality region is

$$\Lambda_\ell(x) := \{\lambda \geq 0 : S_\ell(x;\lambda) \geq S_j(x;\lambda) \text{ for all } j \in \mathcal{L}\}.$$

For each competitor $j \in \mathcal{L}$, the inequality $S_\ell(x;\lambda) \geq S_j(x;\lambda)$ is equivalent to a single linear inequality in λ, hence defines a half-line, the whole line, a singleton, or the empty set. Therefore

$$\Lambda_\ell(x) = \bigcap_{j\in\mathcal{L}} \{\lambda \geq 0 : S_\ell(x;\lambda) \geq S_j(x;\lambda)\}$$

is an intersection of intervals on the real line, and is therefore itself an interval, possibly empty.

Since there are $|\mathcal{L}| = k+1$ candidate levels, there can be at most $k+1$ nonempty optimality intervals.

Part (iii): transition thresholds. Consider two adjacent efficient levels on the upper envelope, say $\ell^{(j-1)}$ and $\ell^{(j)}$. At the value of λ where the optimum switches between them, their score lines must intersect. Hence

$$G_{\ell^{(j)}}(x) - \lambda \Psi_{\ell^{(j)}}(x) = G_{\ell^{(j-1)}}(x) - \lambda \Psi_{\ell^{(j-1)}}(x).$$

By the pre-pruning step, distinct efficient levels cannot have the same displacement coordinate: if two levels shared the same Ψ value, only the one with the larger gain could remain efficient. Thus

$$\Psi_{\ell^{(j)}}(x) \neq \Psi_{\ell^{(j-1)}}(x),$$

and solving for λ yields

$$\bar{\lambda}_j(x) = \frac{G_{\ell^{(j)}}(x) - G_{\ell^{(j-1)}}(x)}{\Psi_{\ell^{(j)}}(x) - \Psi_{\ell^{(j-1)}}(x)}.$$

This is exactly the threshold value stated in the proposition. □

F.22. Proof of Proposition E.2 (Efficient levels and dominated-level pruning)

Proof. Fix an augmented state $x = (s, c) \in \mathcal{X}$. For each level $\ell \in \mathcal{L}$, associate the corresponding point $(\Psi_\ell(x), G_\ell(x))$ in the (Ψ, G)-plane. Recall that level ℓ is optimal at price λ if and only if

$$S_\ell(x;\lambda) \geq S_{\ell'}(x;\lambda) \qquad \text{for all } \ell' \in \mathcal{L},$$

where

$$S_\ell(x;\lambda) = G_\ell(x) - \lambda\Psi_\ell(x).$$

Level ℓ is optimal at price λ if and only if its point maximizes the linear functional $(\Psi, G) \mapsto G - \lambda\Psi$. Maximizing $G - \lambda\Psi$ selects the feasible point(s) supported by the highest line of slope λ in the (Ψ, G)-plane. Hence a level is optimal for some $\lambda \geq 0$ if and only if its point admits a supporting line of slope $\lambda \geq 0$, i.e., lies on the portion of the upper concave envelope exposed by nonnegative slopes; if the supporting line coincides with a flat face, all points on that face are optimal at that same λ.

Next, suppose the point $(\Psi_\ell(x), G_\ell(x))$ lies strictly below the upper concave envelope. Then no supporting line of slope $\lambda \geq 0$ can touch the feasible set first at that point. Equivalently, for every $\lambda \geq 0$, there exists another level $\ell' \in \mathcal{L}$ such that

$$S_{\ell'}(x;\lambda) > S_\ell(x;\lambda).$$

Hence ℓ is never optimal for any λ. Removing such a dominated level does not alter the upper concave envelope of the feasible set. Therefore it does not change the maximizer of $G - \lambda\Psi$, and hence does not change the optimal policy at any λ. This proves the menu-pruning claim.

To complete the proof, we use the monotonicity of adjacent slopes. Let

$$(\Psi_{\ell^{(0)}}(x), G_{\ell^{(0)}}(x)),\ (\Psi_{\ell^{(1)}}(x), G_{\ell^{(1)}}(x)),\ \ldots,\ (\Psi_{\ell^{(m)}}(x), G_{\ell^{(m)}}(x))$$

be the efficient points ordered by increasing displacement coordinate. By part (i), these points lie on the portion of the upper concave envelope exposed by supporting lines of slope $\lambda \geq 0$. The slope of a concave envelope is weakly decreasing as one moves to the right. Therefore the secant slopes between successive efficient points satisfy

$$r_1(x) \geq r_2(x) \geq \cdots \geq r_m(x).$$

It remains to show that each $r_j(x) \geq 0$. Because gains and displacement are defined relative to silence,

$$(\Psi_{\ell^{(0)}}(x), G_{\ell^{(0)}}(x)) = (0, 0).$$

If, for some j, we had $r_j(x) < 0$, then

$$G_{\ell^{(j)}}(x) < G_{\ell^{(j-1)}}(x) \qquad \text{and} \qquad \Psi_{\ell^{(j)}}(x) > \Psi_{\ell^{(j-1)}}(x).$$

It would follow that for every $\lambda \geq 0$,

$$G_{\ell^{(j)}}(x) - \lambda\Psi_{\ell^{(j)}}(x) < G_{\ell^{(j-1)}}(x) - \lambda\Psi_{\ell^{(j-1)}}(x),$$

so $\ell^{(j)}$ could not be efficient. This contradicts the definition of the efficient set. Hence $r_j(x) \geq 0$ for every $j = 1, \ldots, m$. The boundary case $r_j(x) = 0$ can occur when adjacent efficient points lie on a flat face supported at $\lambda = 0$, which is allowed under part (i). □

F.23. Proof of Corollary E.1 (Intermediate signals can be strictly optimal)

Proof. Fix an augmented state $x = (s, c) \in \mathcal{X}$, and let

$$\ell^{(0)} < \ell^{(1)} < \cdots < \ell^{(m)}$$

be the efficient levels identified in Proposition E.2. By Proposition E.2, these are exactly the levels on the portion of the upper concave envelope exposed by supporting lines of nonnegative slope. By Proposition E.1, the adjacent slopes

$$r_j(x) = \frac{G_{\ell^{(j)}}(x) - G_{\ell^{(j-1)}}(x)}{\Psi_{\ell^{(j)}}(x) - \Psi_{\ell^{(j-1)}}(x)}, \qquad j = 1, \ldots, m,$$

are exactly the threshold values at which the optimal level changes as λ varies. Now take any interior efficient level $\ell^{(j)}$ with $0 < j < m$ such that

$$r_{j+1}(x) < r_j(x).$$

The two relevant adjacent thresholds for $\ell^{(j)}$ are therefore $r_{j+1}(x)$ and $r_j(x)$. Since the inequality is strict, the interval $\big(r_{j+1}(x), r_j(x)\big)$ is nonempty.

Fix any $\lambda \in \big(r_{j+1}(x), r_j(x)\big)$. At such a λ, the score line of $\ell^{(j)}$ lies strictly above the score lines of its two adjacent efficient neighbors, $\ell^{(j-1)}$ and $\ell^{(j+1)}$. Because the efficient levels lie on the upper concave envelope, this is enough to identify the unique exposed point of the envelope at that λ: between the two adjacent breakpoints $r_{j+1}(x)$ and $r_j(x)$, the upper envelope is supported uniquely by $\ell^{(j)}$. Hence $\ell^{(j)}$ strictly dominates every other efficient level as well.

By Proposition E.2(ii), every dominated level lies strictly below the upper concave envelope and therefore cannot maximize the score at any $\lambda \geq 0$. In particular, no dominated level can beat $\ell^{(j)}$ on the interval above. It follows that $\ell^{(j)}$ is the unique maximizer for every $\lambda \in \big(r_{j+1}(x), r_j(x)\big)$.

Setting

$$\underline{\lambda}_j := r_{j+1}(x), \qquad \bar{\lambda}_j := r_j(x)$$

gives the claimed nonempty open interval on which the intermediate signal is uniquely optimal. □

F.24. Proof of Proposition B.1 (Approximation error bound)

Proof. Fix (s,c) and compare a generic signal level ℓ to the baseline level 0. Let

$$W := \mathcal{U}(S_{t+1}, C_{t+1}) \in [0,1], \qquad m_\ell := \mathbb{E}[W \mid s,c,\ell], \qquad m_0 := \mathbb{E}[W \mid s,c,0],$$

and define

$$\Psi_\ell(s,c) := m_\ell - m_0.$$

By the bounded-curvature assumption of the proposition, $|g''(w)| \le L$ for all $w \in [0,1]$. Taylor-expanding g around the no-signal mean m_0 gives

$$g(w) = g(m_0) + g'(m_0)(w - m_0) + r(w), \qquad |r(w)| \le \frac{L}{2}(w - m_0)^2.$$

Taking expectations under level ℓ and under silence,

$$\mathbb{E}_\ell[g(W)] - \mathbb{E}_0[g(W)] = g'(m_0)(m_\ell - m_0) + \underbrace{\mathbb{E}_\ell[r(W)] - \mathbb{E}_0[r(W)]}_{\varepsilon_\ell(s,c)}.$$

Let $\lambda^{\mathrm{loc}} := -g'(m_0)$. Then

$$\Delta_\ell(s,c) = G_\ell(s,c) - \lambda^{\mathrm{loc}}\Psi_\ell(s,c) + \varepsilon_\ell(s,c).$$

We now bound the remainder term. By the triangle inequality and the Taylor remainder bound,

$$|\varepsilon_\ell(s,c)| \le |\mathbb{E}_\ell[r(W)]| + |\mathbb{E}_0[r(W)]| \le \frac{L}{2}\mathbb{E}_\ell[(W - m_0)^2] + \frac{L}{2}\mathbb{E}_0[(W - m_0)^2].$$

For the baseline term,

$$\mathbb{E}_0[(W - m_0)^2] = \mathrm{Var}_0(W).$$

For the signal-ℓ term,

$$\mathbb{E}_\ell[(W - m_0)^2] = \mathrm{Var}_\ell(W) + (m_\ell - m_0)^2 = \mathrm{Var}_\ell(W) + \Psi_\ell(s,c)^2.$$

Therefore

$$|\varepsilon_\ell(s,c)| \le \frac{L}{2}\Big(\mathrm{Var}_\ell(W) + \mathrm{Var}_0(W) + \Psi_\ell(s,c)^2\Big).$$

This is the bound stated in the proposition. If a fixed design parameter λ is used instead of λ^{loc}, then

$$\Delta_\ell(s,c) - \big(G_\ell(s,c) - \lambda\Psi_\ell(s,c)\big) = \varepsilon_\ell(s,c) + (\lambda - \lambda^{\mathrm{loc}})\Psi_\ell(s,c).$$

Hence, by the triangle inequality,

$$\big|\Delta_\ell(s,c) - (G_\ell(s,c) - \lambda\Psi_\ell(s,c))\big| \le |\varepsilon_\ell(s,c)| + |\lambda - \lambda^{\mathrm{loc}}|\,|\Psi_\ell(s,c)|.$$

Combining this with the bound above yields

$$\big|\Delta_\ell(s,c) - (G_\ell(s,c) - \lambda\Psi_\ell(s,c))\big| \le \frac{L}{2}\Big(\mathrm{Var}_\ell(W) + \mathrm{Var}_0(W) + \Psi_\ell(s,c)^2\Big) + |\lambda - \lambda^{\mathrm{loc}}|\,|\Psi_\ell(s,c)|.$$

This is exactly the displayed calibration-error bound in the proposition.

Finally, if $g'(m_0) \le 0$, then $\lambda^{\mathrm{loc}} = -g'(m_0) \ge 0$, so the linear term admits the interpretation of a local cost price. This proves the proposition. □

F.25. Proof of Proposition E.3 (Multi-level approximation bound)

Proof. We can use the proof of Proposition B.1, which is carried out level by level relative to the baseline level $\ell = 0$, and therefore extends immediately from the binary case to a finite ordered menu.

Fix an augmented state $x = (s, c) \in \mathcal{X}$. For each signal level $\ell \in \mathcal{L}$, Proposition B.1 gives the expansion

$$\Delta_\ell(x) = G_\ell(x) - \lambda^{\mathrm{loc}} \Psi_\ell(x) + \varepsilon_\ell(x),$$

together with the error bound

$$|\varepsilon_\ell(x)| \le \frac{L}{2}\Big(\mathrm{Var}_\ell(W) + \mathrm{Var}_0(W) + \Psi_\ell(x)^2\Big).$$

Nothing in the derivation of Proposition B.1 uses the fact that the menu has only two levels; it only compares a given level ℓ to the baseline level 0. Hence the same expansion and bound hold simultaneously for every $\ell \in \mathcal{L}$.

We now compare the exact maximizing level to the one-step approximation rule. Let

$$\hat{\ell}(x) \in \arg\max_{\ell \in \mathcal{L}} \{G_\ell(x) - \lambda^{\mathrm{loc}} \Psi_\ell(x)\}$$

be the level selected by the approximate score, and let $\ell^\star(x) \in \arg\max_{\ell \in \mathcal{L}} \Delta_\ell(x)$ be the exact optimal level.

Then

$$\begin{aligned}\Delta_{\ell^\star}(x) - \Delta_{\hat{\ell}}(x) = &\Big(G_{\ell^\star}(x) - \lambda^{\mathrm{loc}} \Psi_{\ell^\star}(x)\Big) - \Big(G_{\hat{\ell}}(x) - \lambda^{\mathrm{loc}} \Psi_{\hat{\ell}}(x)\Big) \\ &+ \varepsilon_{\ell^\star}(x) - \varepsilon_{\hat{\ell}}(x). \end{aligned} \tag{32}$$

By definition of $\hat{\ell}(x)$, it maximizes the approximate score $G_\ell(x) - \lambda^{\mathrm{loc}} \Psi_\ell(x)$. Therefore the first line of (32) is nonpositive:

$$\Big(G_{\ell^\star}(x) - \lambda^{\mathrm{loc}} \Psi_{\ell^\star}(x)\Big) - \Big(G_{\hat{\ell}}(x) - \lambda^{\mathrm{loc}} \Psi_{\hat{\ell}}(x)\Big) \le 0.$$

Using this in (32), we have

$$\begin{aligned}\Delta_{\ell^\star}(x) - \Delta_{\hat{\ell}}(x) &\le |\varepsilon_{\ell^\star}(x)| + |\varepsilon_{\hat{\ell}}(x)| \\ &\le 2 \max_{\ell \in \mathcal{L}} |\varepsilon_\ell(x)|.\end{aligned}$$

□

F.26. Proof of Proposition A.1 (Value robustness to within-cycle behavioral drift)

Proof. We prove the result in three steps: (i) bound the one-step transition distortion induced by replacing μ with $\tilde{\mu}$; (ii) propagate that distortion through the state distribution over time; and (iii) translate the resulting distributional error into a trajectory-level value loss.

One-step transition distortion. For a fixed signal level ℓ, let

$$P_\ell^\mu(x' \mid x) := \sum_a \mu(a \mid x, \ell)\, F(s' \mid s, a)\, K(c' \mid c, s, a, \ell)$$

denote the state-transition kernel induced by the response kernel μ. Define $P_\ell^{\tilde{\mu}}$ analogously using $\tilde{\mu}$.

Fix a state $x = (s, c)$ and a signal ℓ. Then

$$P_\ell^\mu(\cdot \mid x) - P_\ell^{\tilde{\mu}}(\cdot \mid x) = \sum_a \big(\mu(a \mid x, \ell) - \tilde{\mu}(a \mid x, \ell)\big)\, F(\cdot \mid s, a)\, K(\cdot \mid c, s, a, \ell). \tag{33}$$

For each action a, the product

$$F(\cdot \mid s, a)\, K(\cdot \mid c, s, a, \ell)$$

is a probability distribution over next states (s', c'). Therefore, by the triangle inequality,

$$\begin{aligned} \|P_\ell^\mu(\cdot \mid x) - P_\ell^{\tilde{\mu}}(\cdot \mid x)\|_1 &\le \sum_a \big|\mu(a \mid x, \ell) - \tilde{\mu}(a \mid x, \ell)\big| \\ &= \|\mu(\cdot \mid x, \ell) - \tilde{\mu}(\cdot \mid x, \ell)\|_1 \\ &\le \delta. \end{aligned} \tag{34}$$

Propagation through the state distribution. Let $\nu_t^\mu, \nu_t^{\tilde{\mu}}$ be the stage-t state distributions under the same policy π, but under response kernels μ and $\tilde{\mu}$, respectively. Since both systems start from the same initial distribution, we have

$$\nu_1^\mu = \nu_1^{\tilde{\mu}}.$$

Using induction, we now show that

$$\|\nu_t^\mu - \nu_t^{\tilde{\mu}}\|_1 \le (t-1)\delta, \quad \forall t.$$

The claim is trivial at $t = 1$. Suppose it holds at stage t. Let $P_{\pi_t}^\mu$ and $P_{\pi_t}^{\tilde{\mu}}$ denote the controlled transition kernels induced by the stage-t decision rule π_t. Then $\nu_{t+1}^\mu = \nu_t^\mu P_{\pi_t}^\mu, \nu_{t+1}^{\tilde{\mu}} = \nu_t^{\tilde{\mu}} P_{\pi_t}^{\tilde{\mu}}$, and we have

$$\begin{aligned} \|\nu_{t+1}^\mu - \nu_{t+1}^{\tilde{\mu}}\|_1 &= \|\nu_t^\mu P_{\pi_t}^\mu - \nu_t^{\tilde{\mu}} P_{\pi_t}^{\tilde{\mu}}\|_1 \\ &\le \|(\nu_t^\mu - \nu_t^{\tilde{\mu}}) P_{\pi_t}^\mu\|_1 + \|\nu_t^{\tilde{\mu}}(P_{\pi_t}^\mu - P_{\pi_t}^{\tilde{\mu}})\|_1. \end{aligned} \tag{35}$$

As stochastic kernels contract total variation, we have

$$\|(\nu_t^\mu - \nu_t^{\tilde{\mu}}) P_{\pi_t}^\mu\|_1 \le \|\nu_t^\mu - \nu_t^{\tilde{\mu}}\|_1,$$

and by (34),

$$\|\nu_t^{\tilde{\mu}}(P_{\pi_t}^\mu - P_{\pi_t}^{\tilde{\mu}})\|_1 \le \delta.$$

Therefore,

$$\begin{aligned} \|\nu_{t+1}^\mu - \nu_{t+1}^{\tilde{\mu}}\|_1 &\le \|\nu_t^\mu - \nu_t^{\tilde{\mu}}\|_1 + \delta \\ &\le (t-1)\delta + \delta \\ &= t\delta. \end{aligned} \tag{36}$$

Convert distributional distortion into value loss. At stage t, the expected reward difference between the two systems has two sources: they may place different probability mass on states, and at a common state, they may induce different action distributions. The first contribution is bounded by $R_{\max} \|\nu_t^\mu - \nu_t^{\tilde{\mu}}\|_1$. For the second contribution, fix a state $x = (s, c)$. We have

$$\left| \sum_a \mu(a \mid x, \pi_t(x)) R(s, a) - \sum_a \tilde{\mu}(a \mid x, \pi_t(x)) R(s, a) \right|$$

$$\leq R_{\max} \|\mu(\cdot \mid x, \pi_t(x)) - \tilde{\mu}(\cdot \mid x, \pi_t(x))\|_1$$
$$\leq R_{\max}\delta. \tag{37}$$

Thus the stage-t reward difference is at most

$$R_{\max} \|\nu_t^{\mu} - \nu_t^{\tilde{\mu}}\|_1 + R_{\max}\delta \leq R_{\max}(t-1)\delta + R_{\max}\delta$$
$$= tR_{\max}\delta. \tag{38}$$

Summing over $t = 1, \ldots, T$ gives

$$|J_T(\pi;\mu) - J_T(\pi;\tilde{\mu})| \leq \sum_{t=1}^{T} tR_{\max}\delta$$
$$= \frac{T(T+1)}{2} R_{\max}\delta. \tag{39}$$

To compare the optimal policies, let

$$\pi_{\mu}^{\star} \in \arg\max_{\pi} J_T(\pi;\mu), \qquad \pi_{\tilde{\mu}}^{\star} \in \arg\max_{\pi} J_T(\pi;\tilde{\mu}).$$

Then

$$J_T(\pi_{\mu}^{\star};\mu) - J_T(\pi_{\tilde{\mu}}^{\star};\mu) = \Big[J_T(\pi_{\mu}^{\star};\mu) - J_T(\pi_{\mu}^{\star};\tilde{\mu})\Big]$$
$$+ \Big[J_T(\pi_{\mu}^{\star};\tilde{\mu}) - J_T(\pi_{\tilde{\mu}}^{\star};\tilde{\mu})\Big]$$
$$+ \Big[J_T(\pi_{\tilde{\mu}}^{\star};\tilde{\mu}) - J_T(\pi_{\tilde{\mu}}^{\star};\mu)\Big]. \tag{40}$$

The middle term is nonpositive by optimality of $\pi_{\tilde{\mu}}^{\star}$ under $\tilde{\mu}$. Each outer term is bounded in absolute value by (39). Therefore

$$J_T(\pi_{\mu}^{\star};\mu) - J_T(\pi_{\tilde{\mu}}^{\star};\mu) \leq T(T+1)R_{\max}\delta.$$

This is exactly the stated robustness bound. □